



Group Number : R25-068

**Ayurvedic Medicine Identification and
Health Management App**

Introduction

Our Ayurvedic Medicine Identification and Health Management App combines the ancient wisdom of Ayurveda with modern technology.

By leveraging

- > Artificial intelligence.**
- > Image processing**
- > Data analytics**

This app makes

- > Ayurvedic care more accessible and personalized.**
- > Helps users identify medicinal plants**
- > Gain health insights**

By focusing demand for holistic and natural health solutions, this app offers an accurate, user-friendly way to bridge traditional practices and modern needs.

Mainly Focusing on,

- > Tongue Disease identification and treatment suggestions**
- > Skin disease identification and treatment Suggestions.**
- > Ayurvedic medicinal plant identification with detailed usage guidelines.**
- > Personalized yoga and lifestyle guidance based on wearable health data.**

Research Questions

- Ayurvedic diagnosis is limited by practitioner dependency, delaying treatment and reducing accessibility.

Skin diseases lack integrated solutions combining Ayurvedic and modern approaches. Health tools fail to offer personalized care by excluding Ayurvedic data, while difficulties in identifying medicinal plants further restrict remedy access.

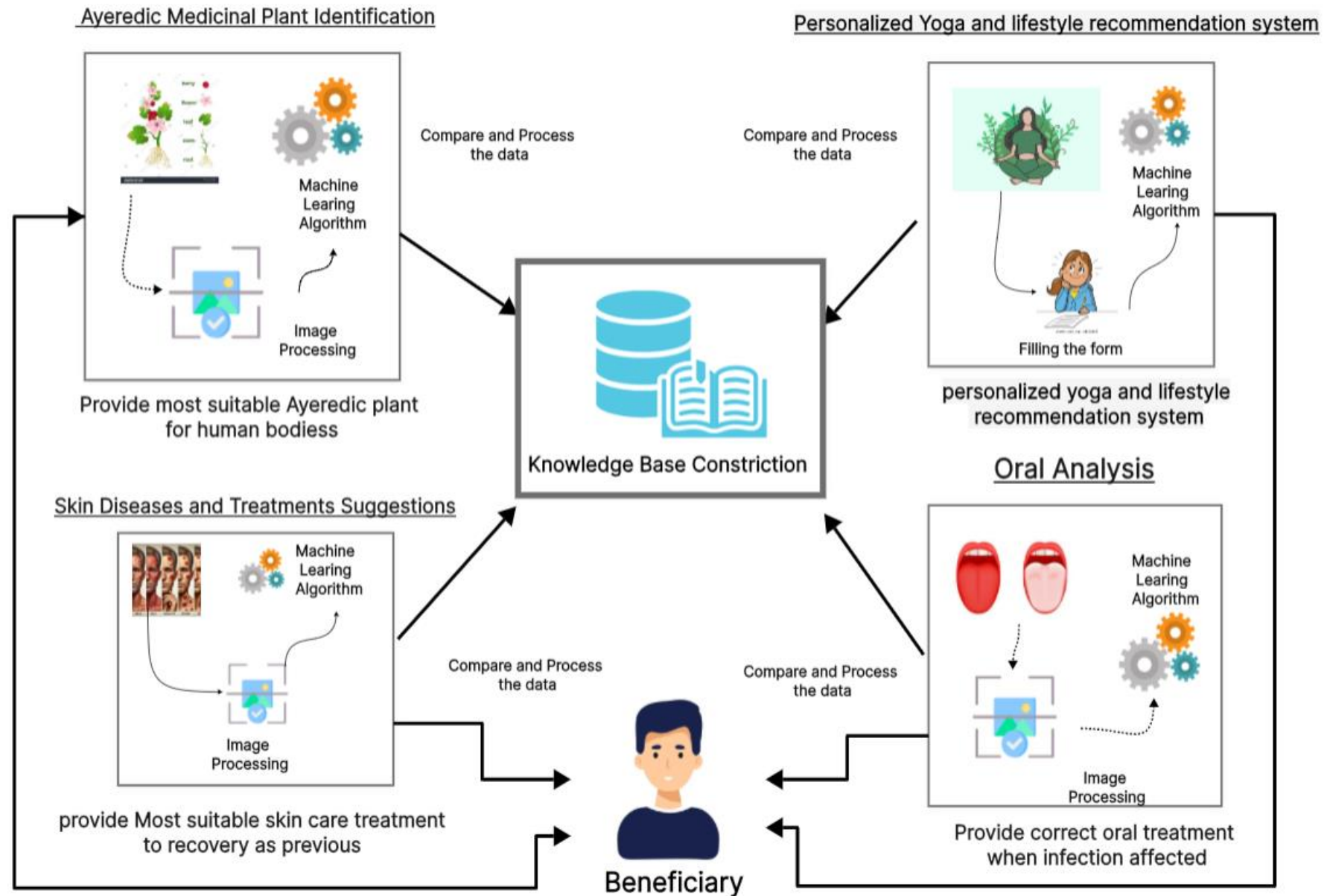


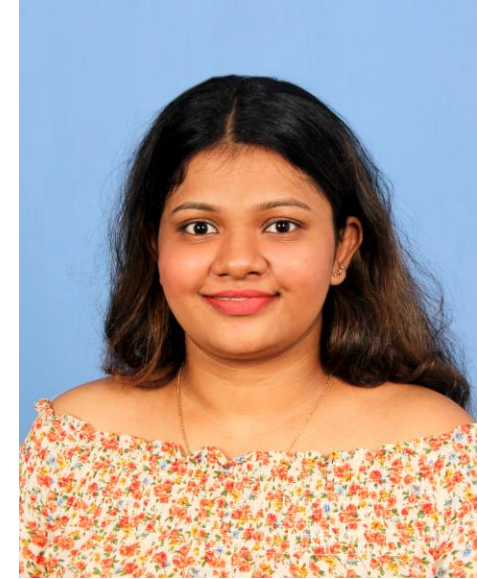
Objectives

- The **Primary Goal** is to create an AI-driven mobile app that combines Ayurvedic principles with modern technology to provide personalized health management and treatment recommendations.
- The objectives include
 - > Tongue imaging analysis for personalized Ayurvedic advice
 - > Skin disease identification framework blending Ayurvedic and modern diagnostics
 - > Tailored yoga and lifestyle recommendations using wearable health data
 - > Medicinal plant identification with detailed usage guidance.



System Diagram





PEIRIS P.H.R - IT21584718

AI-POWERED TONGUE DIAGNOSTICS FOR HEALTH MONITORING, WELLNESS GUIDANCE AND NUTRITION SUGGESTING

Information Technology (IT)

Introduction

Background



- **Tongue diagnosis in Ayurveda:** The condition of the tongue (its color, texture, and coating) is used in Ayurvedic healthcare to assess overall health.
 1. **Detects health issues:** Identify problems like dehydration, nutrient deficiencies, and inflammation.
 2. **AI enhances accuracy :** detecting subtle changes in tongue images.
 3. **Combining AI with Ayurveda Home Remedies:** Integrating AI with Ayurvedic methods makes health assessments more personalized and accessible.
 4. **Reduces clinic visits:** AI-powered tools allow users to monitor their health at home,

Selected Diseases

Black hairy diseases	Fissured Tongue



Black hairy tongue is caused by a buildup of dead skin cells on the tiny bumps on the tongue called papillae



Fissured tongue is a harmless condition that causes grooves or furrows on the tongue



Research Gap

Limited AI Integration in Tongue Diagnosis – Existing healthcare apps do not fully utilize AI for accurate tongue-based health assessments.

No Personalized Wellness Tracking – Most apps lack personalized progress tracking based on tongue health changes.

Lack of Digital Ayurvedic Insights – Ayurvedic tongue diagnosis is not well-represented in modern health apps.

Insufficient Tongue Image Datasets – There is a shortage of annotated tongue images linked to different health conditions.

Limited Access to Remote Health Monitoring – People, especially in rural areas, struggle to access Ayurvedic experts, needing AI-driven remote solutions.

Research Problems

- **No scalable AI-based tongue diagnosis tools** exist for health assessments.
- **Challenges in combining AI with Ayurvedic principles** for accurate analysis.
- **Lack of sufficient and structured tongue image datasets** for AI training.
- **No personalized health tracking** based on tongue condition changes.
- **Limited access to AI-driven Ayurvedic health solutions**, especially in rural areas.



Research Objectives

Specific Objective

- Develop an AI- Powered mobile application to detect diseases on human tongue and suggest home ramadies and nutritions that humen need until they recovery.

Sub Objectives

- Collect a diverse dataset of tongue images** labeled with different health conditions (e.g., normal, dehydrated, inflamed, and infected) to improve AI accuracy.
- Train and deploy a deep learning model** to analyze tongue images, detect health risks, and track changes over time.
- Develop a personalized wellness recommendation system** based on AI analysis and Ayurvedic insights to suggest hydration, nutrition, and lifestyle improvements.
- Integrate a notification system** to remind users about hydration, nutrition intake, and health progress monitoring.
- Ensure secure data storage and privacy measures** to protect user health information.

Requirements Specification

Functional Requirements

- **Health Risk Detection and Staging**

- The app must analyze tongue images to **detect health conditions** (e.g., dehydration, inflammation, infections) and classify them into **mild, moderate, or severe stages** using AI.

- **Personalized Wellness Recommendations**

- Instead of direct treatment, the app should provide **general Ayurvedic-based wellness advice**, including **hydration tips, dietary suggestions, and lifestyle recommendations** based on the detected health condition.

Non - Functional Requirements

- **Data Security and Privacy.**

Ensure secure storage and processing of user data, complying with privacy regulations and safeguarding sensitive health information.

- **Performance.**

The app should deliver real-time results, with image analysis and treatment recommendations completed within a few seconds.

Achievement



- Successful data collection for model training
- Trained two models for Selected Diseases
- Highest accuracy models up to now
- Developed the mobile application (Emulator)

Fissured tongue

- A **fissured tongue** is when the tongue has deep grooves or cracks on its surface.
- It's usually **harmless** and can happen as
 - people get older.
 - food or bacteria can get stuck in these cracks, which might cause irritation or bad breath. Cleaning the tongue regularly can help keep it clean and reduce any issues.





Stages of Fissured tongue

1. Successful data collection for model training

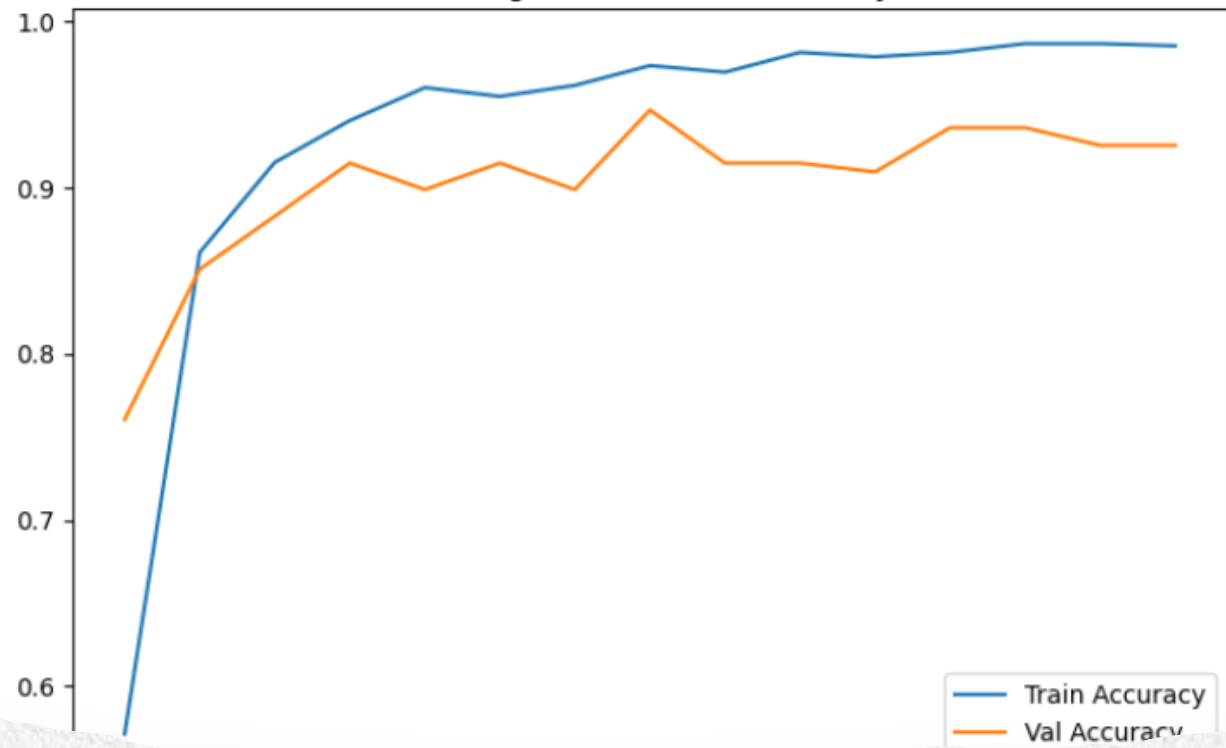


Accuracy

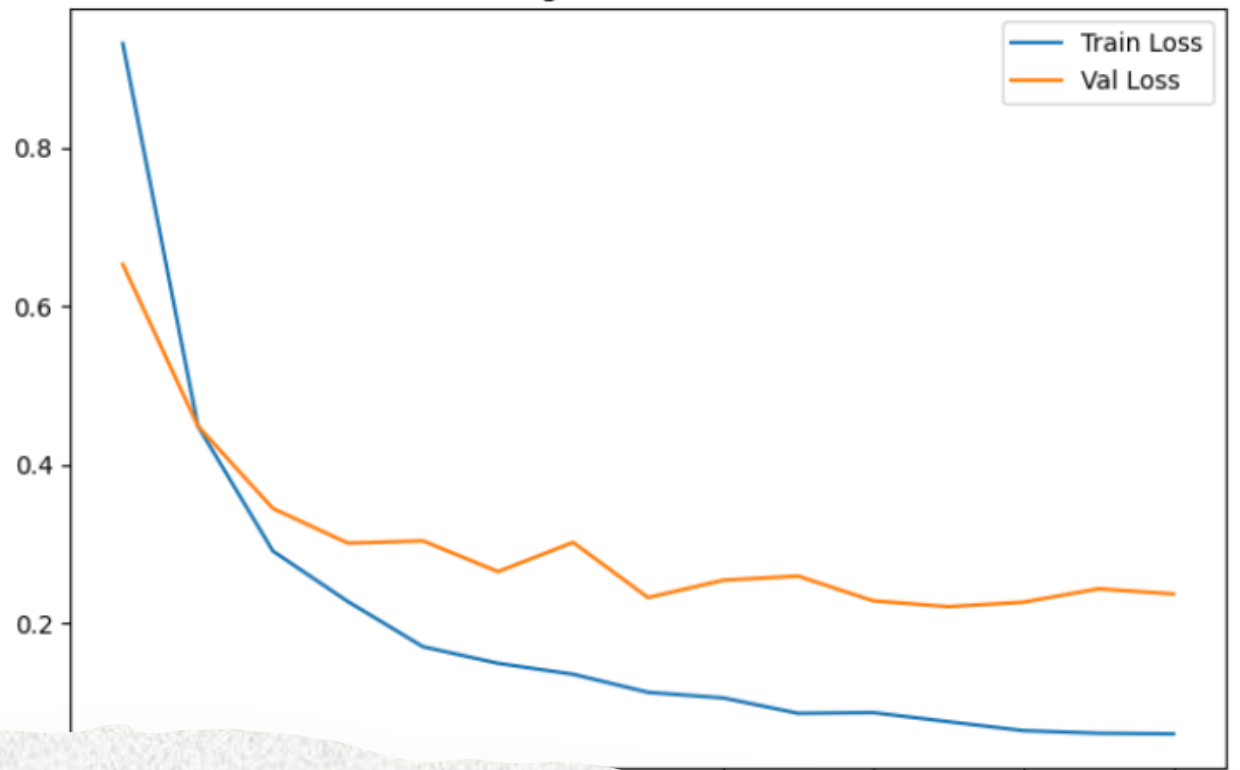
- Accuracy
97%

```
self._warn_if_super_not_called()
Epoch 1/15
24/24 ————— 287s 12s/step - accuracy: 0.4273 - loss: 1.1334 - val_accuracy: 0.7606 - val_loss: 0.6530
Epoch 2/15
24/24 ————— 16s 650ms/step - accuracy: 0.8231 - loss: 0.5062 - val_accuracy: 0.8511 - val_loss: 0.4492
Epoch 3/15
24/24 ————— 20s 641ms/step - accuracy: 0.9114 - loss: 0.3028 - val_accuracy: 0.8830 - val_loss: 0.3452
Epoch 4/15
24/24 ————— 15s 632ms/step - accuracy: 0.9284 - loss: 0.2507 - val_accuracy: 0.9149 - val_loss: 0.3015
Epoch 5/15
24/24 ————— 15s 641ms/step - accuracy: 0.9613 - loss: 0.1757 - val_accuracy: 0.8989 - val_loss: 0.3043
Epoch 6/15
24/24 ————— 17s 735ms/step - accuracy: 0.9450 - loss: 0.1526 - val_accuracy: 0.9149 - val_loss: 0.2655
Epoch 7/15
24/24 ————— 15s 631ms/step - accuracy: 0.9570 - loss: 0.1384 - val_accuracy: 0.8989 - val_loss: 0.3024
Epoch 8/15
24/24 ————— 17s 740ms/step - accuracy: 0.9668 - loss: 0.1224 - val_accuracy: 0.9468 - val_loss: 0.2328
Epoch 9/15
24/24 ————— 18s 741ms/step - accuracy: 0.9691 - loss: 0.1062 - val_accuracy: 0.9149 - val_loss: 0.2547
Epoch 10/15
24/24 ————— 16s 663ms/step - accuracy: 0.9731 - loss: 0.0941 - val_accuracy: 0.9149 - val_loss: 0.2599
Epoch 11/15
24/24 ————— 15s 639ms/step - accuracy: 0.9776 - loss: 0.0852 - val_accuracy: 0.9096 - val_loss: 0.2288
Epoch 12/15
24/24 ————— 15s 636ms/step - accuracy: 0.9789 - loss: 0.0772 - val_accuracy: 0.9362 - val_loss: 0.2214
Epoch 13/15
...
Epoch 14/15
24/24 ————— 15s 644ms/step - accuracy: 0.9854 - loss: 0.0641 - val_accuracy: 0.9255 - val_loss: 0.2438
Epoch 15/15
24/24 ————— 15s 631ms/step - accuracy: 0.9791 - loss: 0.0692 - val_accuracy: 0.9255 - val_loss: 0.2375
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings.
```

Training and Validation Accuracy

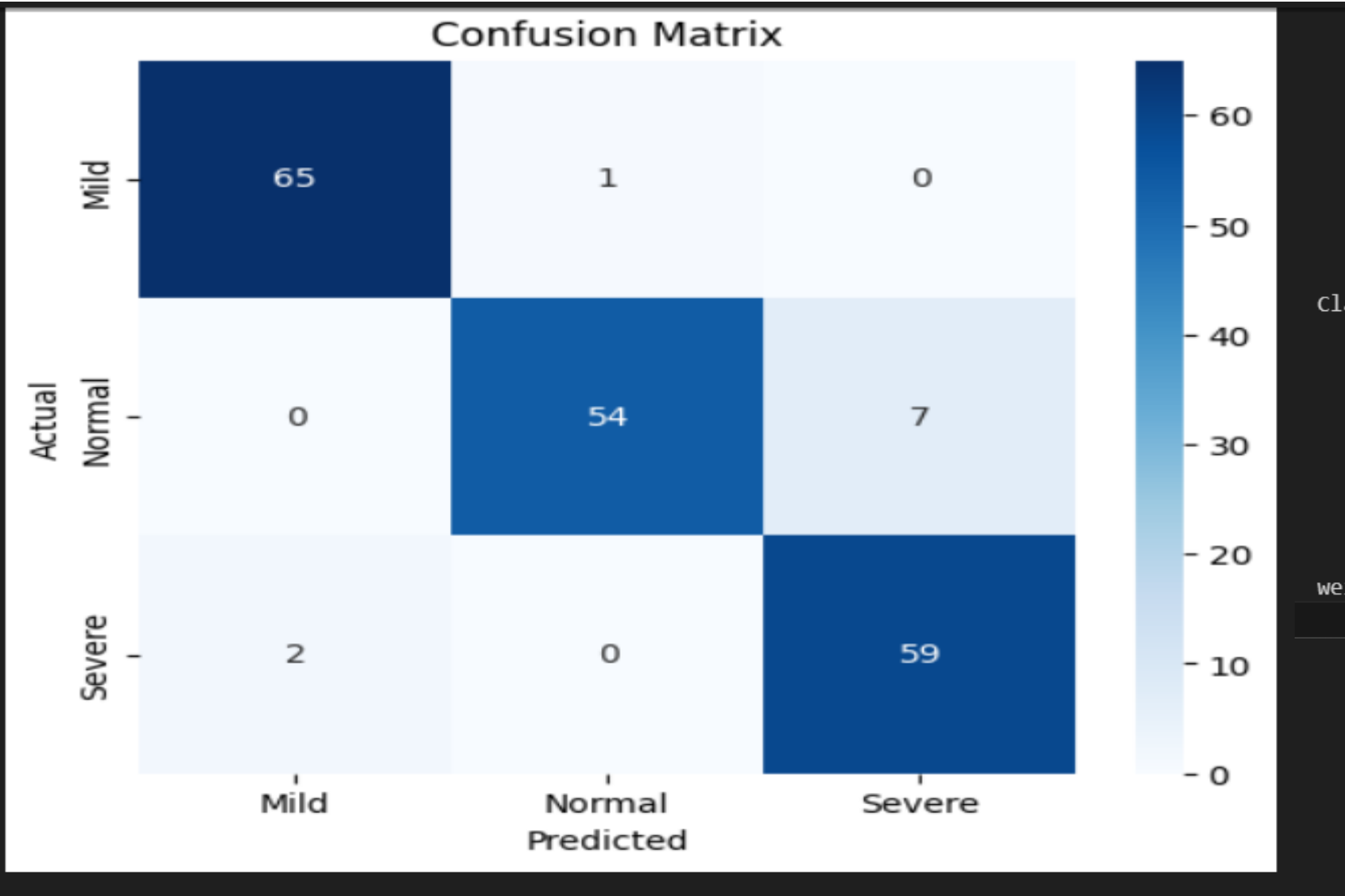


Training and Validation Loss



Testing Validation Accuracy & Lost

Confusion Matrix



Classification Report:

	precision	recall	f1-score	support
Mild	0.97	0.98	0.98	66
Normal	0.98	0.89	0.93	61
Severe	0.89	0.97	0.93	61
accuracy			0.95	188
macro avg	0.95	0.95	0.95	188
weighted avg	0.95	0.95	0.95	188

Layers

Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, 224, 224, 3)	0	-
Conv1 (Conv2D)	(None, 112, 112, 32)	864	input_layer[0][0]
bn_Conv1 (BatchNormalization)	(None, 112, 112, 32)	128	Conv1[0][0]
Conv1_relu (ReLU)	(None, 112, 112, 32)	0	bn_Conv1[0][0]
expanded_conv_depthwise (DepthwiseConv2D)	(None, 112, 112, 32)	288	Conv1_relu[0][0]
expanded_conv_depthwise_... (BatchNormalization)	(None, 112, 112, 32)	128	expanded_conv_depthwi...
expanded_conv_depthwise_... (ReLU)	(None, 112, 112, 32)	0	expanded_conv_depthwi...
expanded_conv_project (Conv2D)	(None, 112, 112, 16)	512	expanded_conv_depthwi...
expanded_conv_project_BN (BatchNormalization)	(None, 112, 112, 16)	64	expanded_conv_project...
block_1_expand (Conv2D)	(None, 112, 112, 96)	1,536	expanded_conv_project...
block_1_expand_BN (BatchNormalization)	(None, 112, 112, 96)	384	block_1_expand[0][0]
block_1_expand_relu (ReLU)	(None, 112, 112, 96)	0	block_1_expand_BN[0][...]
block_1_pad (ZeroPadding2D)	(None, 113, 113, 96)	0	block_1_expand_relu[0...]
block_1_depthwise (DepthwiseConv2D)	(None, 56, 56, 96)	864	block_1_pad[0][0]

Black Hairy Tongue

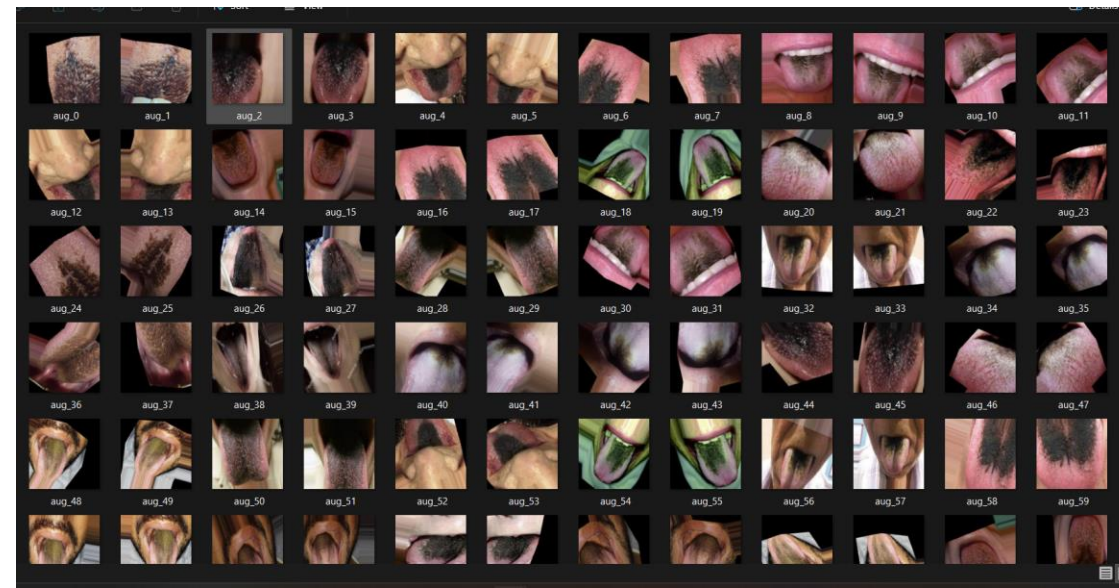
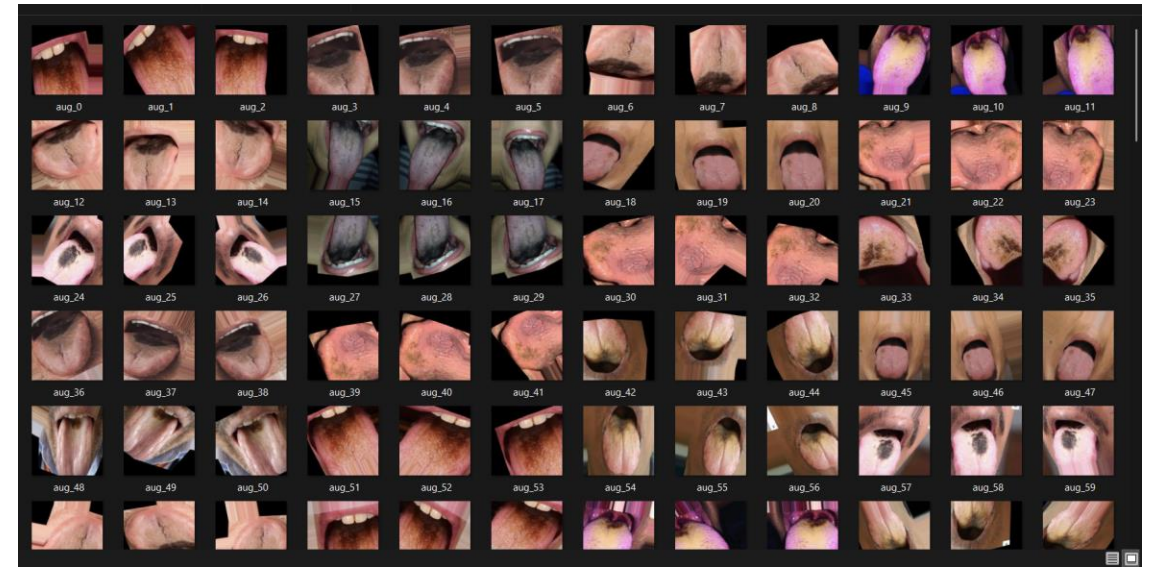
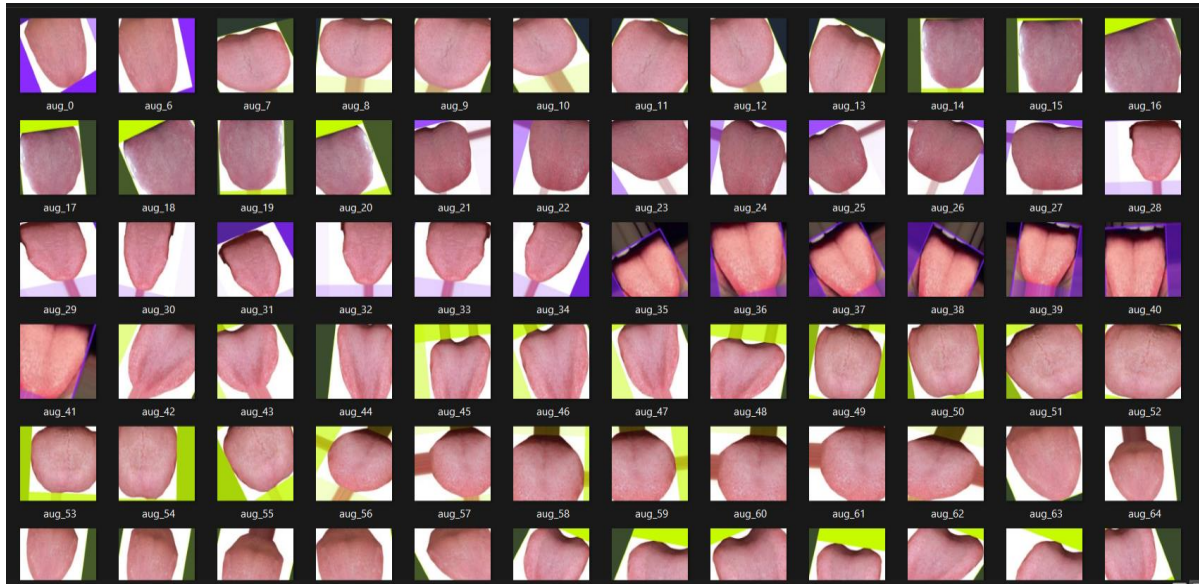
- **Black hairy tongue** is a condition where the tongue looks dark and fuzzy due to buildup of dead skin cells and bacteria. It's usually harmless and can be caused by poor oral hygiene, smoking, or certain medications. Regular brushing can help clear it up.





Stages of Black hairy

1.Successful data collection for model training

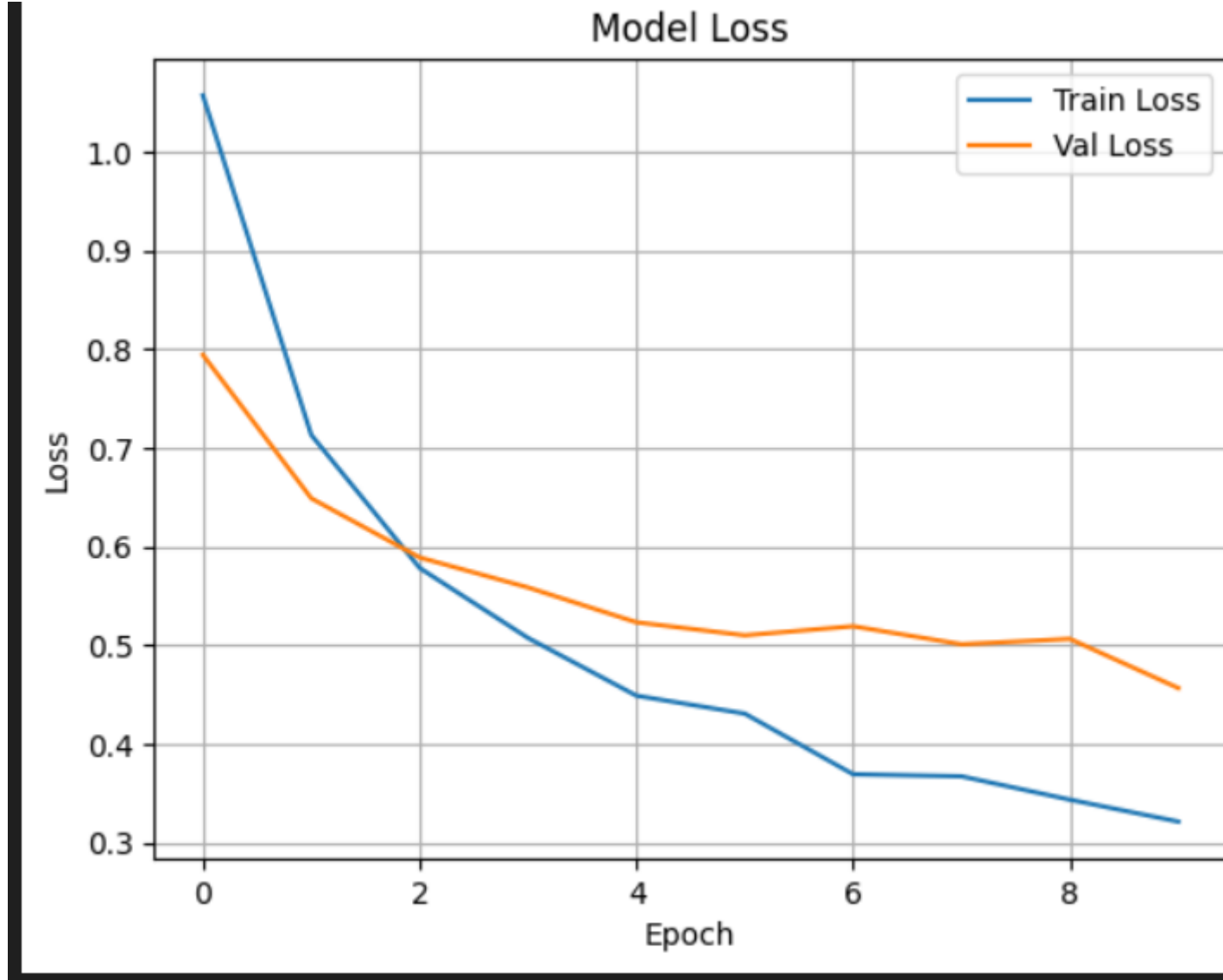
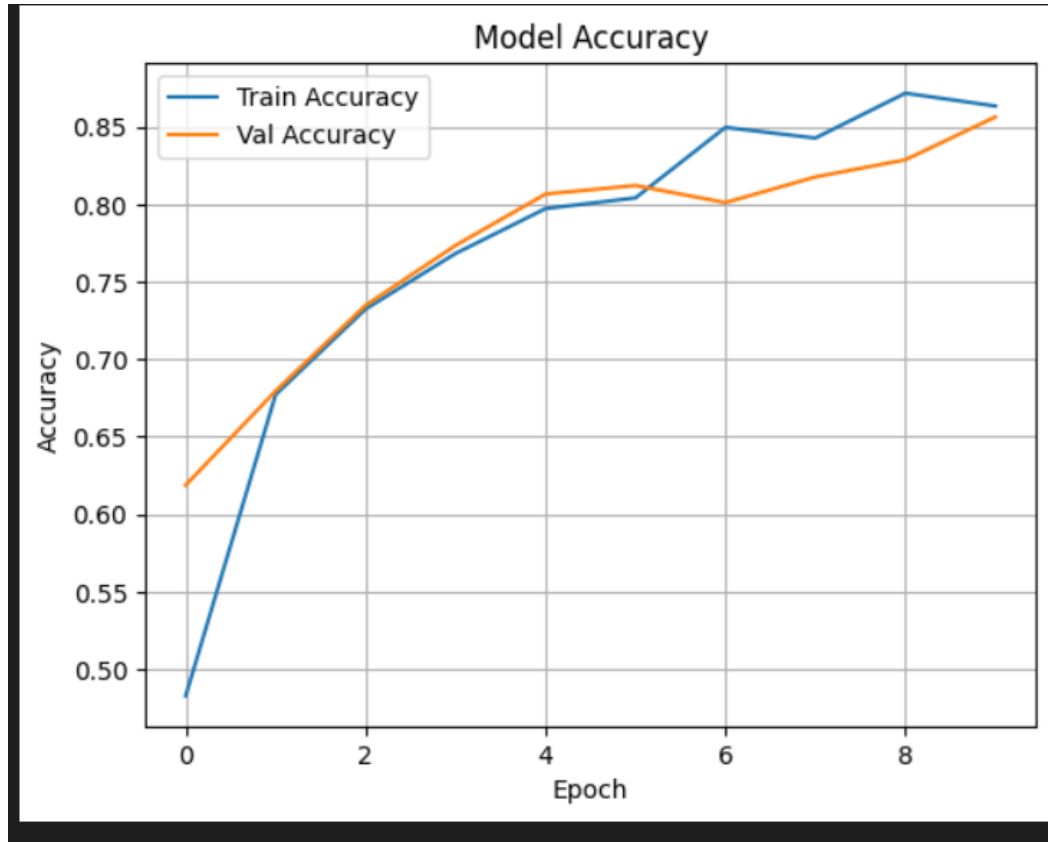


Accuracy – 84%

```
/usr/local/lib/python3.11/dist-packages/keras/src/trainers/data_adapters/py_dataset_adapter.py:121: UserWarning: Your `PyDataset`
  self._warn_if_super_not_called()
Epoch 1/10
23/23 ————— 311s 13s/step - accuracy: 0.3800 - loss: 1.2113 - val_accuracy: 0.6188 - val_loss: 0.7943
Epoch 2/10
23/23 ————— 13s 562ms/step - accuracy: 0.6488 - loss: 0.7824 - val_accuracy: 0.6796 - val_loss: 0.6490
Epoch 3/10
23/23 ————— 15s 643ms/step - accuracy: 0.7330 - loss: 0.5855 - val_accuracy: 0.7348 - val_loss: 0.5890
Epoch 4/10
23/23 ————— 13s 565ms/step - accuracy: 0.7636 - loss: 0.5266 - val_accuracy: 0.7735 - val_loss: 0.5587
Epoch 5/10
23/23 ————— 13s 566ms/step - accuracy: 0.7968 - loss: 0.4534 - val_accuracy: 0.8066 - val_loss: 0.5234
Epoch 6/10
23/23 ————— 13s 573ms/step - accuracy: 0.8130 - loss: 0.4351 - val_accuracy: 0.8122 - val_loss: 0.5102
Epoch 7/10
23/23 ————— 20s 562ms/step - accuracy: 0.8546 - loss: 0.3644 - val_accuracy: 0.8011 - val_loss: 0.5194
Epoch 8/10
23/23 ————— 13s 559ms/step - accuracy: 0.8480 - loss: 0.3620 - val_accuracy: 0.8177 - val_loss: 0.5011
Epoch 9/10
23/23 ————— 13s 569ms/step - accuracy: 0.8685 - loss: 0.3518 - val_accuracy: 0.8287 - val_loss: 0.5066
Epoch 10/10
23/23 ————— 13s 566ms/step - accuracy: 0.8449 - loss: 0.3358 - val_accuracy: 0.8564 - val_loss: 0.4568
```

Validation Accuracy & Lost

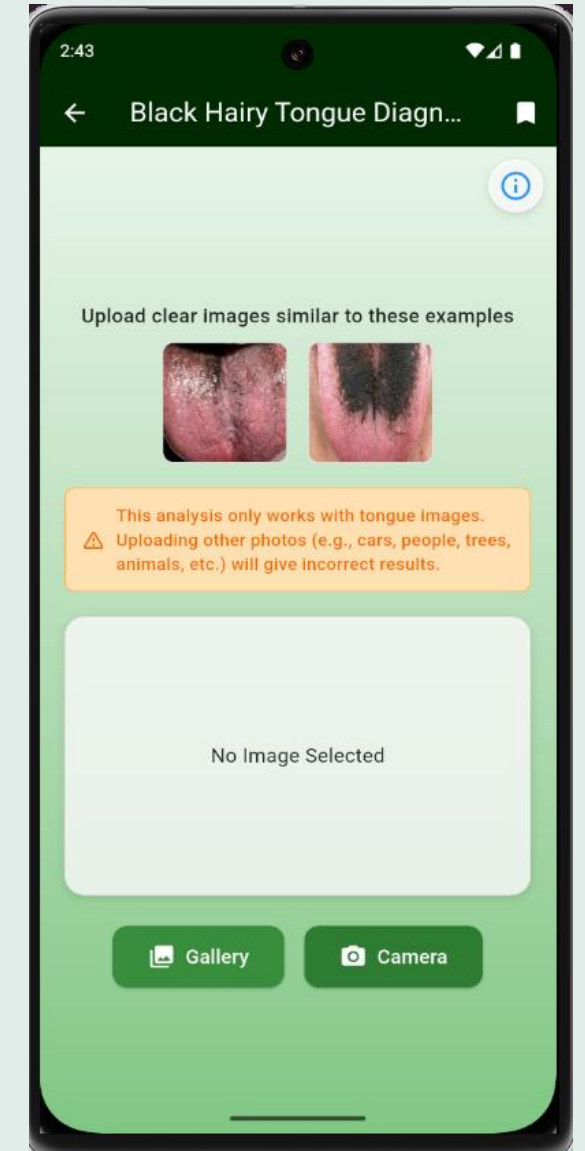
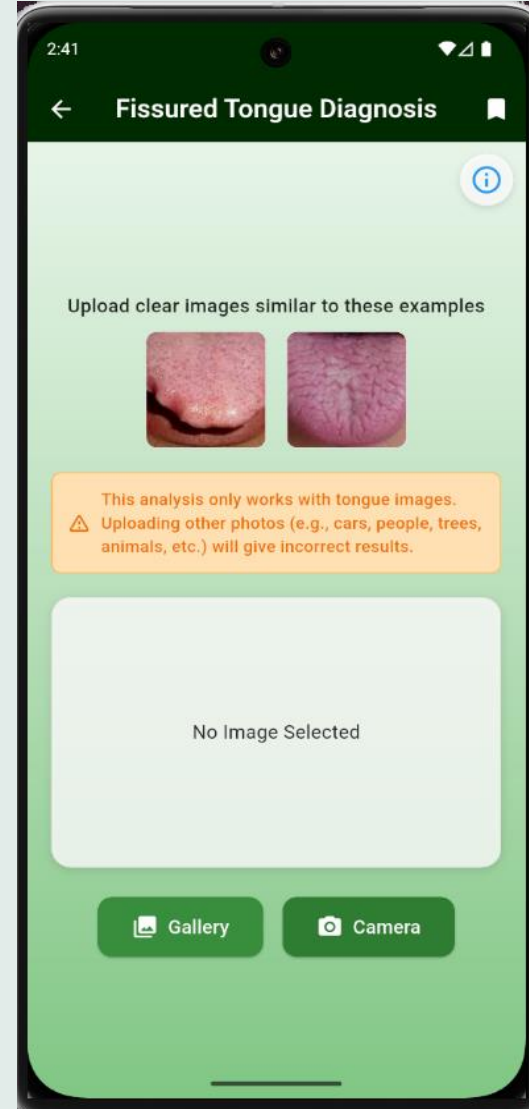
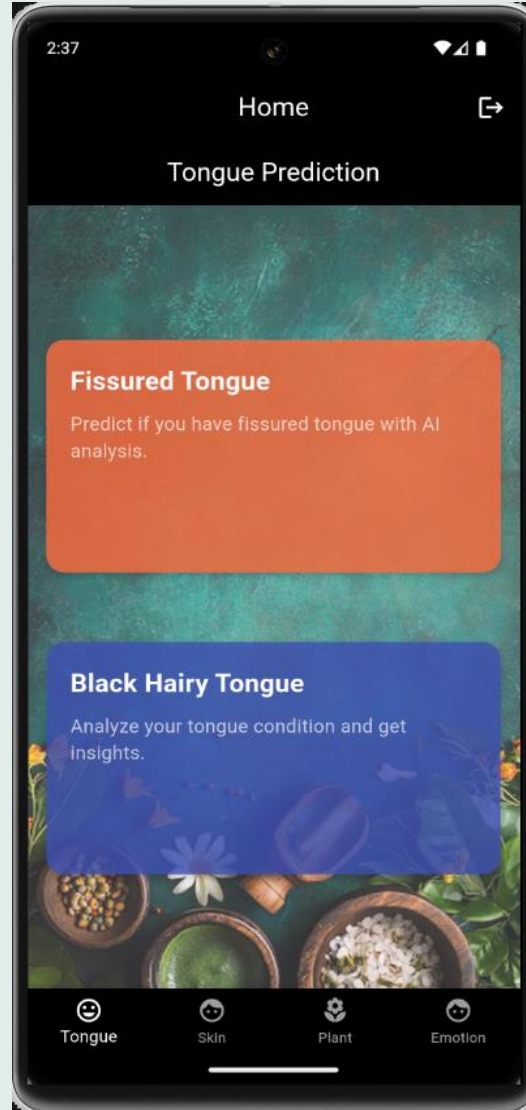
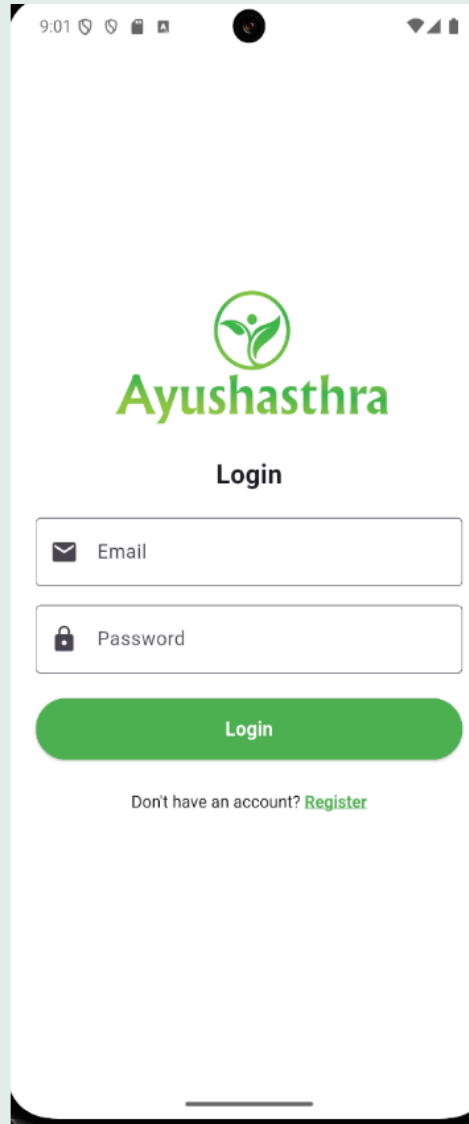
Testing and Validation

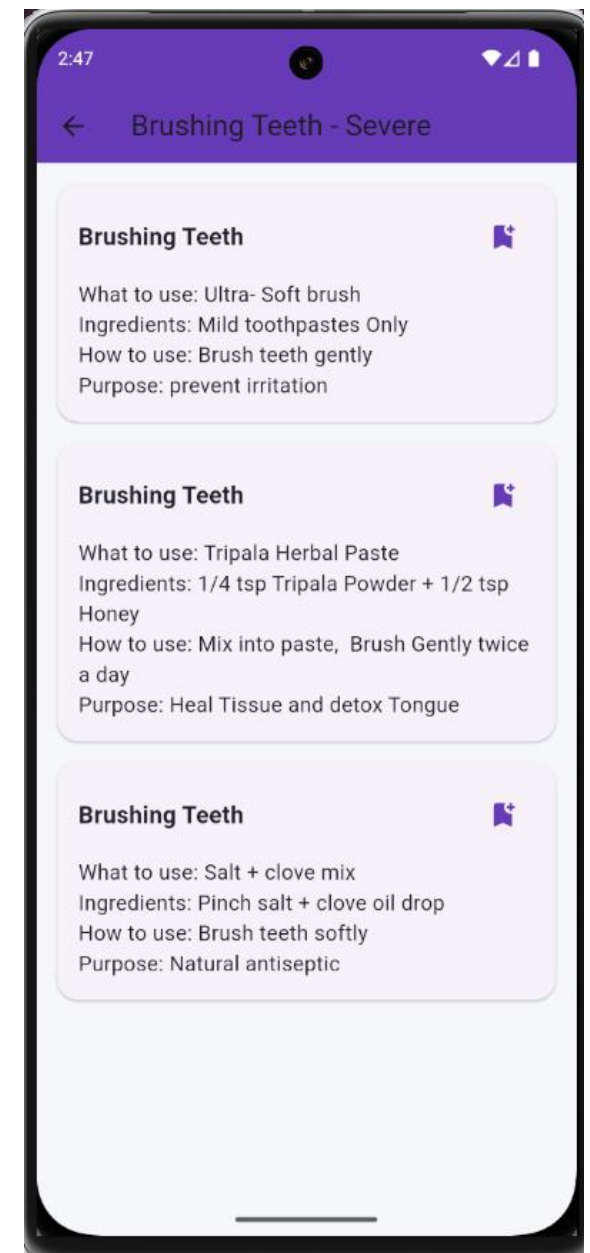
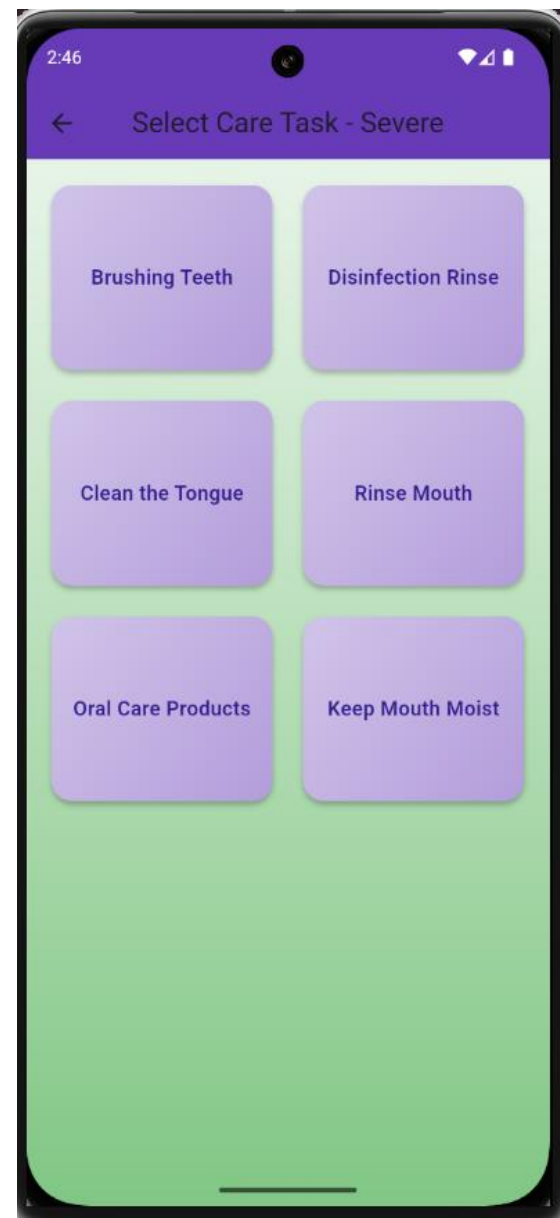
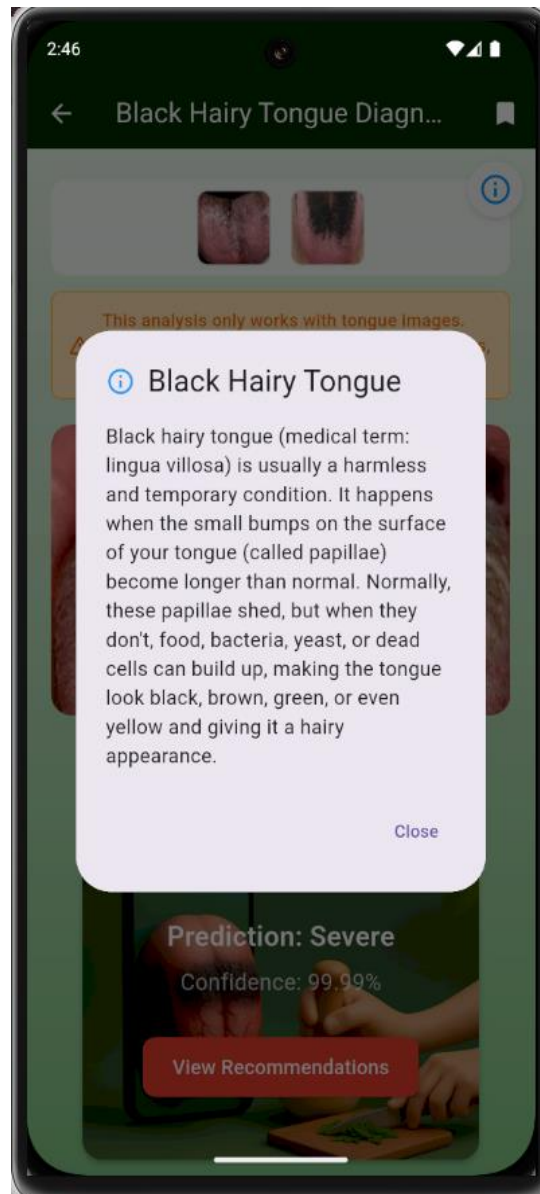
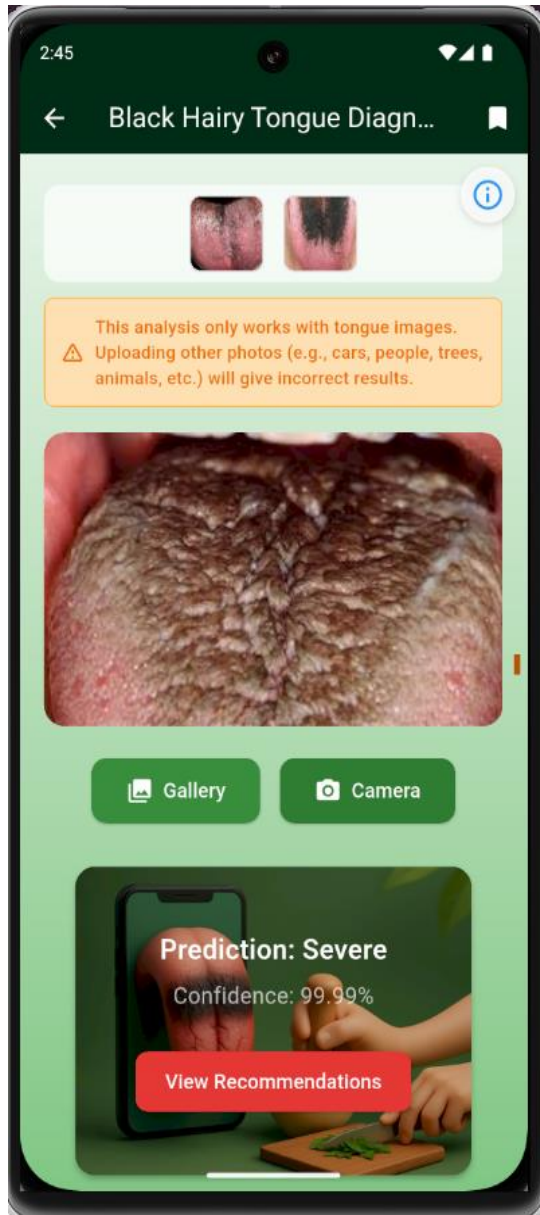


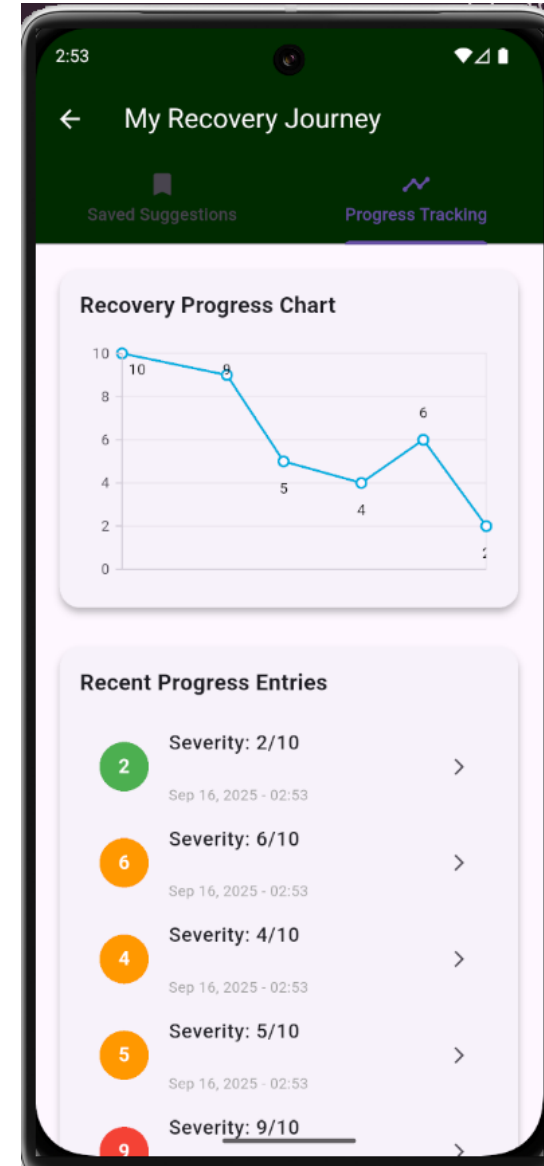
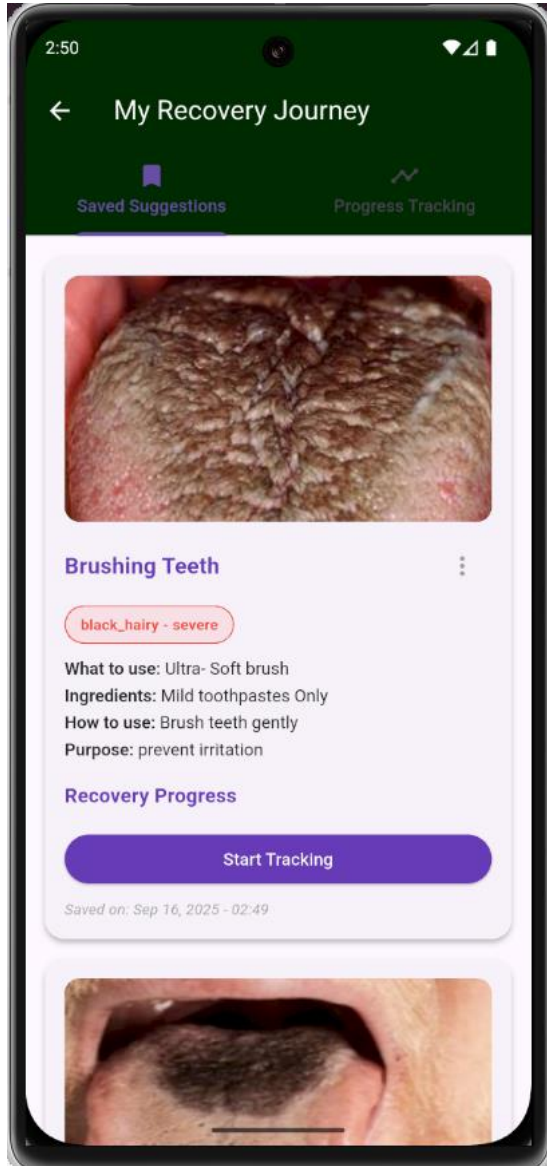
Layers

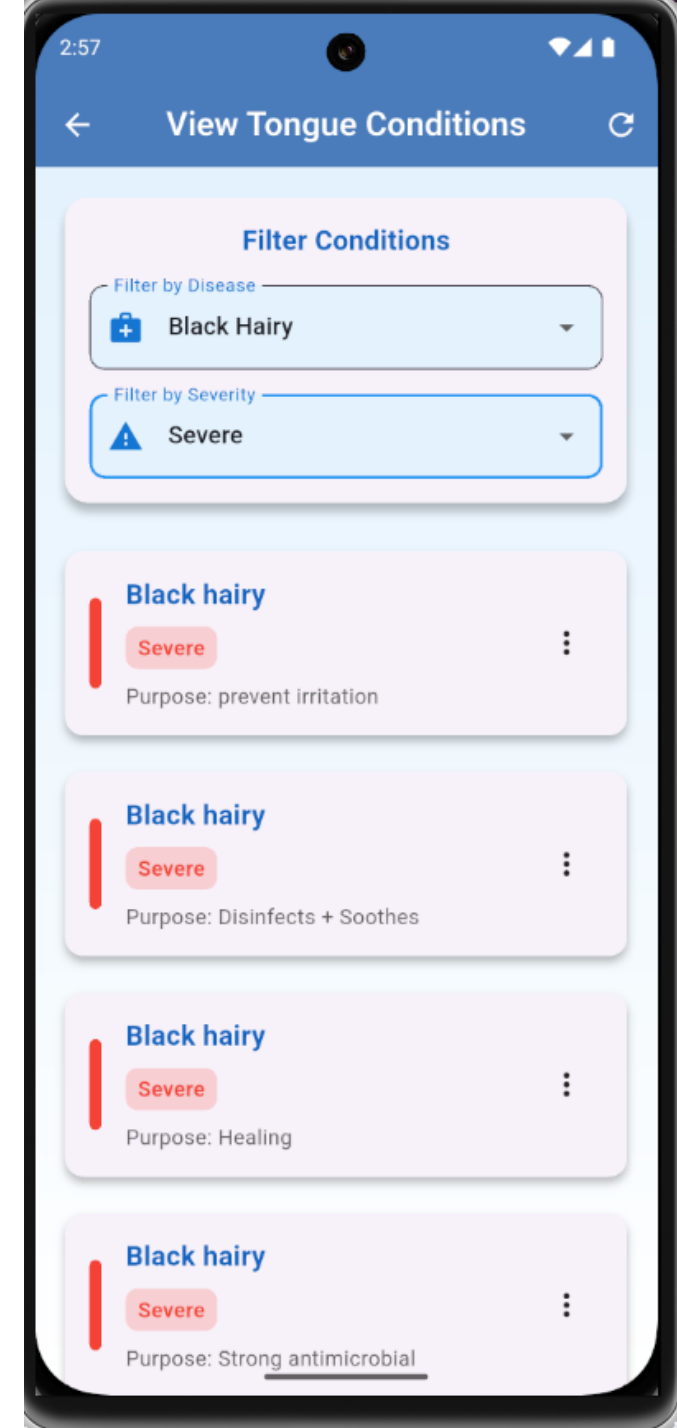
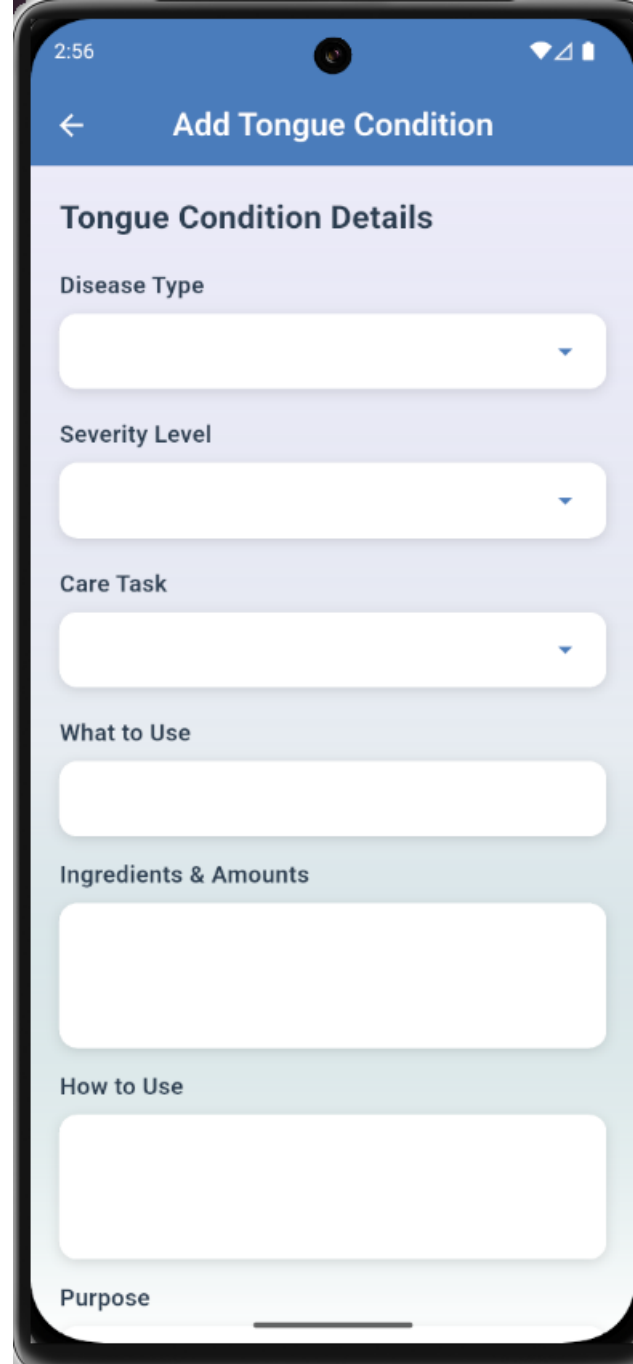
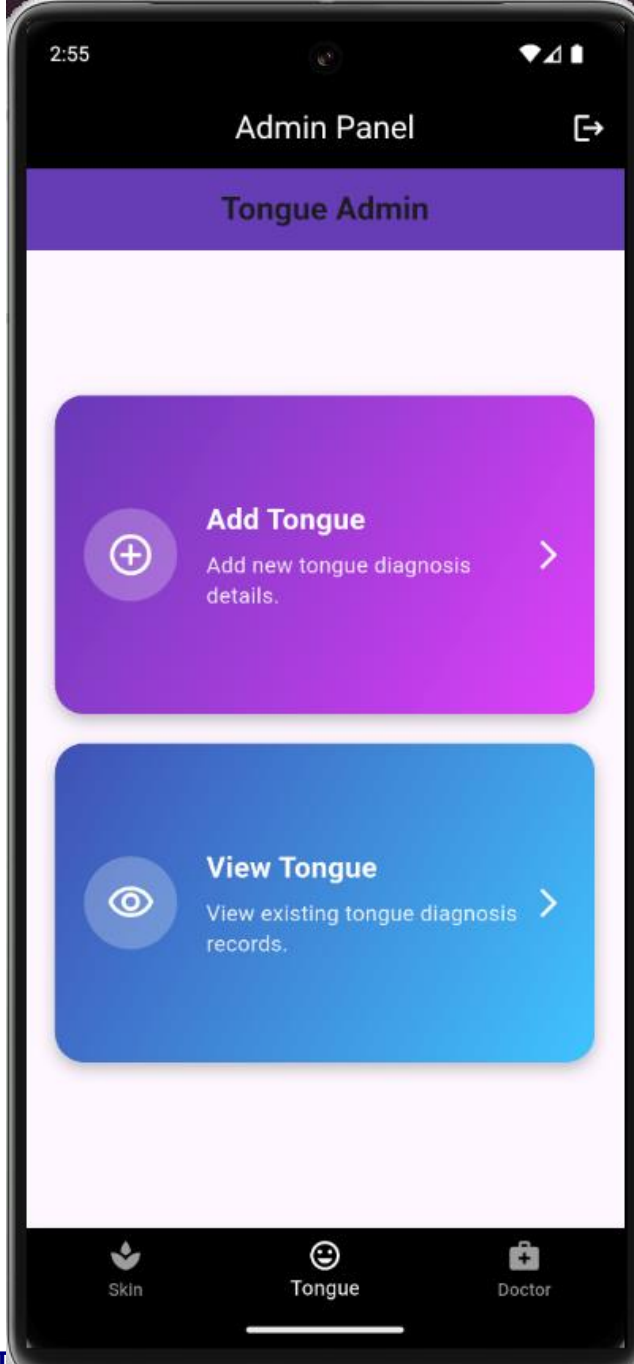
Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, 224, 224, 3)	0	-
Conv1 (Conv2D)	(None, 112, 112, 32)	864	input_layer[0][0]
bn_Conv1 (BatchNormalization)	(None, 112, 112, 32)	128	Conv1[0][0]
Conv1_relu (ReLU)	(None, 112, 112, 32)	0	bn_Conv1[0][0]
expanded_conv_depthwise (DepthwiseConv2D)	(None, 112, 112, 32)	288	Conv1_relu[0][0]
expanded_conv_depthwise_... (BatchNormalization)	(None, 112, 112, 32)	128	expanded_conv_depthwi...
expanded_conv_depthwise_... (ReLU)	(None, 112, 112, 32)	0	expanded_conv_depthwi...
expanded_conv_project (Conv2D)	(None, 112, 112, 16)	512	expanded_conv_depthwi...
expanded_conv_project_BN (BatchNormalization)	(None, 112, 112, 16)	64	expanded_conv_project...
block_1_expand (Conv2D)	(None, 112, 112, 96)	1,536	expanded_conv_project...
block_1_expand_BN (BatchNormalization)	(None, 112, 112, 96)	384	block_1_expand[0][0]
block_1_expand_relu (ReLU)	(None, 112, 112, 96)	0	block_1_expand_BN[0][...
block_1_pad (ZeroPadding2D)	(None, 113, 113, 96)	0	block_1_expand_relu[0...
block_1_depthwise (DepthwiseConv2D)	(None, 56, 56, 96)	864	block_1_pad[0][0]
block_1_depthwise_BN	(None, 56, 56, 96)	384	block_1_depthwise[0][...

Ui design prototype









Next expected Achievement



More enhance the UI
s



Set Reminders



User Management

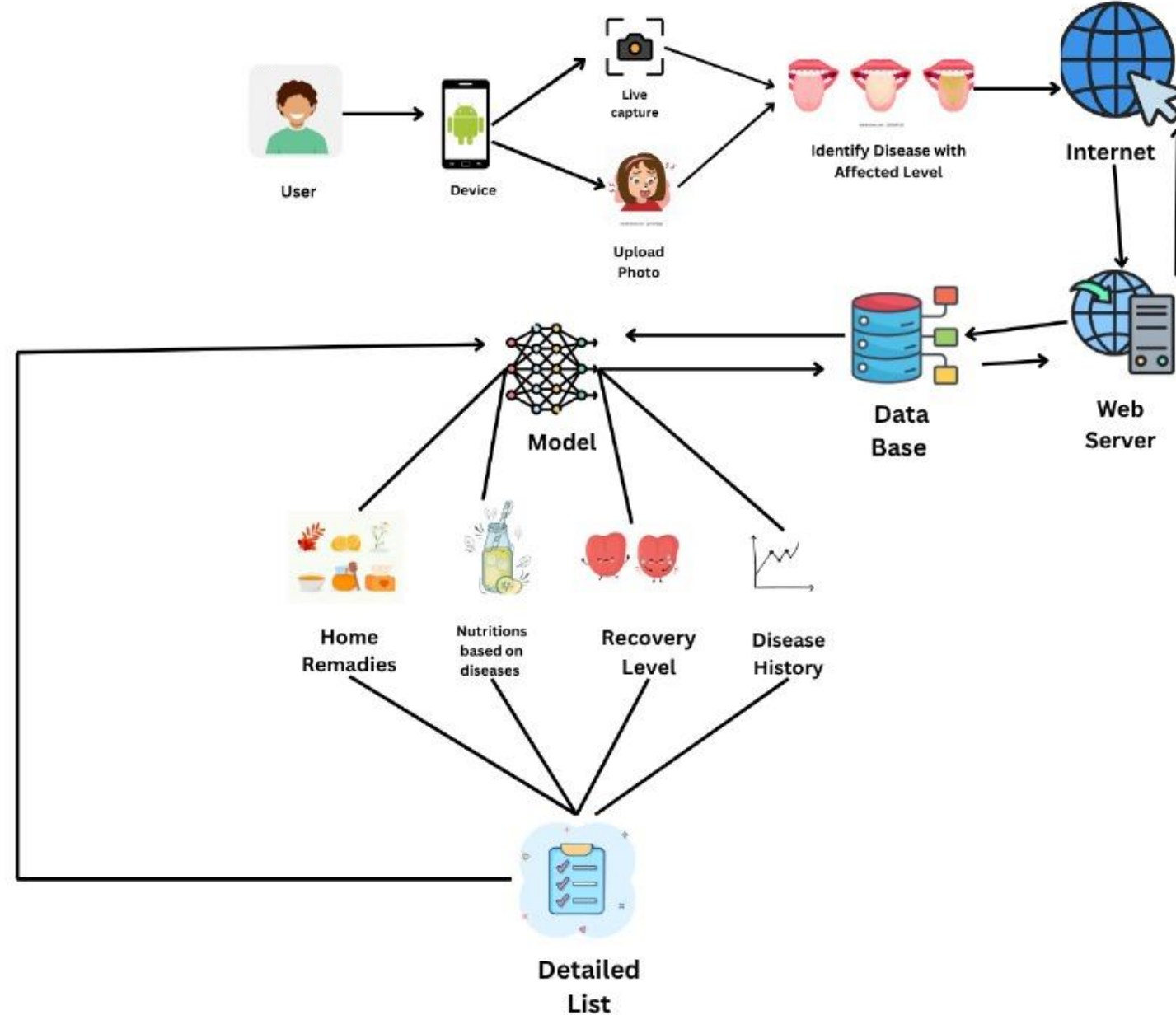


Healthy Nutritional
suggesting to prevent
the disease



Adding avoid Life styles

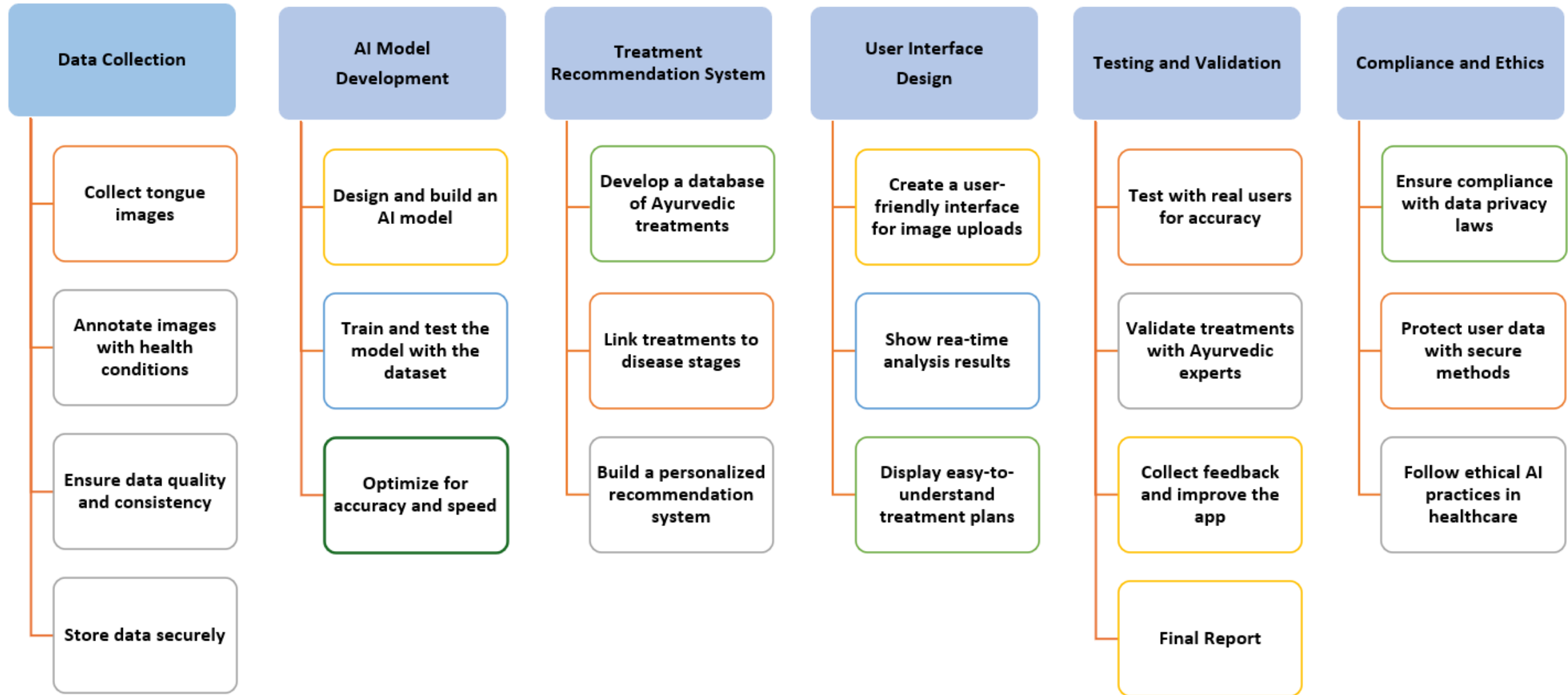
SYSTEM DIAGRAM



Features Comparison Table

Features	Research 1 [1]	Research 2 [2]	Research 3 [3]	Research 4 [4]	Proposed Solution
User Friendly	Low	Low	Medium	Low	High
Mobile Solution	✗	✗	✗	✗	✓
Integration of AI with Ayurvedic Tongue Diagnostics	✗	✗	✗	✓	✓
Innovative Image Processing for Tongue Analysis (Color, Texture, Hydration)	✗	✗	✓	✓	✓
Recovery Tracking System	✗	✗	✗	✗	✓
Recovery tracking system and progress monitoring					
Real-Time Health Risk Assessment	✗	✗	✗	✗	✓
Varied AI Model: CNNs, Computer Vision, ML for Disease Detection	✗	✗	✗	✓	✓

WBS - AI-Powered Tongue Imaging App Component



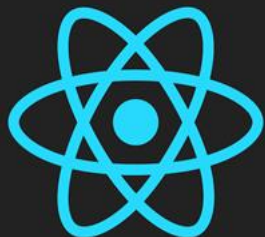
Technologies to be used



Keras
A deep learning library



TensorFlow



React Native



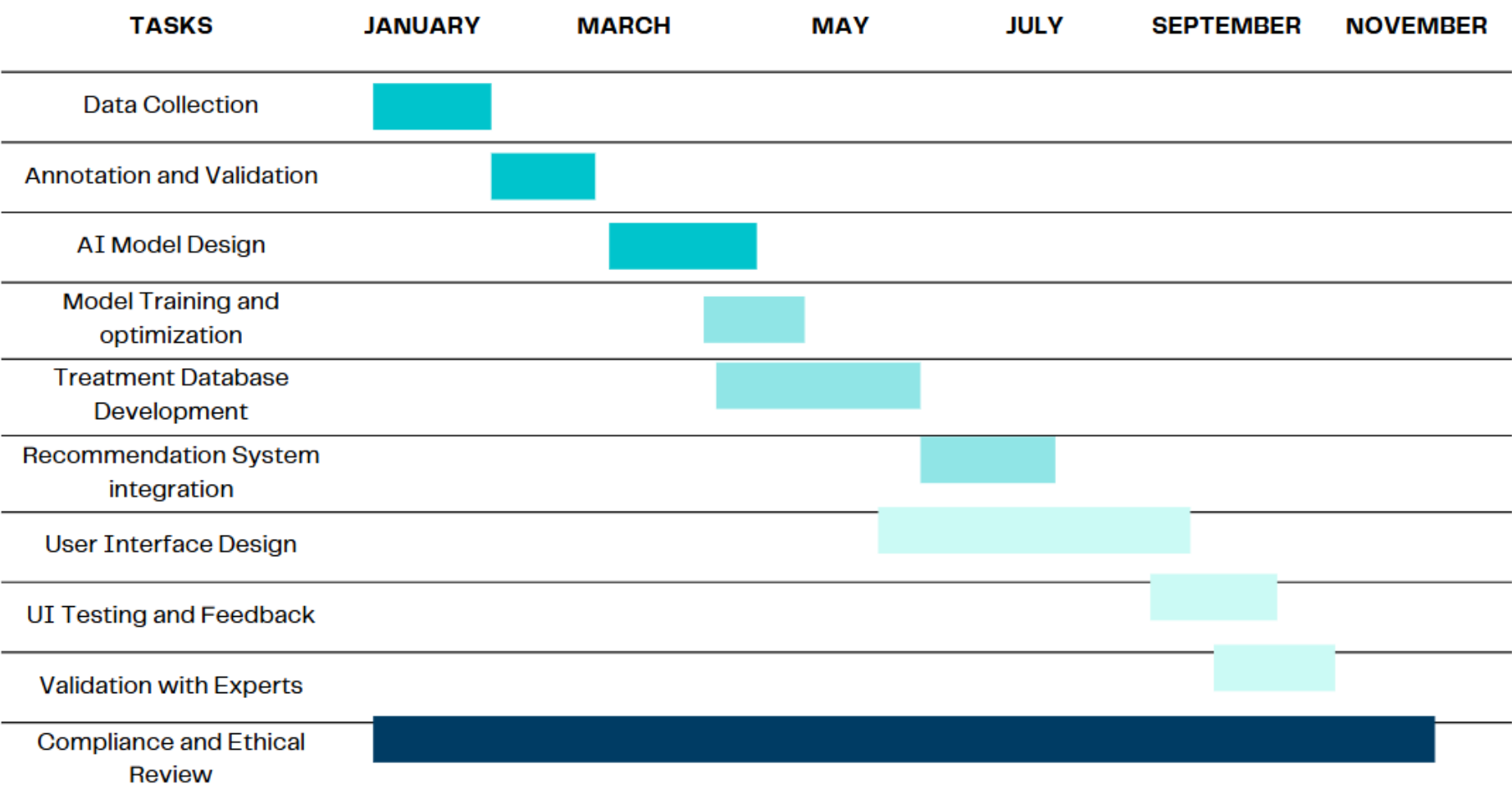
Firebase



GitHub

Gantt chart

AI-POWERED TONGUE IMAGING APP



Supporting Items (Commercialization)

Target Market:

- Focus on health-conscious individuals seeking personalized Ayurvedic solutions.
- Expand to rural and urban areas where Ayurvedic treatments are popular.



Partnerships:

- Collaborate with Ayurvedic practitioners, clinics, and wellness centers to validate and promote the app.
- Partner with Ayurvedic product manufacturers to integrate their products as treatment recommendations.

References

- [1] J. Zhang, L. Wang, and Y. Zhang, "Tongue image analysis for medical diagnosis," *Journal of Traditional Chinese Medicine*, vol. 40, no. 3, pp. 312–319, 2021. [Online]. Available:
- [2] M. T. Ribeiro, S. Singh, and C. Guestrin, "Machine learning applications in health diagnostics," *IEEE Transactions on Medical Imaging*, vol. 38, no. 5, pp. 1203–1212, May 2020. DOI: 10.1109/TMI.2020.123456.
- [3] K. Smith and J. Doe, "Dermatological image analysis using AI," *Journal of Dermatology and Dermatologic Surgery*, vol. 15, no. 2, pp. 85–90, Apr. 2022. DOI: 10.1016/j.jdds.2022.03.007.



IT21813184 - FERNANDO W.M.S. K

AI-POWERED SKIN DISEASE IDENTIFICATION & PERSONALIZED AYURVEDIC SOLUTIONS

Information Technology (IT)



INTRODUCTION

BACKGROUND

- Detect skin conditions through image analysis.
- Classify the disease into stages: **Normal, Moderate, and Severe.**
- Offer personalized recommendations to users, including:
- Preventive lifestyle recommendation .
- Home Dietary Plan and stage.
- Recovery

selected diseases for the treatment suggestions

Diseases	Stage	Home Dietary Plan and Lifestyle Recommendations
Melanoma	Moderate	Hydration: 2.5–3 liters of water daily Avoid direct sun exposure (use wide-brimmed hats, long clothing)
	Severe	Strengthen immune system: Garlic, ginger, turmeric in daily meals Ensure regular rest and proper sleep (7–9 hrs)
Atopic Dermatitis	Moderate	Anti-inflammatory foods: Fresh fruits (papaya, banana)Vegetables (carrot, pumpkin) Lukewarm water for bathing, avoid hot showers
	Severe	Sleep early; avoid late-night screen exposure Herbal compress or cooling packs for flare-ups

Research Gap

Features	Research Paper				Proposed
	A	B	C	D	
AI-based skin disease diagnosis	True	False	True	True	True
Integration of Ayurvedic lifestyle recommendations	False	True	False	False	True
Skin Diseases Categorized stages	True	False	True	False	True
Mobile application for integrated AI-Ayurveda skin health management	False	True	False	True	True

Research Problems

- Data Integration and Standardization
- Personalization of Recommendations
- Validation and Efficacy Assessment
- User Engagement and Compliance



Research Objectives

Specific Objective

To detect skin diseases through image analysis, classify them into stages :Normal, Moderate, and Severe and provide personalized recommendations, including preventive lifestyle tips and home dietary plans based on each stage.



Sub Objectives

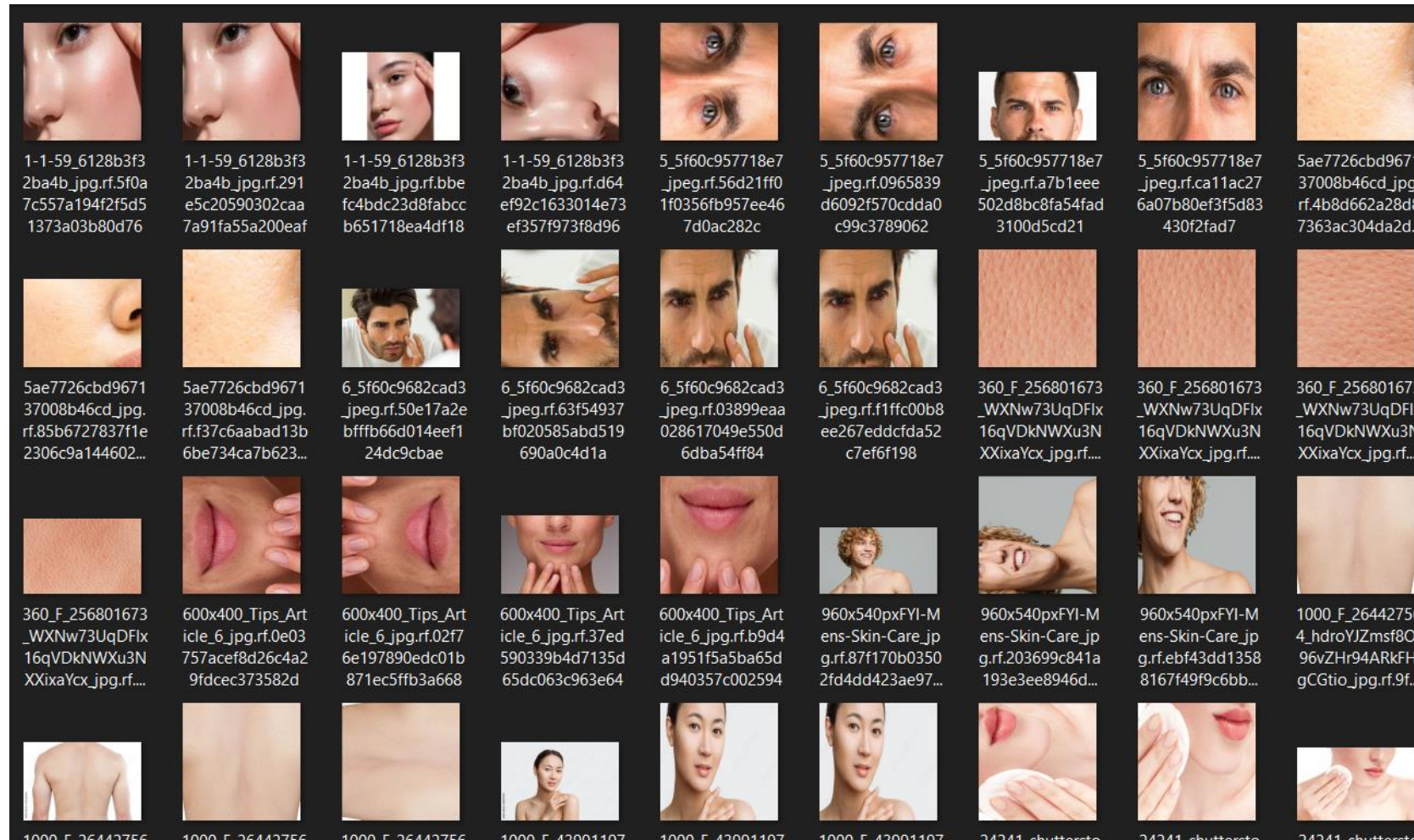
- To train an AI model to identify Melanoma and Atopic Dermatitis from skin images.
- To provide stage-based Ayurvedic home remedies for each skin condition.
- To develop dietary plans tailored to the stage and type of skin disease.
- To recommend preventive lifestyle practices for long-term skin health.
- To present the information in a user-friendly format within the mobile app.
- To classify each disease into stages: Normal, Moderate, and Severe.

ACHIEVEMENTS

- Successful data collection for model training
- Trained two models for selected disease
- Highest accuracy models up to now
- Developed the mobile application
- Suggested the home remedies
- Tracking health recovery
- Admin pannel

ATOPIC DERMATITIS

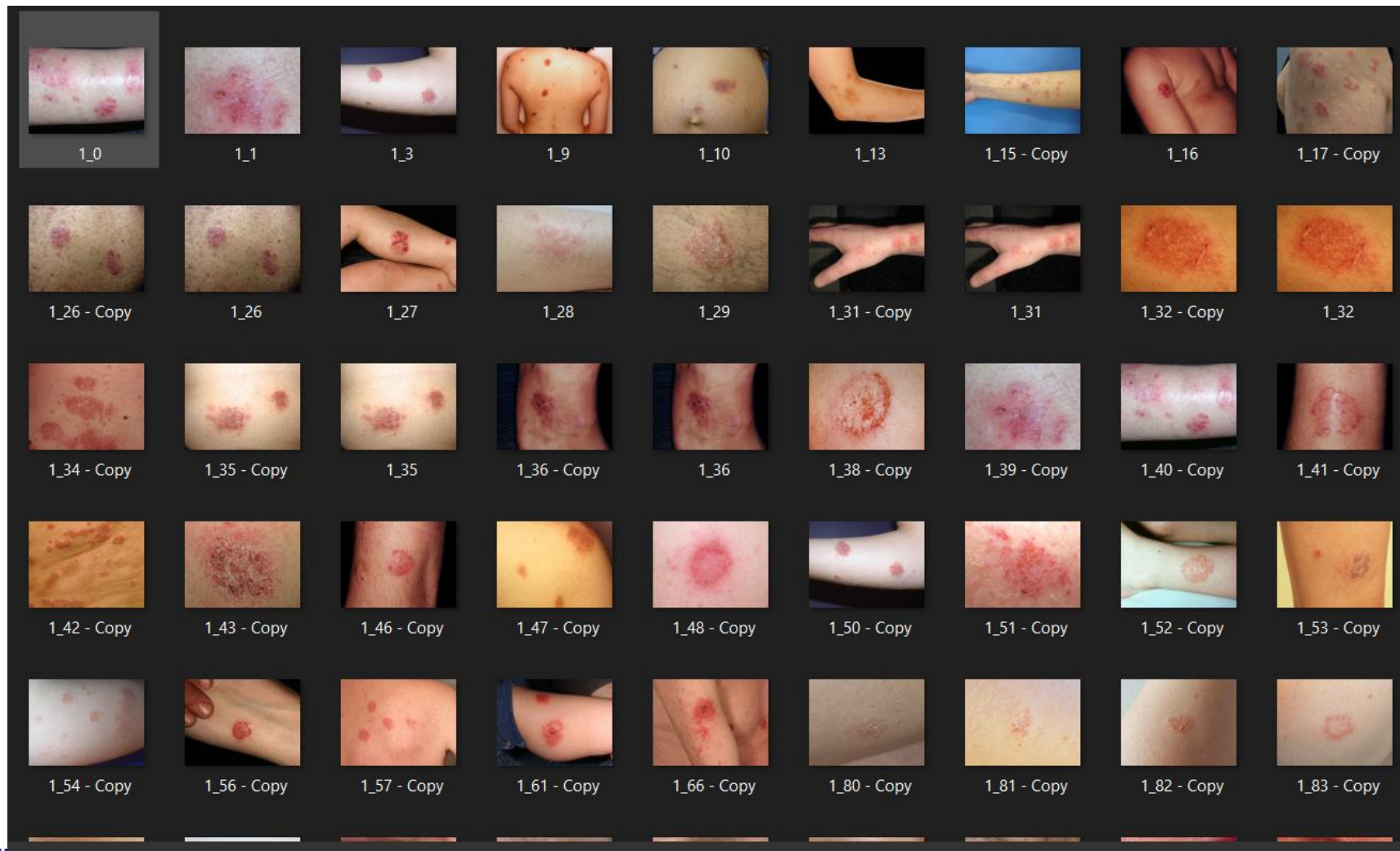
SUCCESSFUL DATA COLLECTION FOR MODEL TRAINING



Normal Stage

ATOPIC DERMATITIS

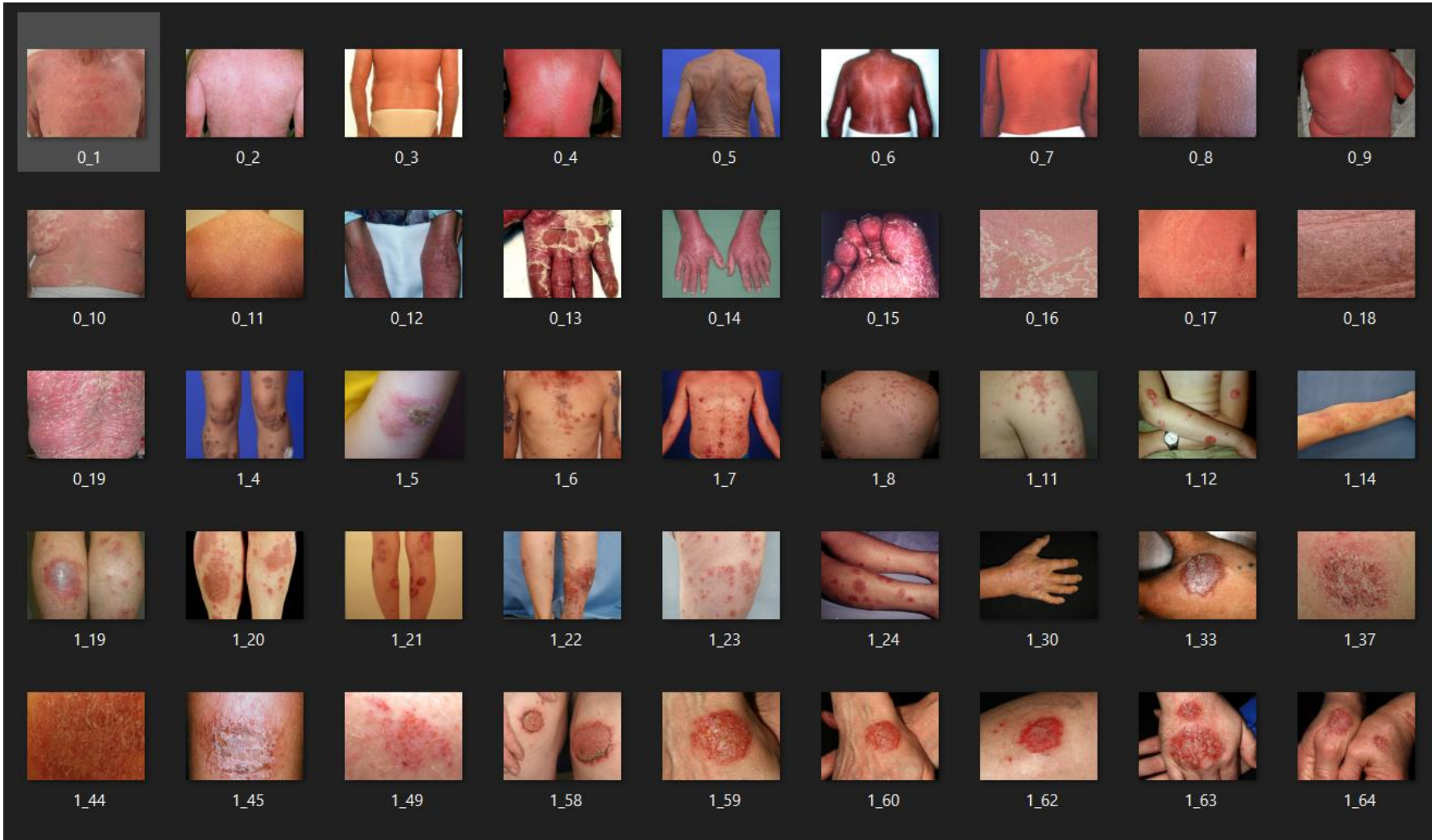
SUCCESSFUL DATA COLLECTION FOR MODEL TRAINING



Moderate Stage

ATOPIC DERMATITIS

SUCCESSFUL DATA COLLECTION FOR MODEL TRAINING



Severe stage

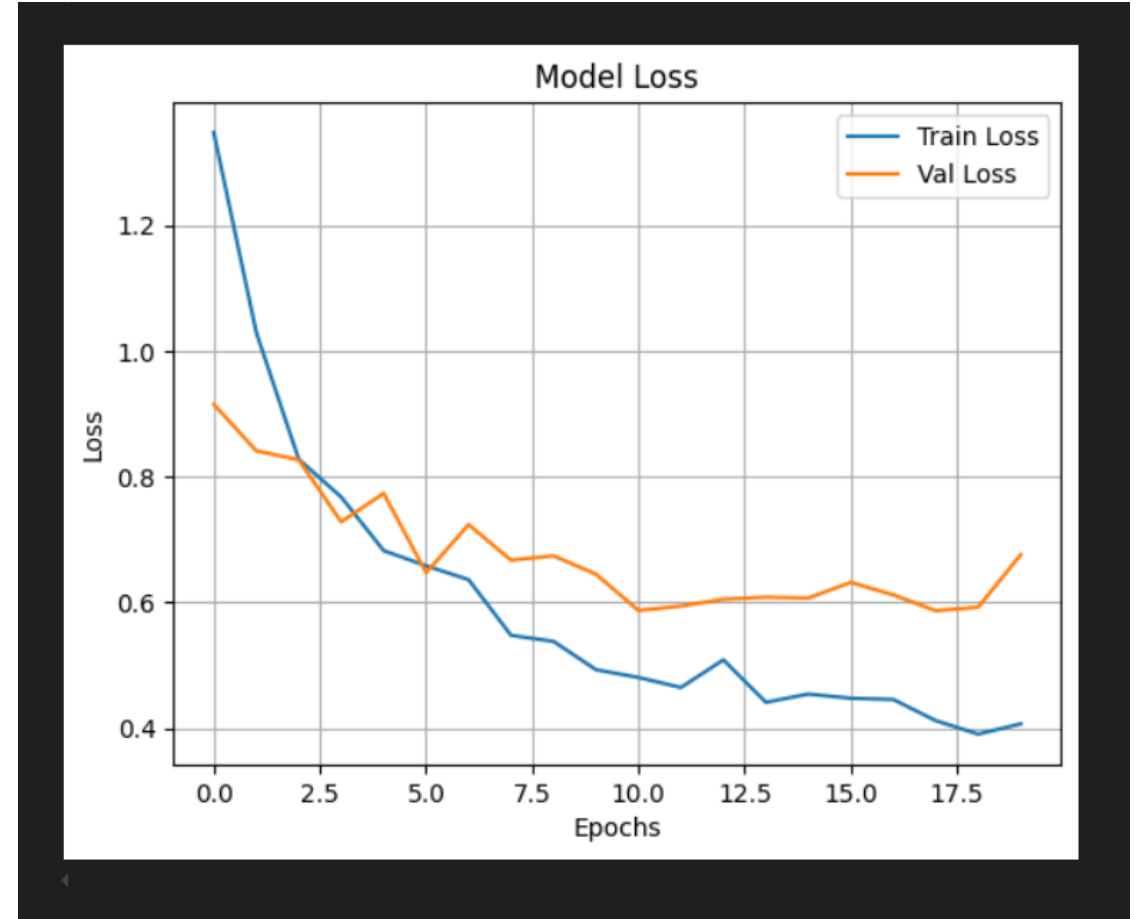
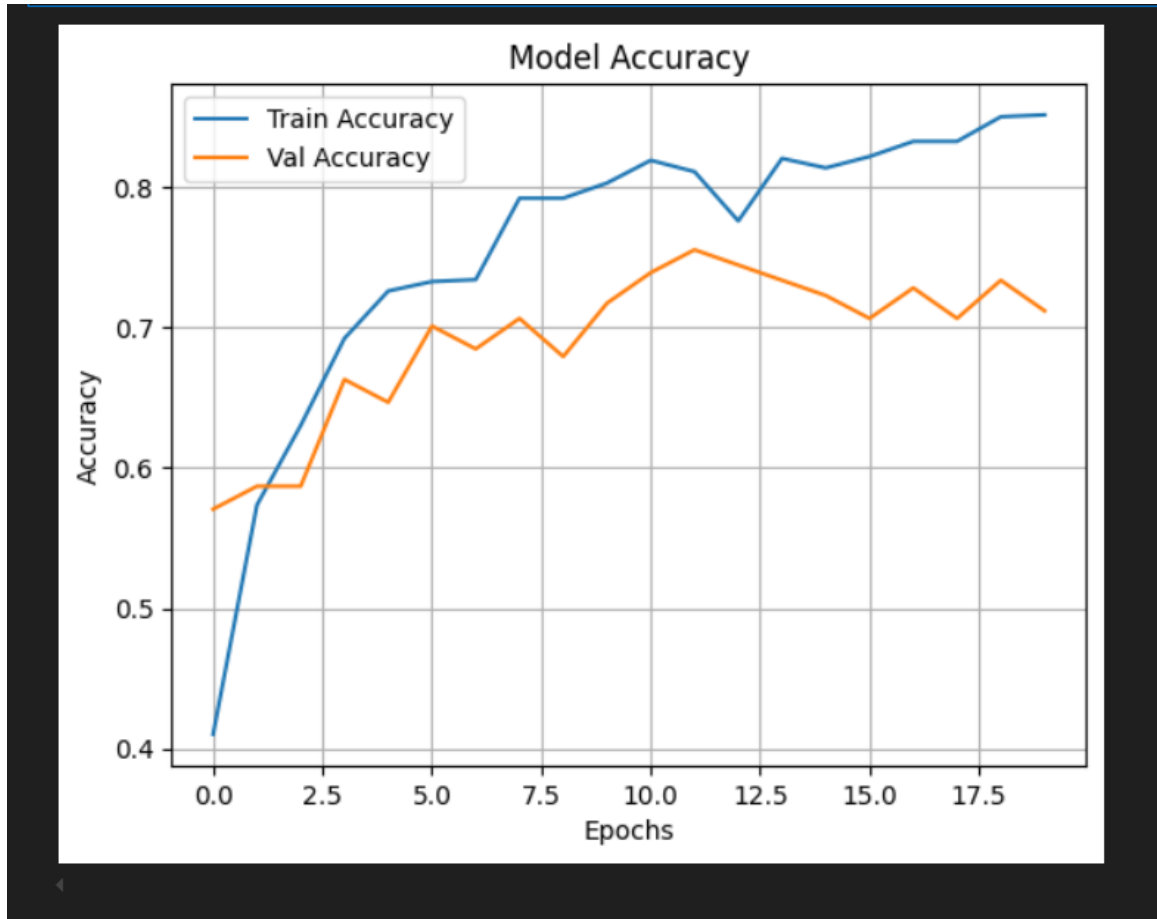
ACCURACY

```
+ Code + Markdown | ▶ Run All ⌵ Clear All Outputs | ☰ Outline ...

Epoch 6/20
24/24 ██████████ 18s 733ms/step - accuracy: 0.7386 - loss: 0.6603 - val_accuracy: 0.7011 - val_loss: 0.6470
Epoch 7/20
24/24 ██████████ 15s 636ms/step - accuracy: 0.7336 - loss: 0.6428 - val_accuracy: 0.6848 - val_loss: 0.7237
Epoch 8/20
24/24 ██████████ 23s 737ms/step - accuracy: 0.7983 - loss: 0.5397 - val_accuracy: 0.7065 - val_loss: 0.6671
Epoch 9/20
24/24 ██████████ 15s 627ms/step - accuracy: 0.7808 - loss: 0.5433 - val_accuracy: 0.6793 - val_loss: 0.6739
Epoch 10/20
24/24 ██████████ 15s 621ms/step - accuracy: 0.7896 - loss: 0.5068 - val_accuracy: 0.7174 - val_loss: 0.6447
Epoch 11/20
24/24 ██████████ 15s 629ms/step - accuracy: 0.8194 - loss: 0.4960 - val_accuracy: 0.7391 - val_loss: 0.5870
Epoch 12/20
24/24 ██████████ 15s 642ms/step - accuracy: 0.8018 - loss: 0.5174 - val_accuracy: 0.7554 - val_loss: 0.5940
Epoch 13/20
...
Epoch 19/20
24/24 ██████████ 15s 625ms/step - accuracy: 0.8358 - loss: 0.4258 - val_accuracy: 0.7337 - val_loss: 0.5924
Epoch 20/20
24/24 ██████████ 15s 618ms/step - accuracy: 0.8573 - loss: 0.4013 - val_accuracy: 0.7120 - val_loss: 0.6758
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
```

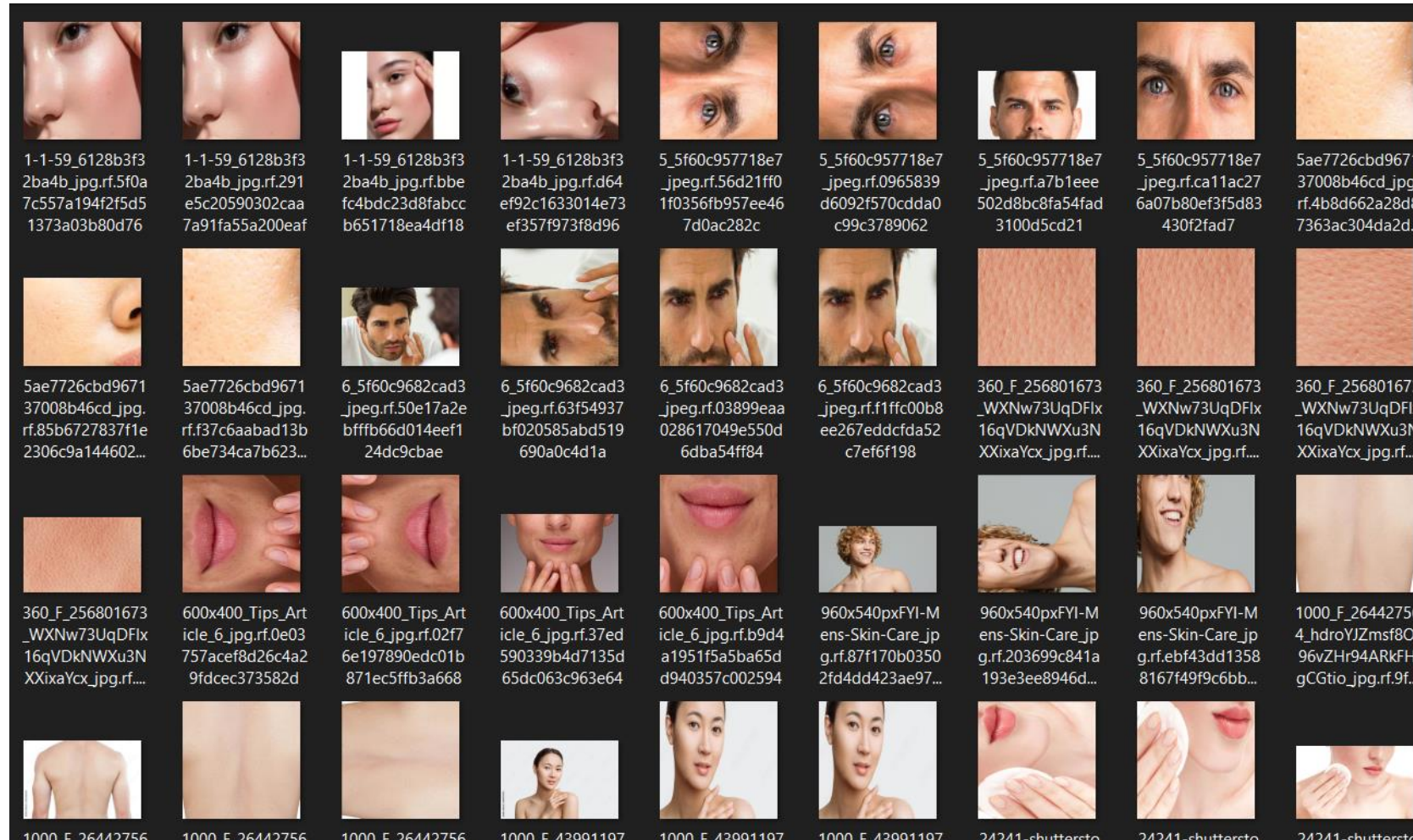
Accuracy 85%

ACCURACY



MELANOMA

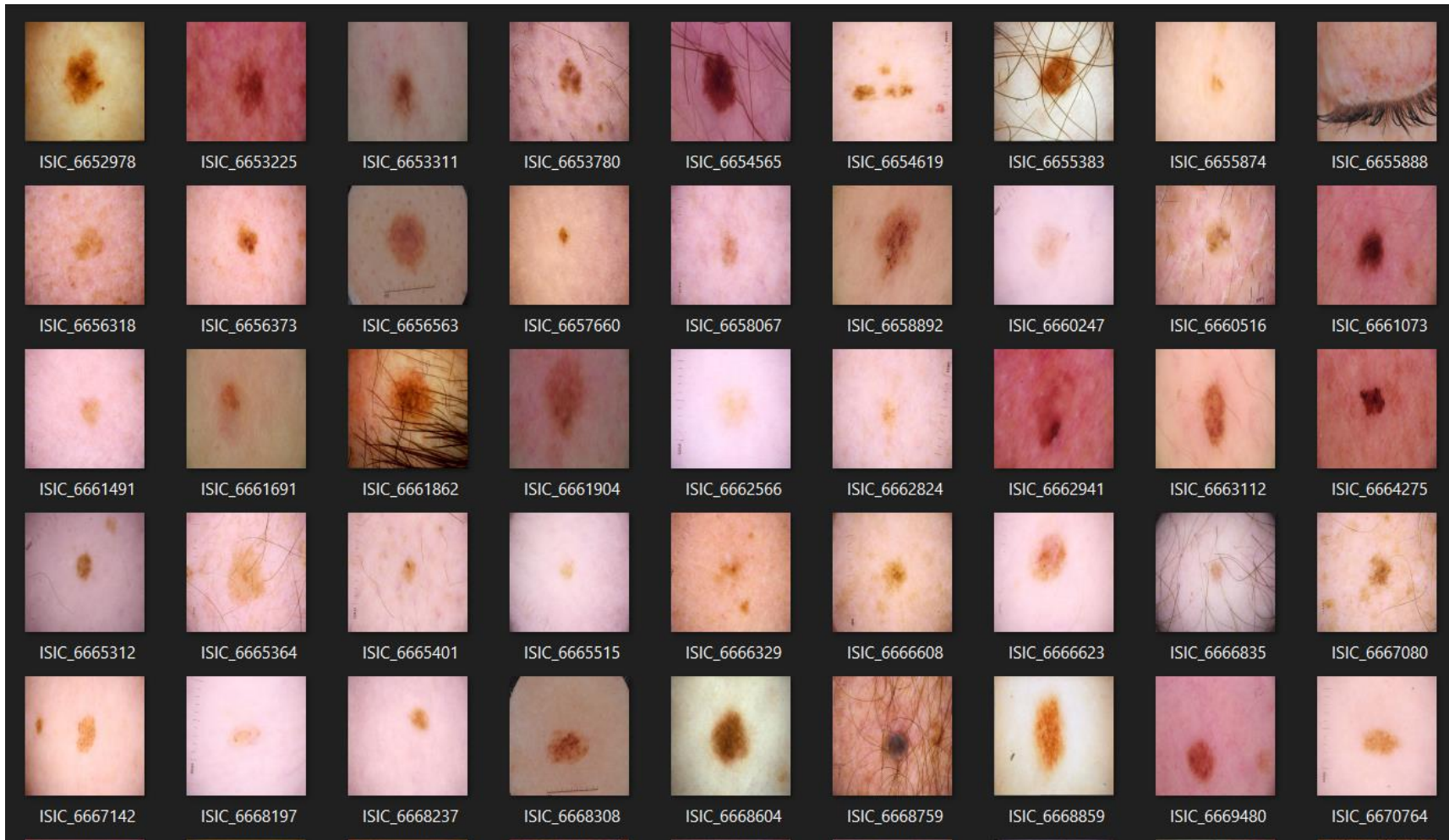
SUCCESSFUL DATA COLLECTION FOR MODEL TRAINING



Normal Stage

MELANOMA

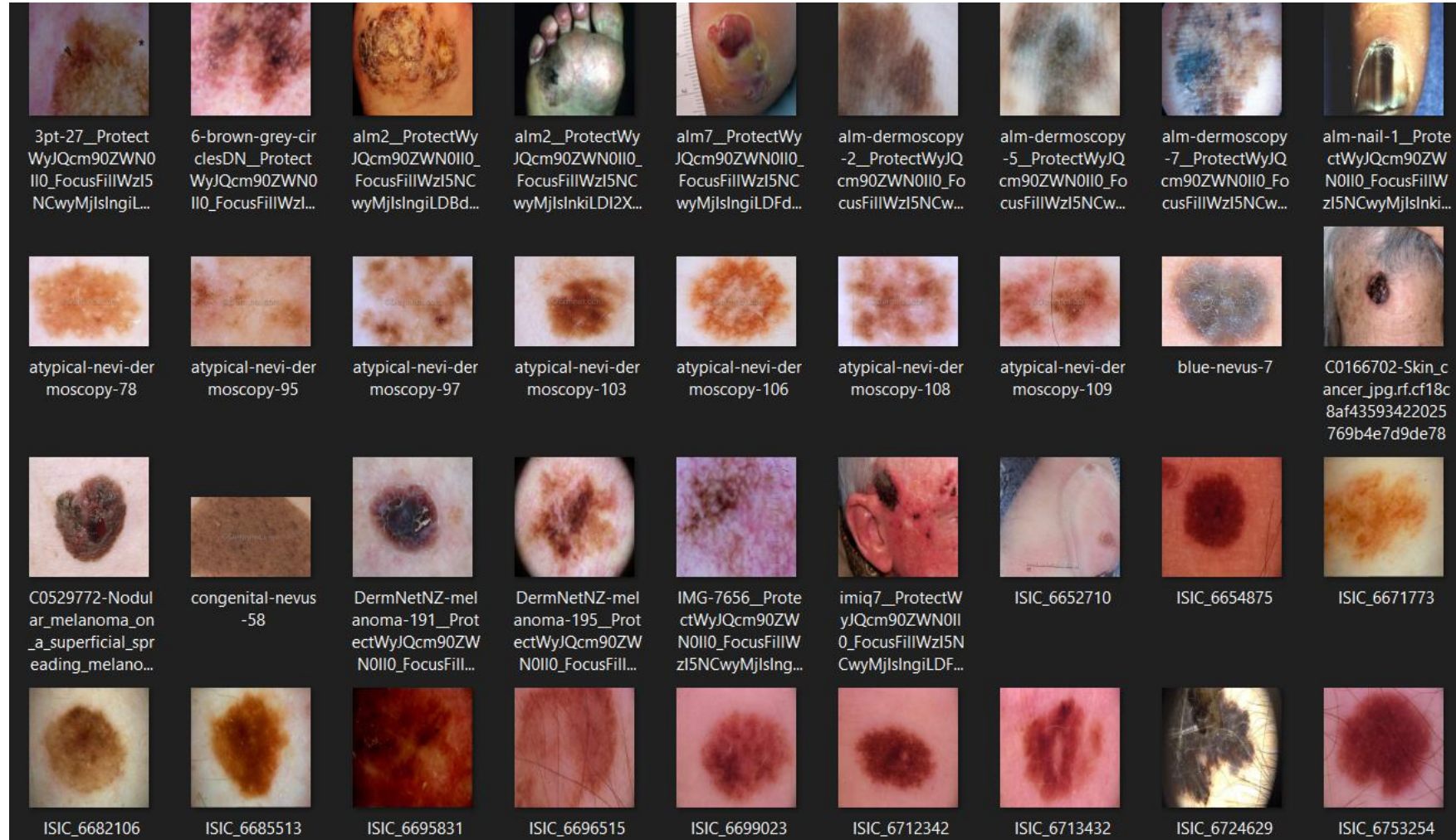
SUCCESSFUL DATA COLLECTION FOR MODEL TRAINING



Moderate Stage

MELANOMA

SUCCESSFUL DATA COLLECTION FOR MODEL TRAINING



Severe Stage

ACCURACY

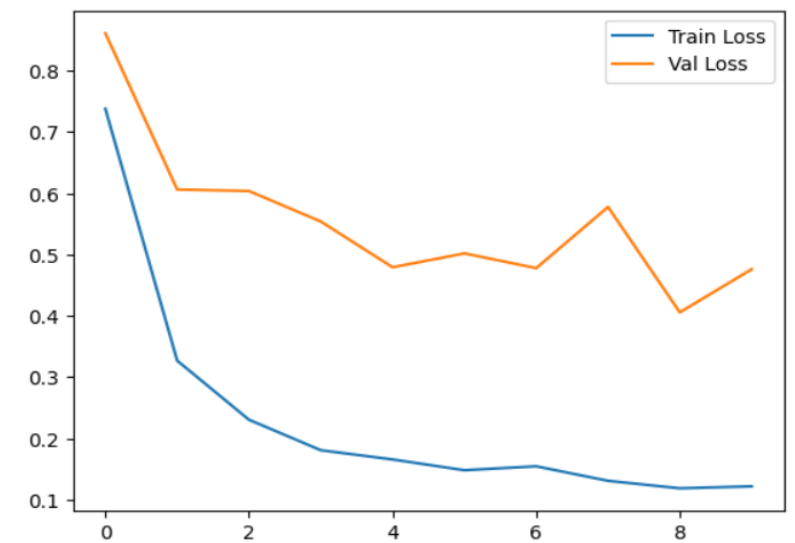
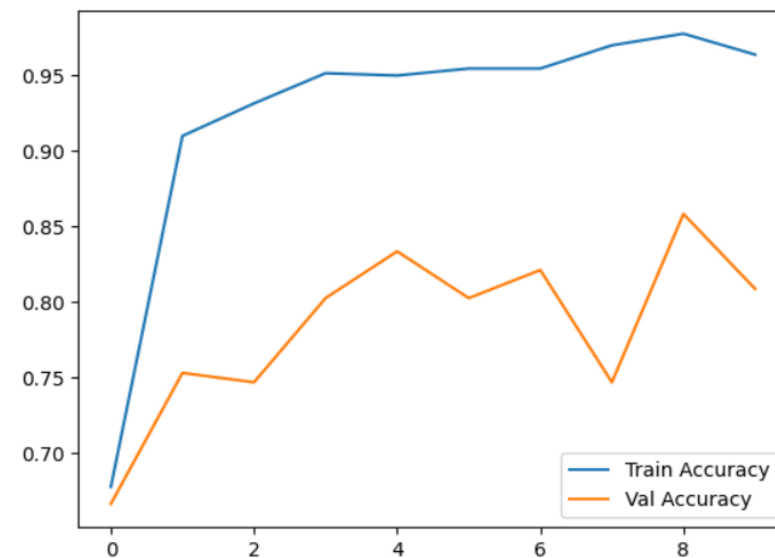
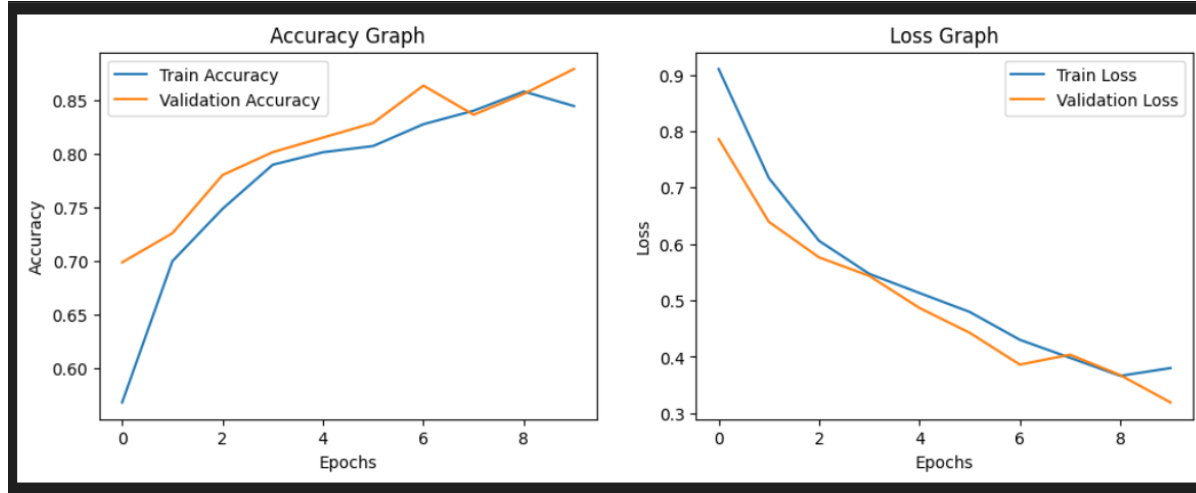
```
/usr/local/lib/python3.11/dist-packages/keras/src/trainers/data_adapters/py_dataset_adapter.py:121: UserWarning: Your `PyDatasetAdapter` class does not implement the `warn_if_super_not_called` method. This may lead to unexpected behavior.
self.warn_if_super_not_called()
Epoch 1/10
65/65 ————— 821s 13s/step - accuracy: 0.4651 - loss: 1.0084 - val_accuracy: 0.6990 - val_loss: 0.7861
Epoch 2/10
65/65 ————— 29s 441ms/step - accuracy: 0.6863 - loss: 0.7404 - val_accuracy: 0.7262 - val_loss: 0.6394
Epoch 3/10
65/65 ————— 29s 438ms/step - accuracy: 0.7457 - loss: 0.6102 - val_accuracy: 0.7806 - val_loss: 0.5764
Epoch 4/10
65/65 ————— 29s 442ms/step - accuracy: 0.7957 - loss: 0.5378 - val_accuracy: 0.8019 - val_loss: 0.5436
Epoch 5/10
65/65 ————— 28s 438ms/step - accuracy: 0.7904 - loss: 0.5428 - val_accuracy: 0.8155 - val_loss: 0.4867
Epoch 6/10
65/65 ————— 41s 439ms/step - accuracy: 0.8215 - loss: 0.4654 - val_accuracy: 0.8291 - val_loss: 0.4427
Epoch 7/10
65/65 ————— 28s 437ms/step - accuracy: 0.8317 - loss: 0.4449 - val_accuracy: 0.8641 - val_loss: 0.3862
Epoch 8/10
65/65 ————— 28s 438ms/step - accuracy: 0.8403 - loss: 0.4156 - val_accuracy: 0.8369 - val_loss: 0.4038
Epoch 9/10
65/65 ————— 27s 416ms/step - accuracy: 0.8574 - loss: 0.3778 - val_accuracy: 0.8563 - val_loss: 0.3675
Epoch 10/10
65/65 ————— 26s 403ms/step - accuracy: 0.8436 - loss: 0.3883 - val_accuracy: 0.8796 - val_loss: 0.3192
```

Custom CNN
Accuracy 84%

VGG-16
Accuracy 96%

```
/usr/local/lib/python3.12/dist-packages/keras/src/trainers/data_adapters/py_dataset_adapter.py:121: UserWarning: Your `PyDatasetAdapter` class does not implement the `warn_if_super_not_called` method. This may lead to unexpected behavior.
self.warn_if_super_not_called()
Epoch 1/10
21/21 ————— 365s 17s/step - accuracy: 0.5487 - loss: 0.9435 - val_accuracy: 0.6667 - val_loss: 0.8611
Epoch 2/10
21/21 ————— 14s 679ms/step - accuracy: 0.8767 - loss: 0.3893 - val_accuracy: 0.7531 - val_loss: 0.6063
Epoch 3/10
21/21 ————— 14s 683ms/step - accuracy: 0.9328 - loss: 0.2480 - val_accuracy: 0.7469 - val_loss: 0.6038
Epoch 4/10
21/21 ————— 15s 696ms/step - accuracy: 0.9415 - loss: 0.2087 - val_accuracy: 0.8025 - val_loss: 0.5541
Epoch 5/10
21/21 ————— 15s 716ms/step - accuracy: 0.9557 - loss: 0.1624 - val_accuracy: 0.8333 - val_loss: 0.4795
Epoch 6/10
21/21 ————— 14s 686ms/step - accuracy: 0.9396 - loss: 0.1719 - val_accuracy: 0.8025 - val_loss: 0.5021
Epoch 7/10
21/21 ————— 17s 812ms/step - accuracy: 0.9351 - loss: 0.1878 - val_accuracy: 0.8210 - val_loss: 0.4780
Epoch 8/10
21/21 ————— 15s 703ms/step - accuracy: 0.9749 - loss: 0.1222 - val_accuracy: 0.7469 - val_loss: 0.5778
Epoch 9/10
21/21 ————— 14s 675ms/step - accuracy: 0.9791 - loss: 0.1116 - val_accuracy: 0.8580 - val_loss: 0.4059
Epoch 10/10
21/21 ————— 14s 684ms/step - accuracy: 0.9612 - loss: 0.1207 - val_accuracy: 0.8086 - val_loss: 0.4762
```

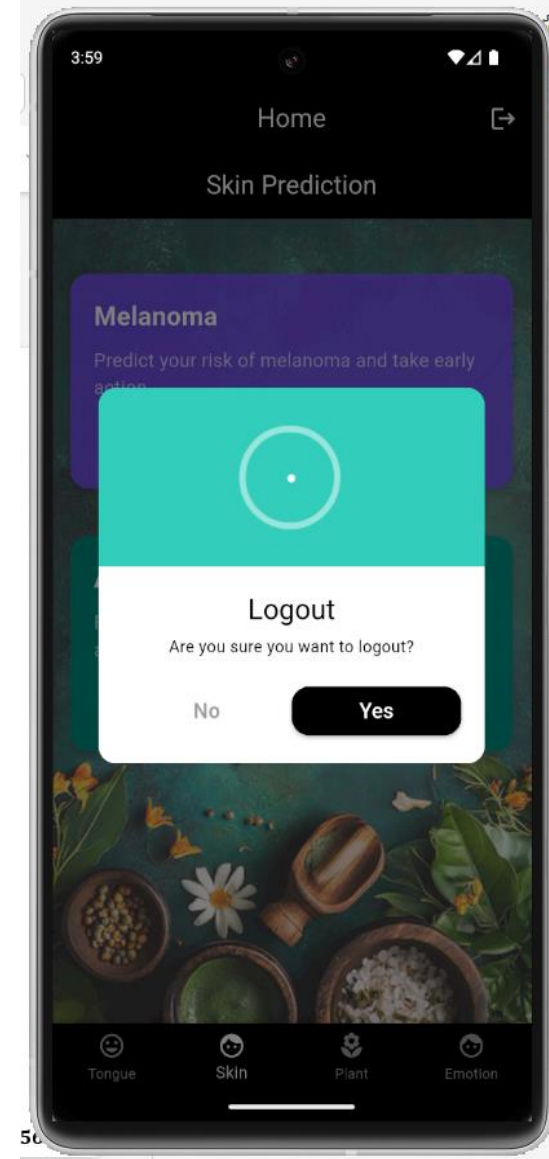
ACCURACY



Developed mobile app UI's

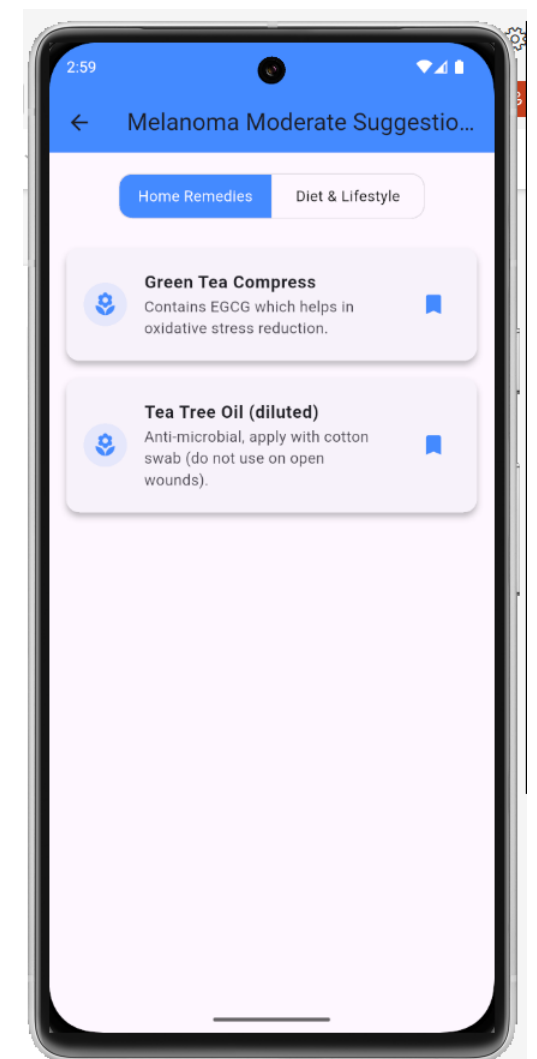
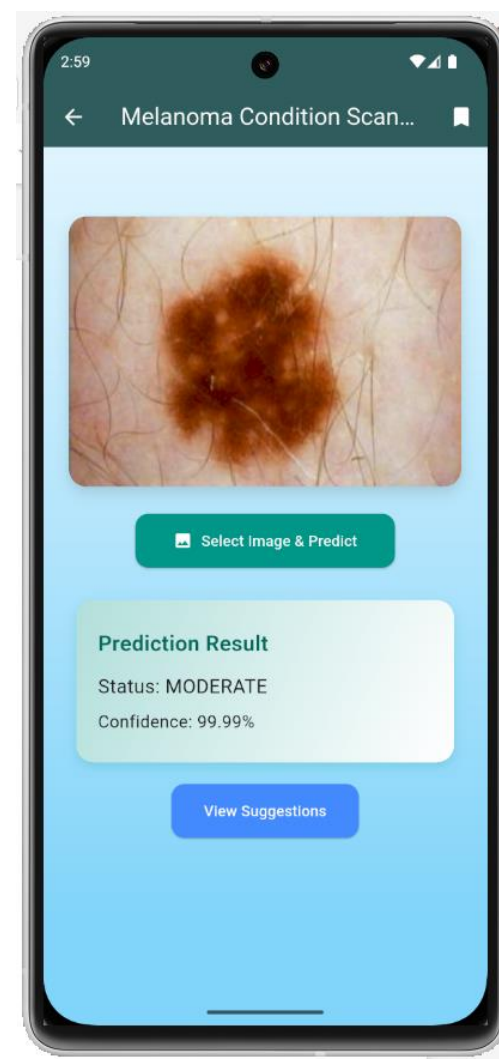
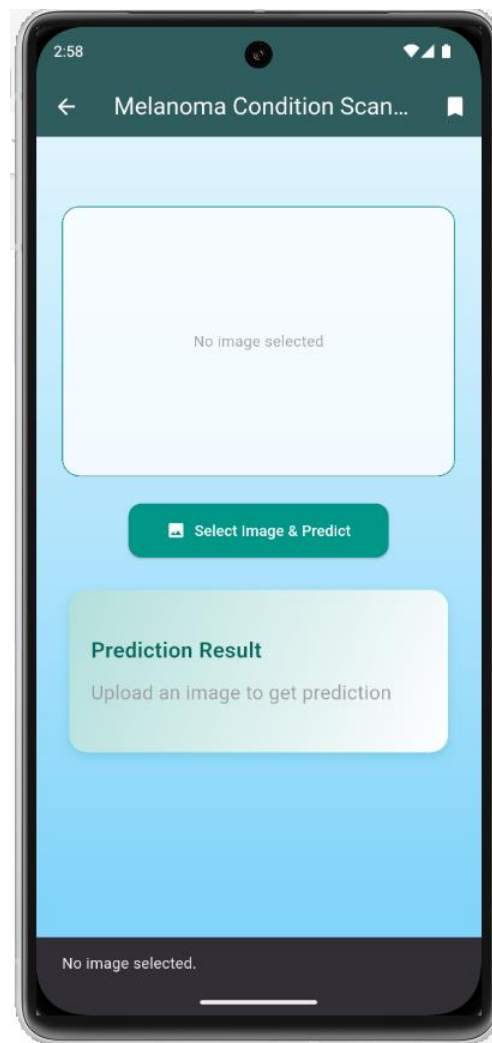
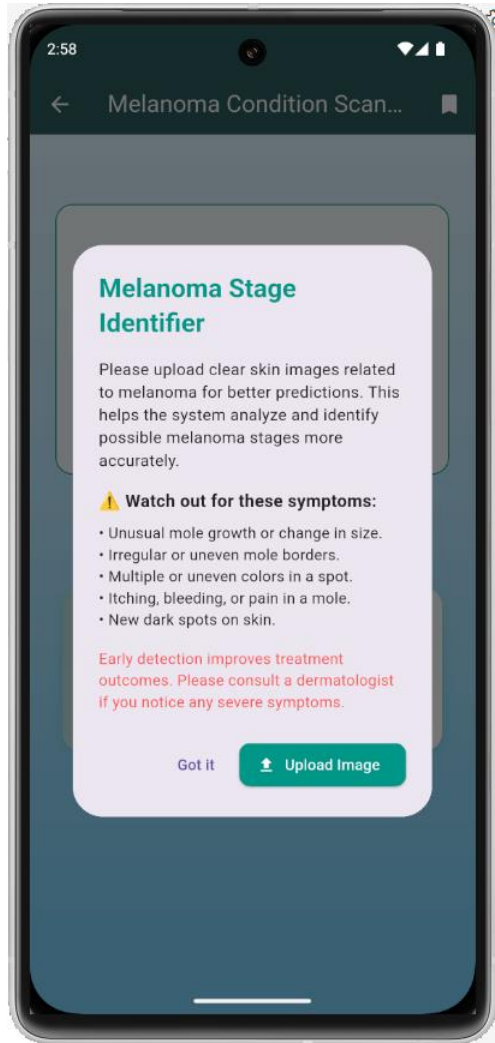


Login /Logout

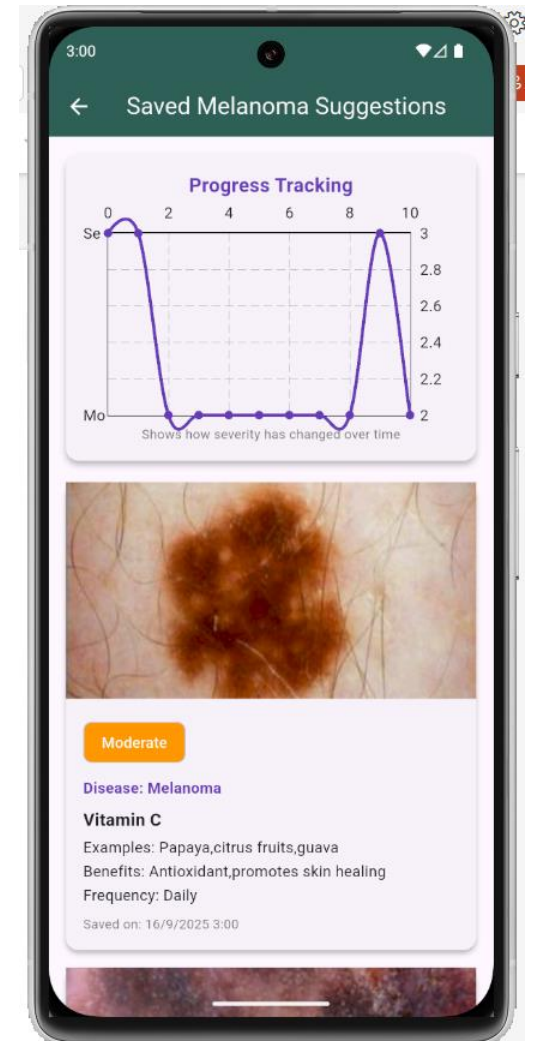
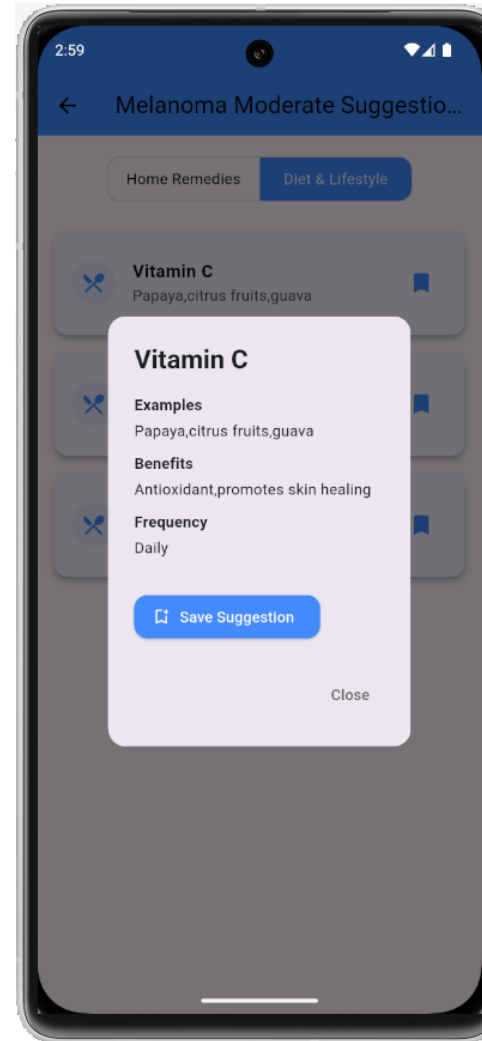
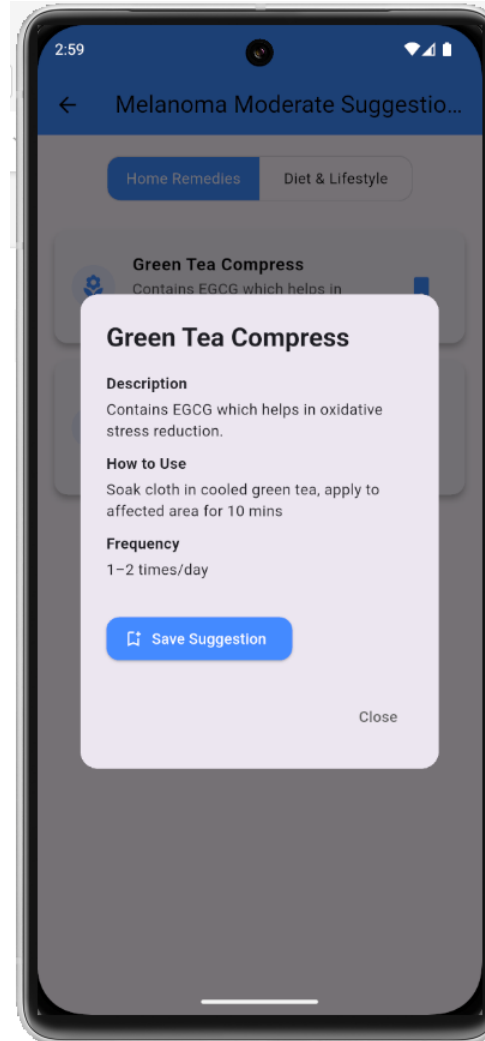
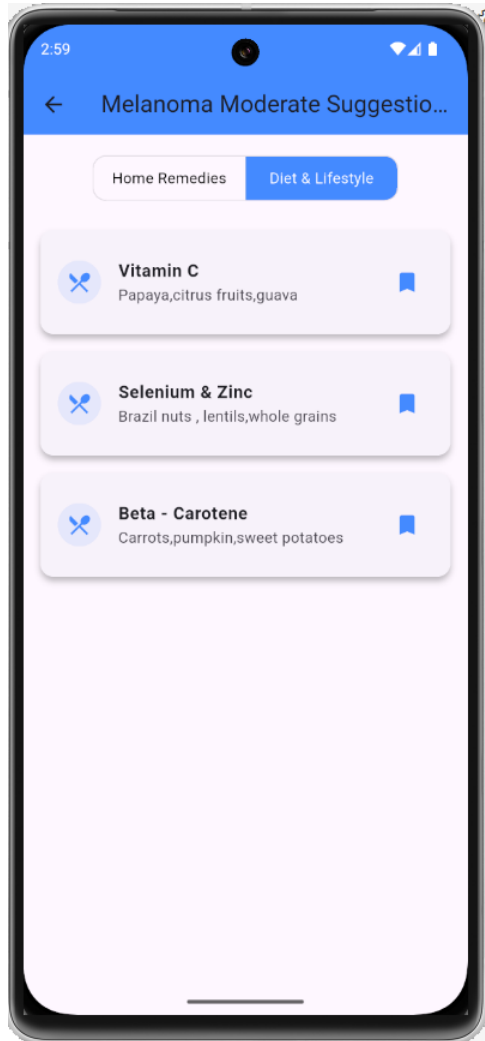


Developed mobile app UI's

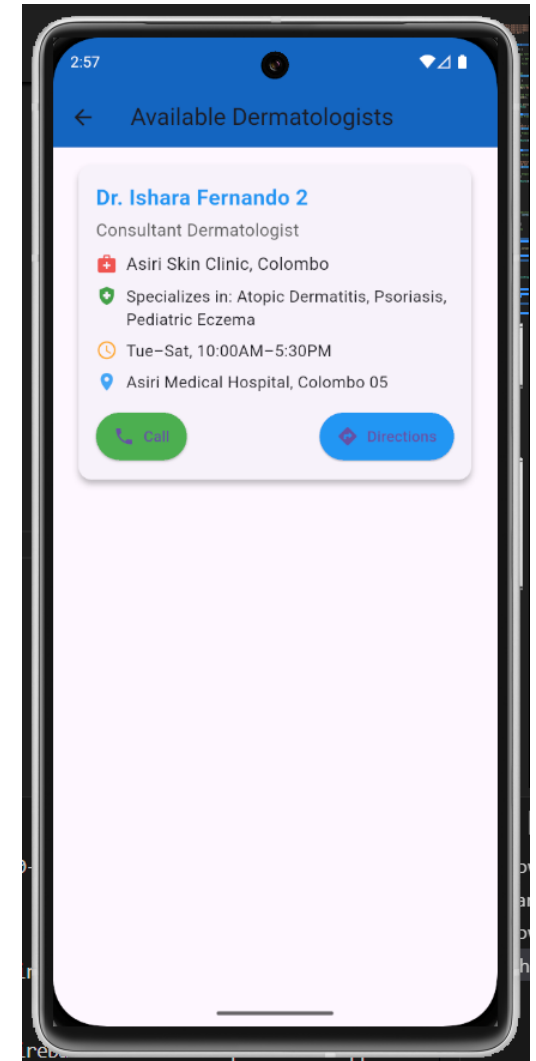
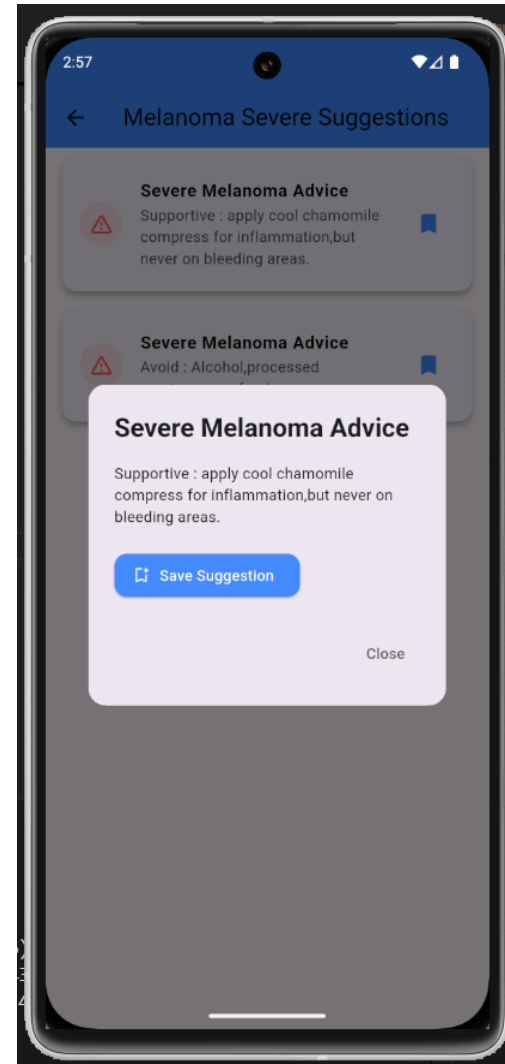
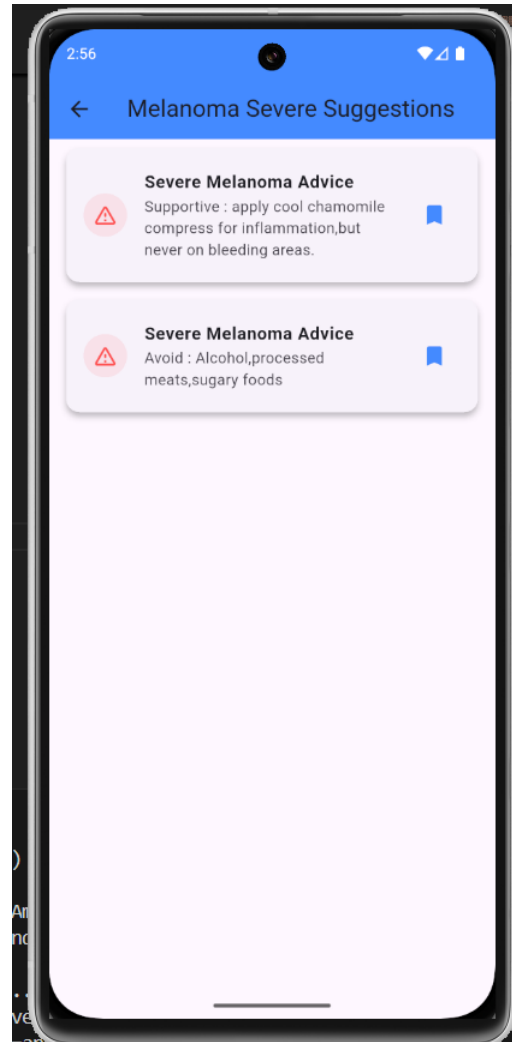
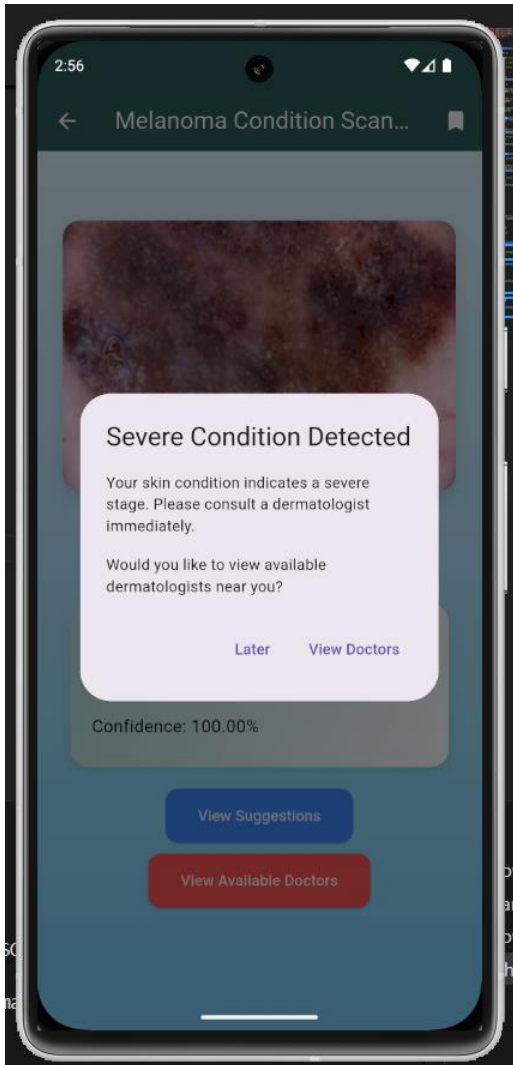
Melanoma



Developed mobile app UI's

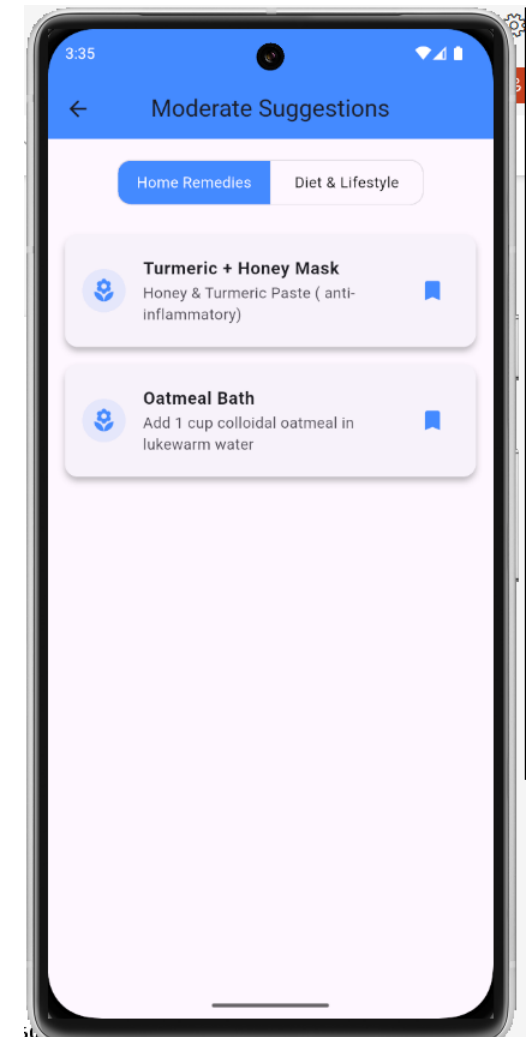
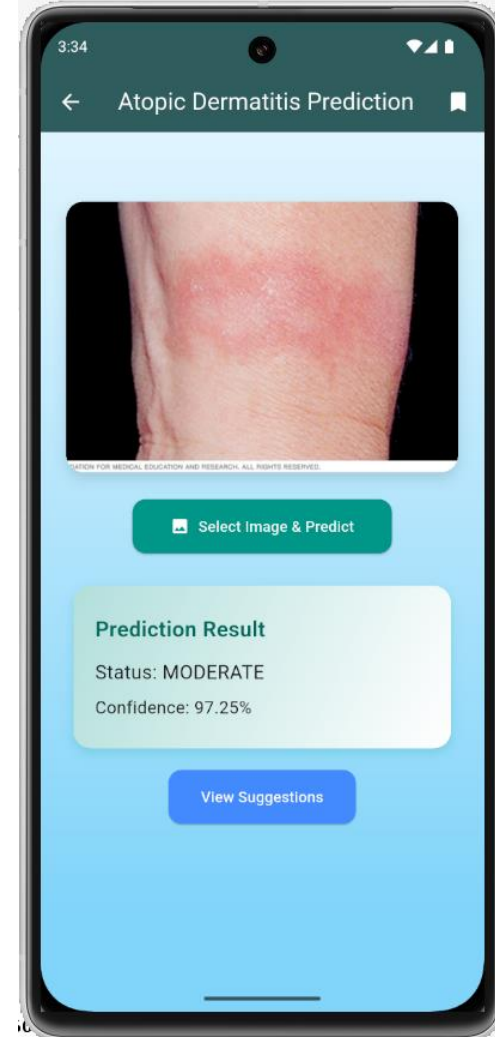
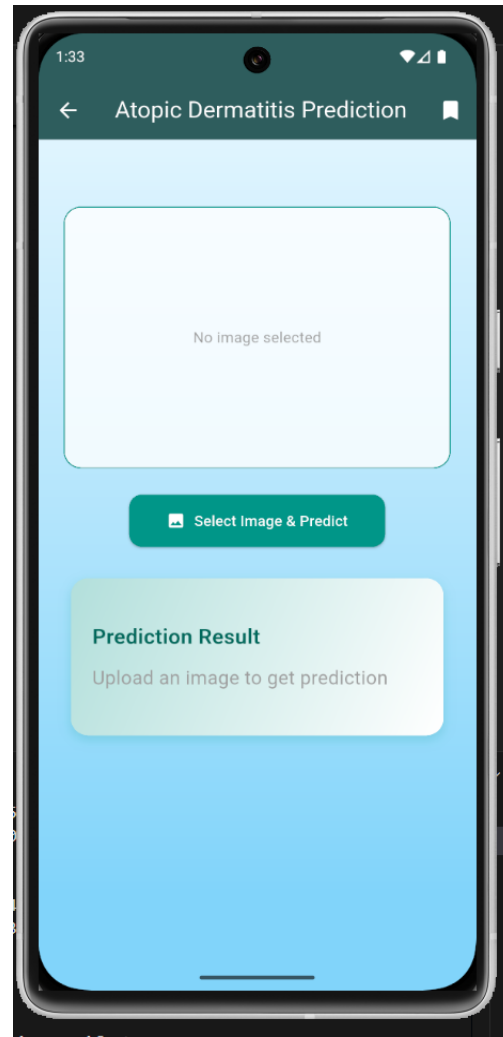
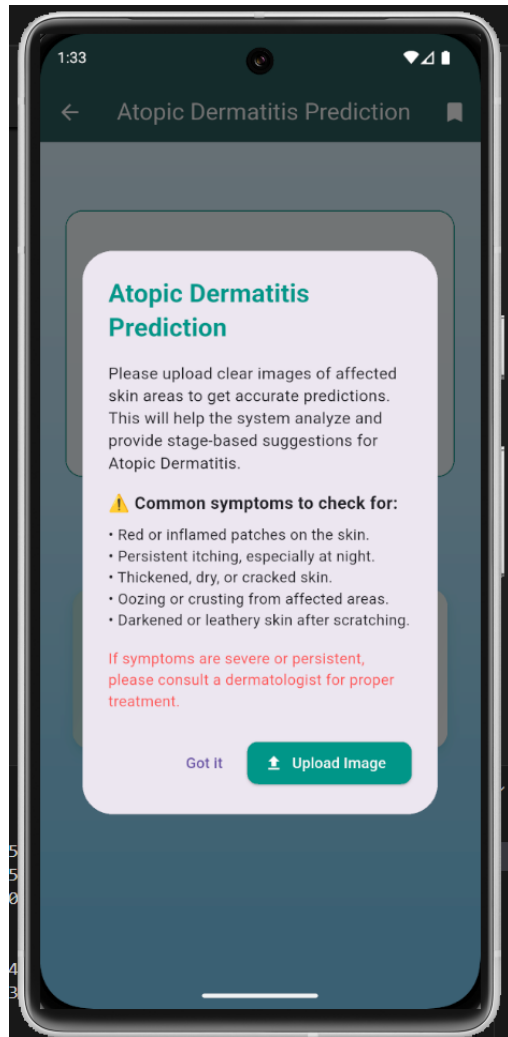


Developed mobile app UI's

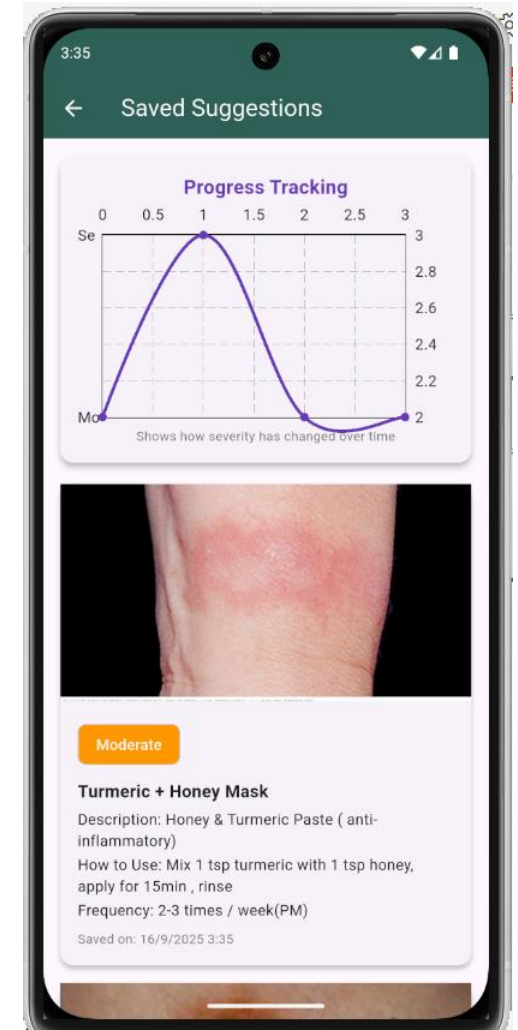
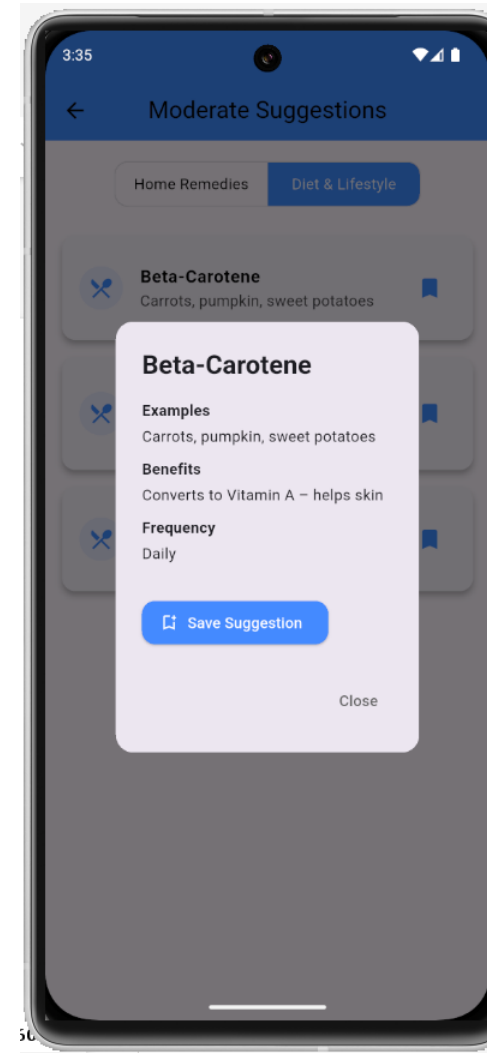
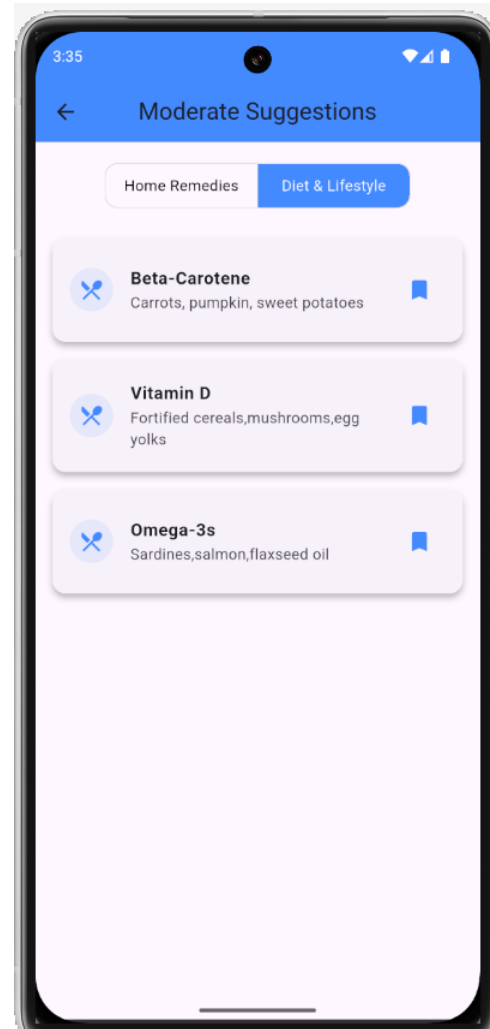
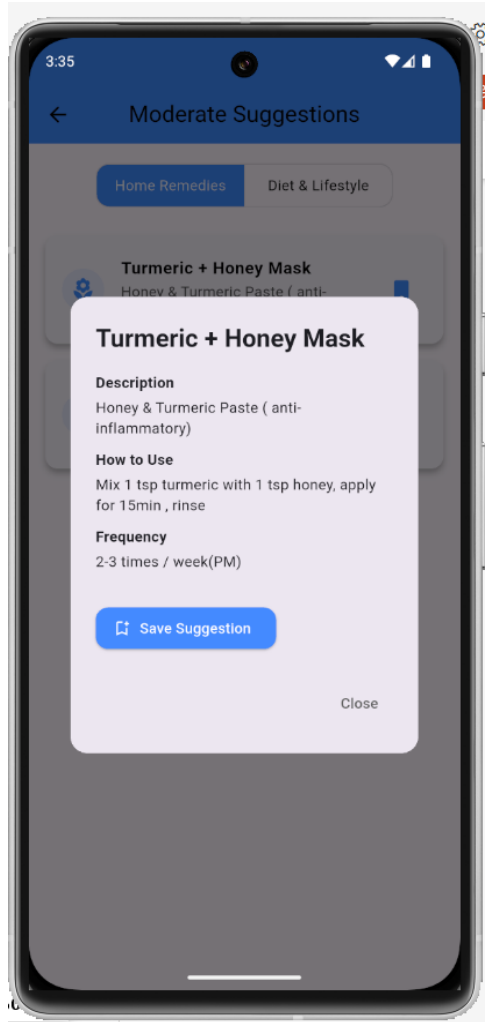


Developed mobile app UI's

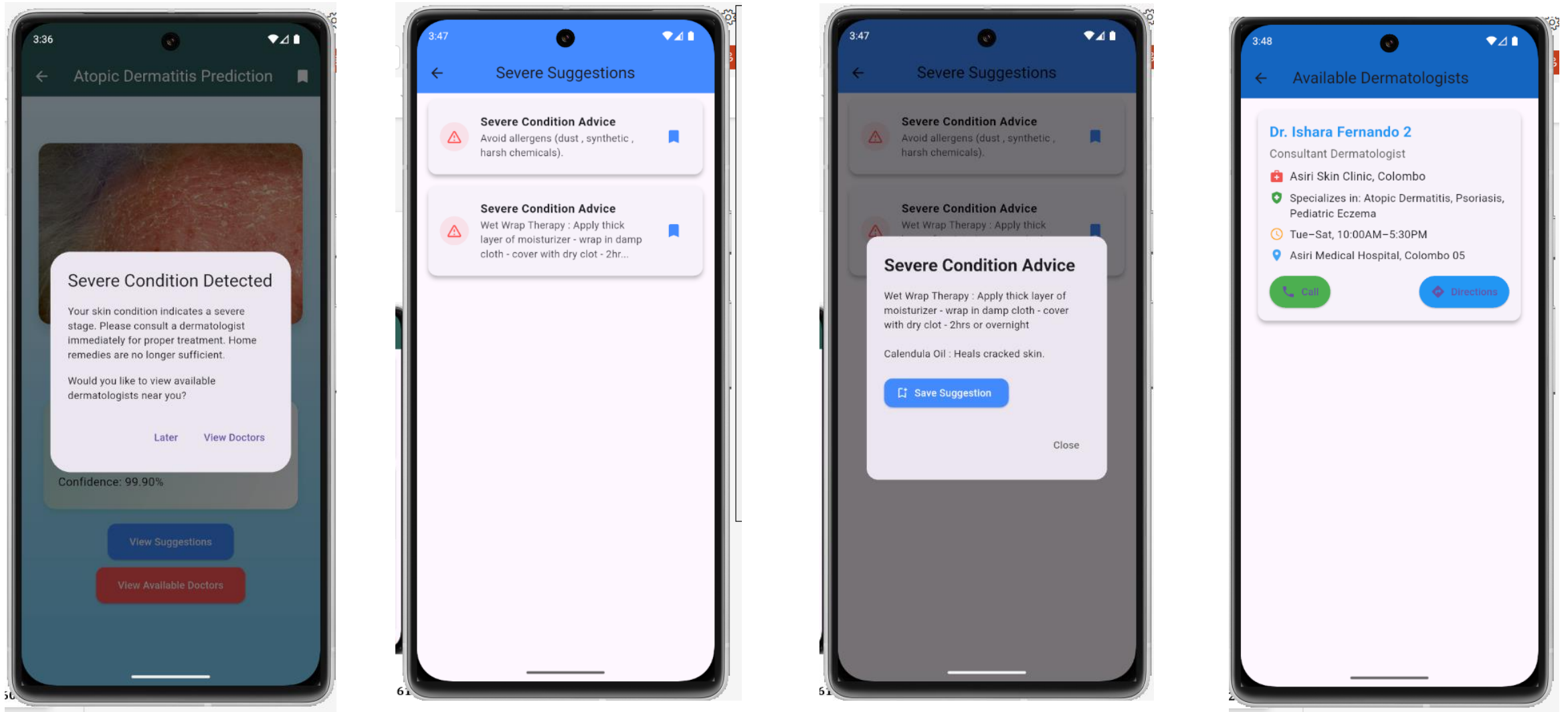
Atopic Dermatitis



Developed mobile app UI's

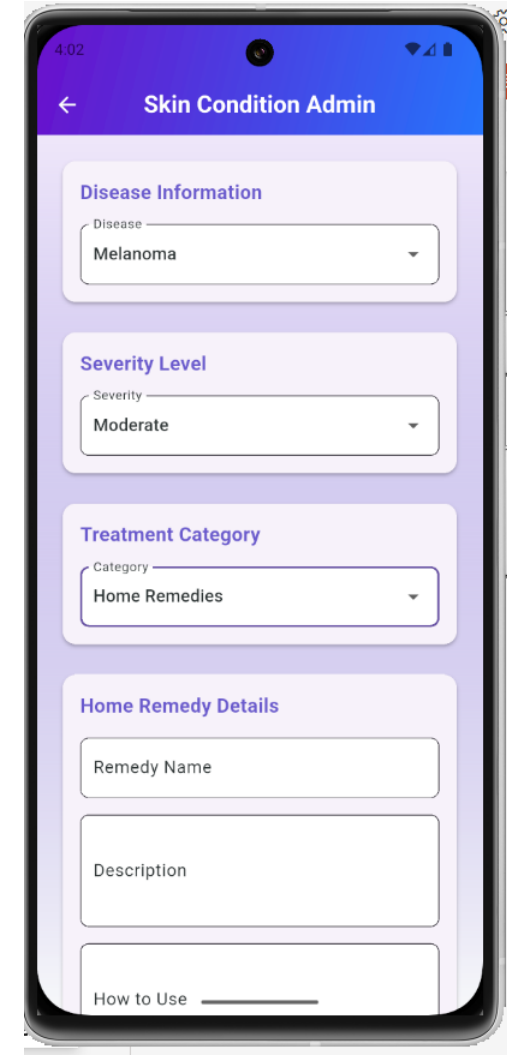
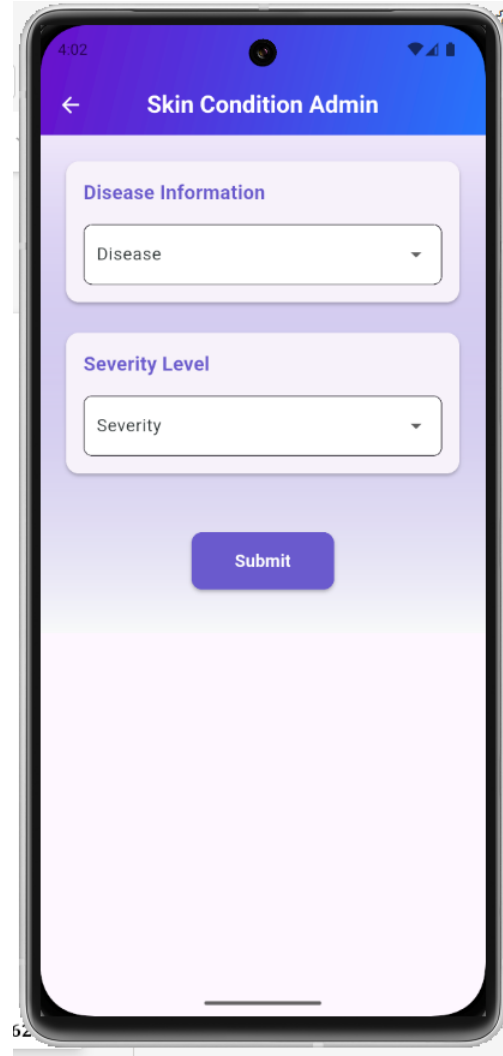
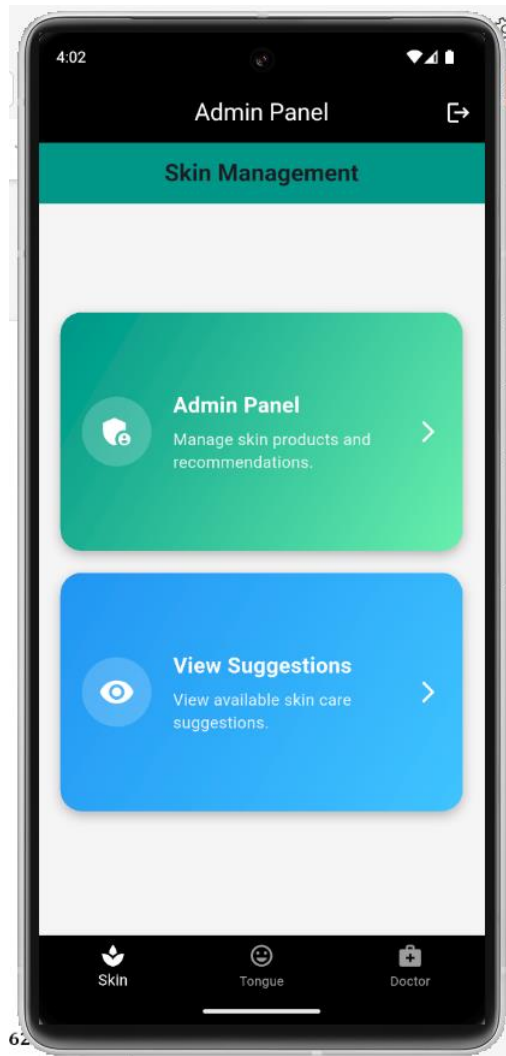


Developed mobile app UI's

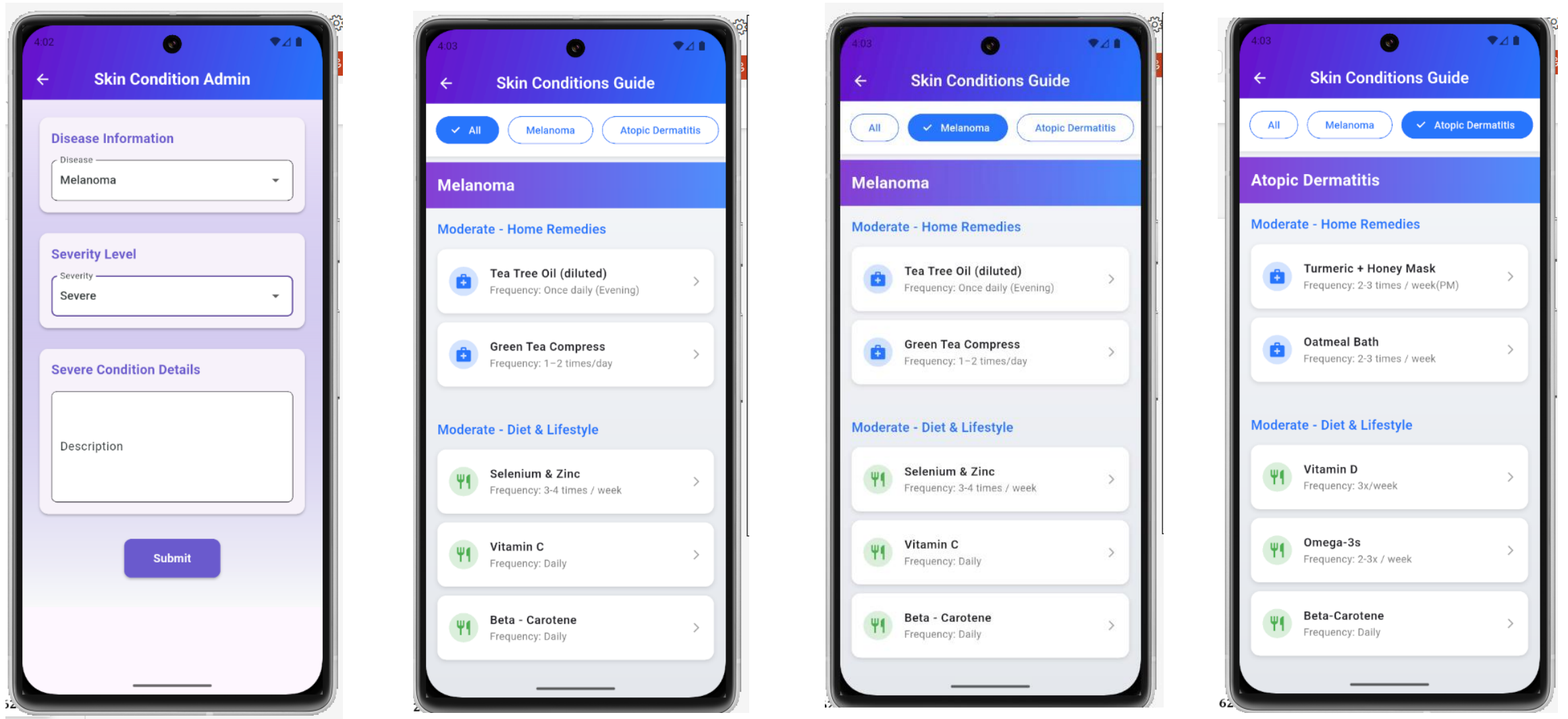


Developed mobile app UI's

Admin Pannel



Developed mobile app UI's



NEXT EXPECTED ACHIEVEMENTS

- ❖ More enhance the Frontend user interfaces
- ❖ discuss with the doctor and increase the home remedies



Requirements Specification

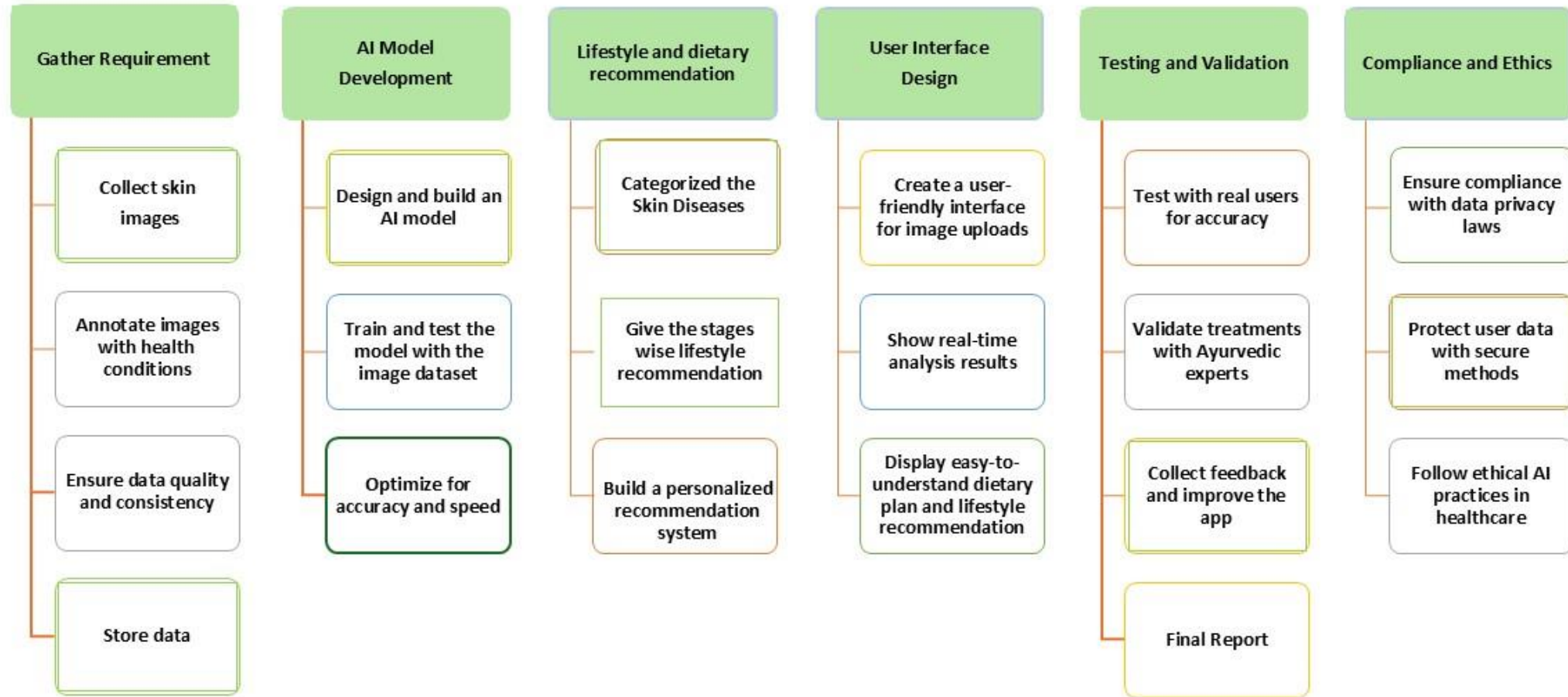
Functional Requirements

- Skin Disease Identification: Classify skin diseases into normal, moderate, and severe stages based on image analysis.
- Stage-Specific Dietary Recommendations and Lifestyle Recommendations
- User Interface: Provide an intuitive interface for users to upload images and receive results.

Non - Functional Requirements

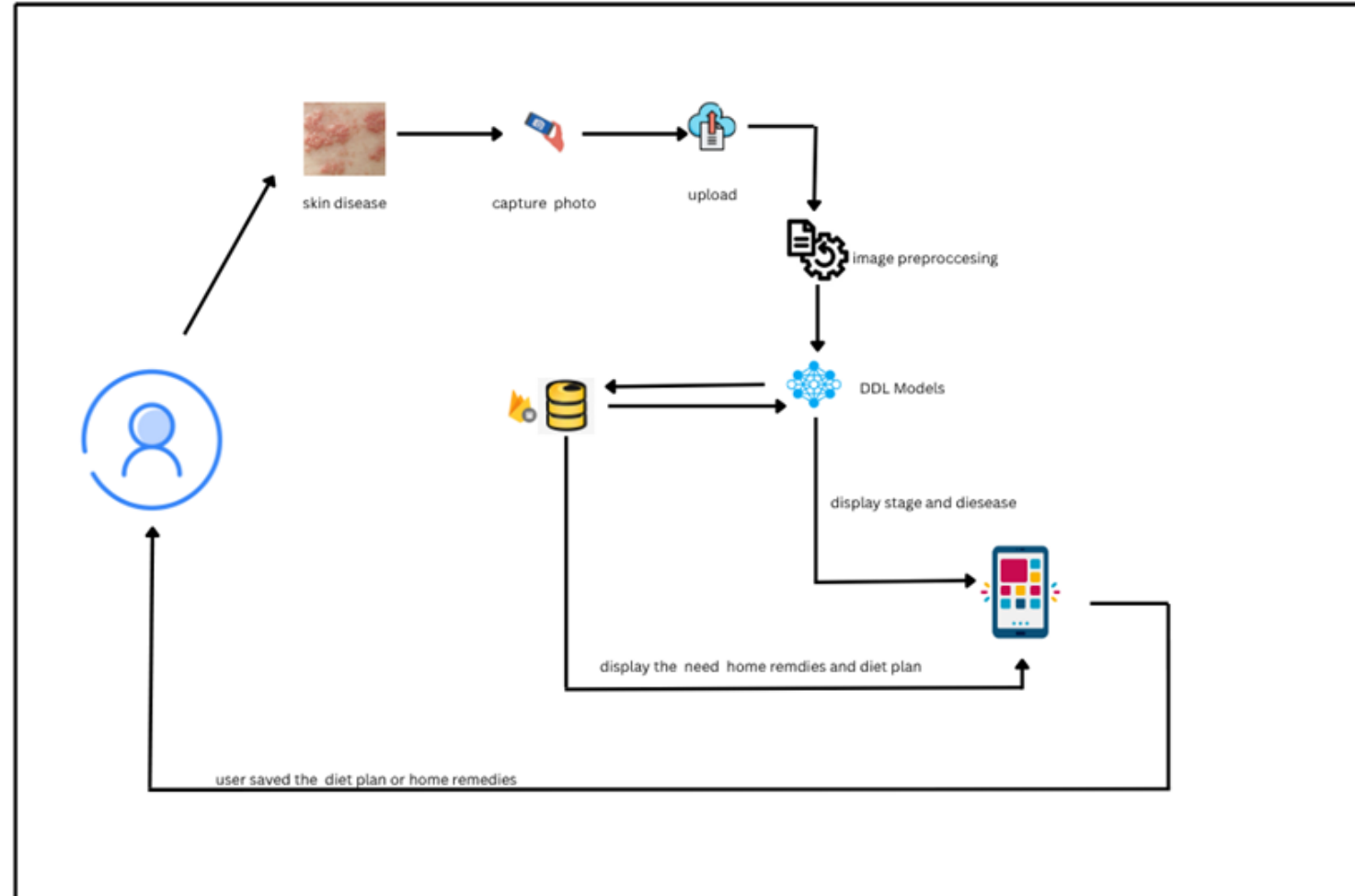
- Performance: Ensure real-time processing of images with minimal latency.
- Scalability: Design the system to handle a growing number of users and data.
- Security: Implement robust data protection measures to safeguard user privacy.
- Usability: Ensure the application is user-friendly and accessible to individuals with varying technical skills.

WBS - AI-powered identification of skin diseases



METHODOLOGY

SYSTEM DIAGRAM

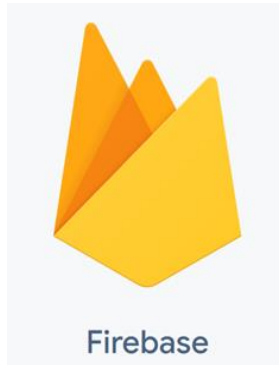


Technologies to be used



Keras

A deep learning library



Firebase

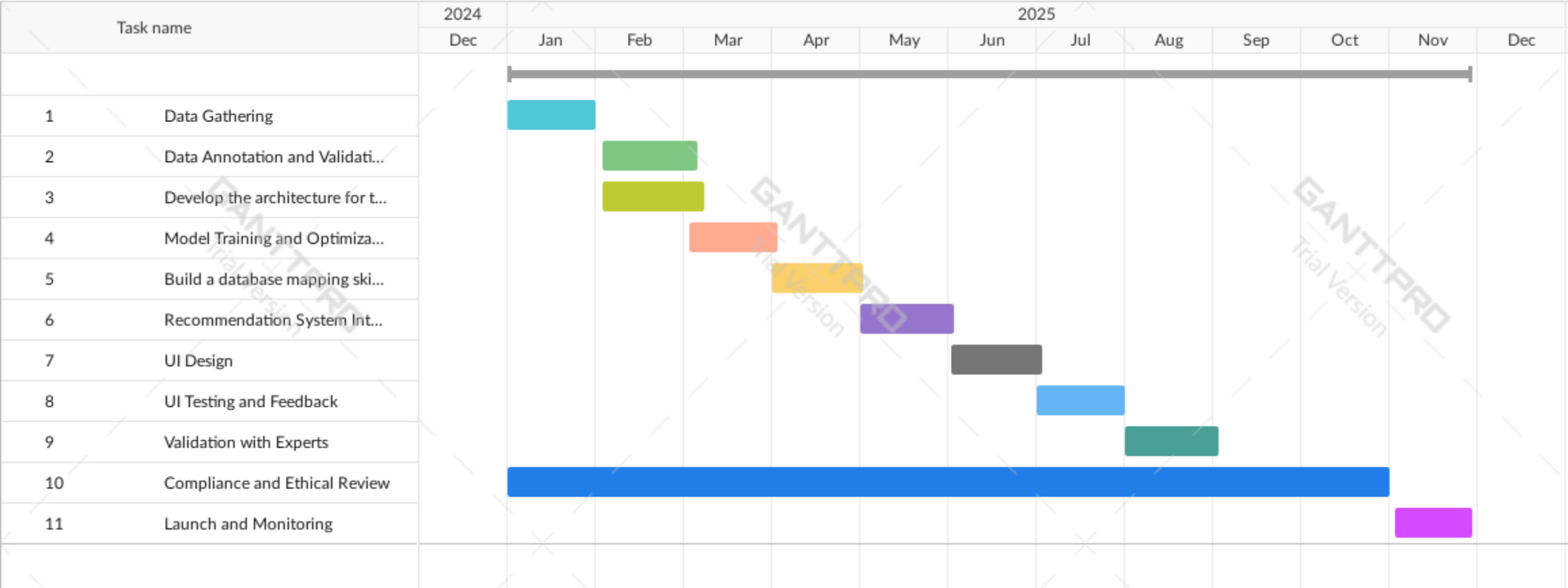


GitHub



TensorFlow

Gantt chart



Supporting Items (Commercialization)

Target Market:

- Health-Conscious Individuals (Ages 18-45)
- People with Skin Conditions (Acne, Eczema, etc.)
- Ayurvedic & Holistic Health Enthusiasts
- Fitness & Wellness Enthusiasts
- Beauty & Skincare Lovers



Partnerships:

- Ayurvedic Clinics & Experts
- Dieticians & Nutritionists
- Skincare Brands & Wellness Apps
- Herbal & Pharmaceutical Companies
- Tech Companies for AI & Image Recognition
- Health Influencers & Bloggers

References

- A. Smith and B. Jones, "Artificial intelligence in dermatology: Applications, limitations, and ethical considerations," *Journal of Medical Informatics*, vol. 47, no. 3, pp. 221–230, May 2022.
- R. Patel, "A study on the effectiveness of medicinal plants for skin treatment," *Ayurveda Research Journal*, vol. 15, no. 2, pp. 85–92, Mar. 2020.
- M. Chopra and A. Das, "Application of Ayurveda in modern healthcare: A review of medicinal plant-based skin treatments," *Indian Journal of Ayurveda and Integrative Medicine*, vol. 12, no. 1, pp. 15–22, Jan. 2021.



IT21802812 | HUZNADH M.N.M

MENTAL WELL-BEING ASSESSMENT THROUGH FACIAL EMOTION ANALYSIS

Information Technology (IT)

INTRODUCTION

- Mental health is crucial for overall well-being.
- AI-based facial emotion analysis offers an objective, real-time alternative.
- This research integrates AI with Ayurvedic principles for holistic well-being

Research Questions

- How can AI-based facial emotion analysis assess mental well-being?
- What Ayurvedic techniques are most effective for stress management?
- How can real-time emotional tracking improve mental health?



Research Gap

Lack of personalized well-being recommendations in AI emotion detection

Limited accessibility of mental health tools

Absence of long-term emotional state monitoring

Ethical and privacy concerns in facial recognition technology



OBJECTIVES

- Build an AI model to accurately detect and classify facial emotions.
- Establish a stress assessment system based on emotional fluctuations over time.
- Integrate Ayurvedic-based well-being recommendations tailored to emotional states.
- Ensure accessibility, usability, and data privacy in the system design.

METHODOLOGY

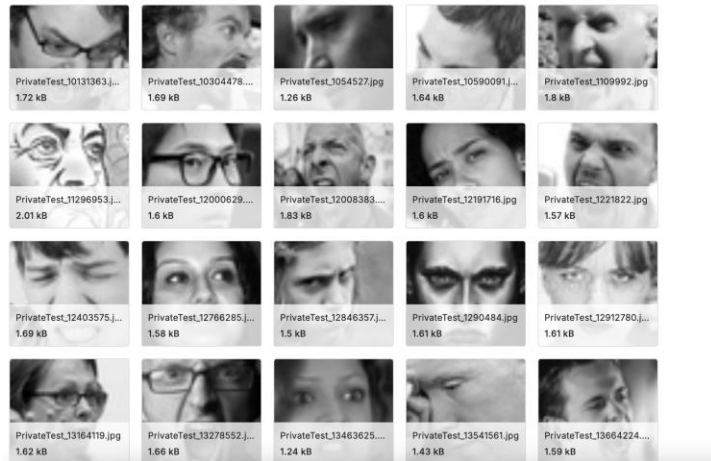
- **Image Acquisition** – Captures and preprocesses user facial images for clarity and accuracy.
- **Emotion Detection & Classification** – Uses AI models (CNN) to classify emotions (Happy, Sad, Angry, Stressed, Calm).
- **Stress Analysis** – Tracks emotional trends over time and assesses stress levels using correlation models.
- **Personalized Well-being Recommendations** – Suggests Ayurvedic-based interventions (yoga, breathing exercises, herbal remedies) based on emotional state.
- **System Integration & Testing** – Conducts unit testing, performance optimization

ACHIEVEMENTS

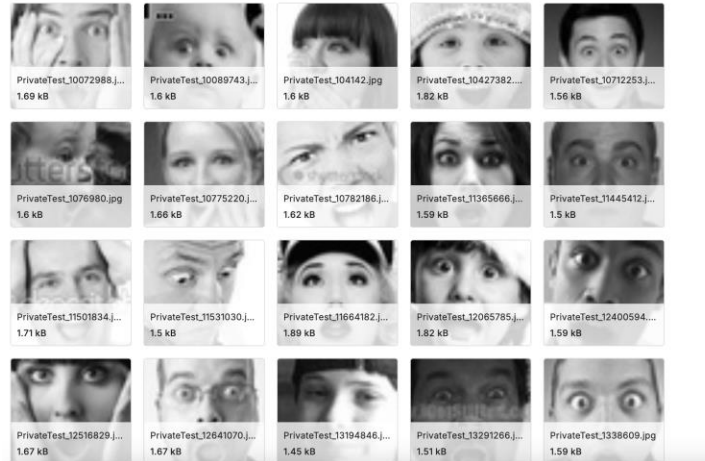
- Successful data collection for model training.
- Trained a one model for specific task.
- Completed primary UI designs prototypes.
- Highest accuracy model up to now.

Successful Data Collection for Model Training

angry (958 files)



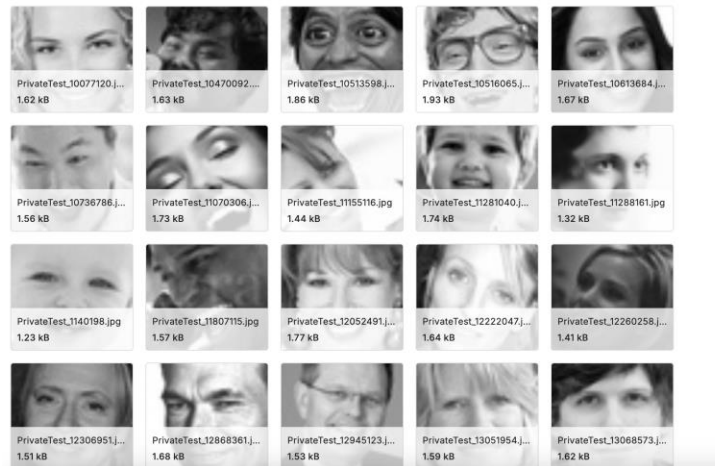
surprise (831 files)



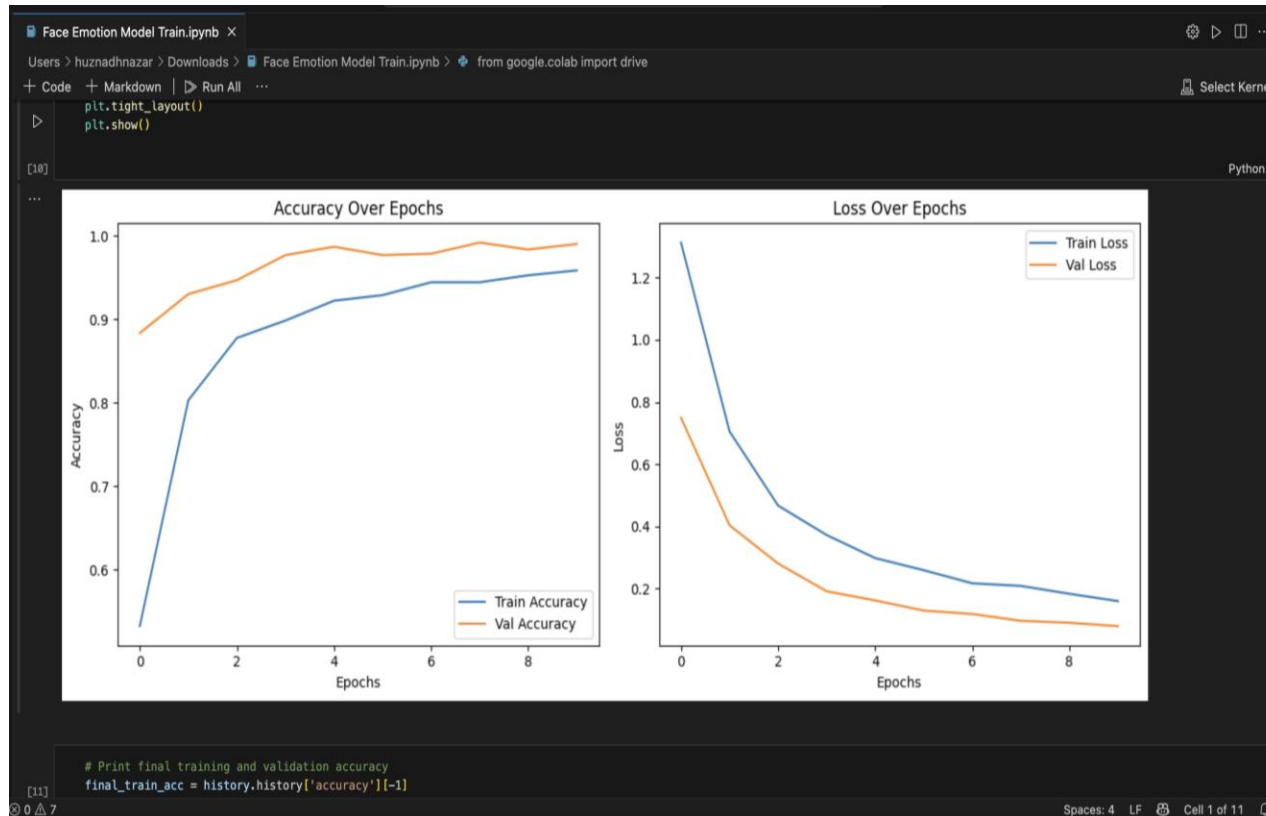
sad (1247 files)



happy (1774 files)



Highest accuracy model up to now



```
# Print final training and validation accuracy
final_train_acc = history.history['accuracy'][-1]
final_val_acc = history.history['val_accuracy'][-1]

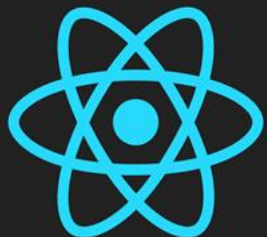
print(f"Final Training Accuracy: {final_train_acc * 100:.2f}%")
print(f"Final Validation Accuracy: {final_val_acc * 100:.2f}%")
```

... Final Training Accuracy: 95.83%
Final Validation Accuracy: 99.00%

Technologies to be used



Keras
A deep learning library



React Native



Firebase



GitHub

Functional Requirements

- Users can upload or capture real-time facial images.
- Track emotional trends over time.
- Identify and classify emotions

Non-Functional Requirements

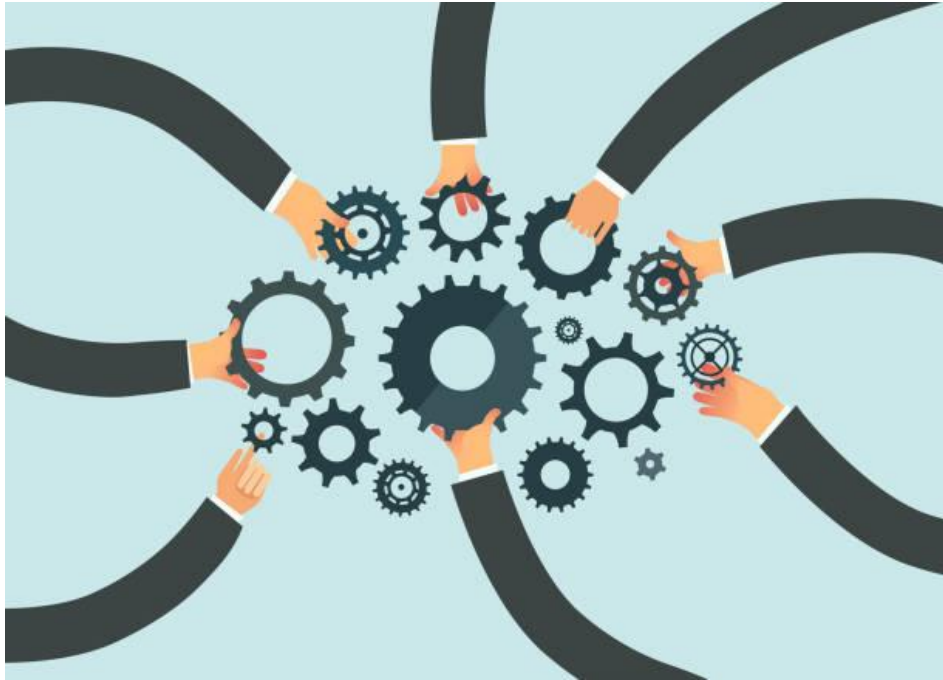
- Handle many users and allow future updates.
- Detect emotions within 2 seconds with 90%+ accuracy.



Supporting Items (Commercialization)

Target Market:

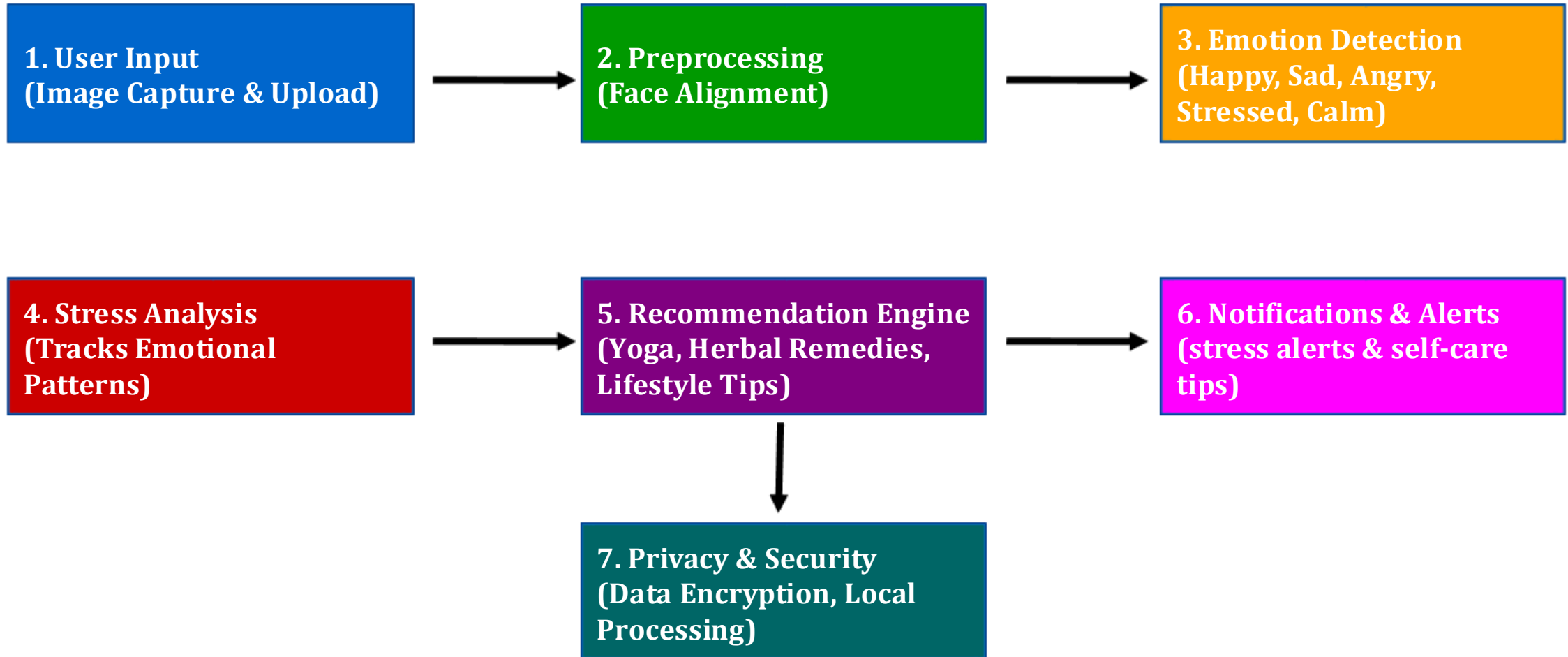
- Health-conscious individuals.
- Urban and rural Ayurveda enthusiasts.



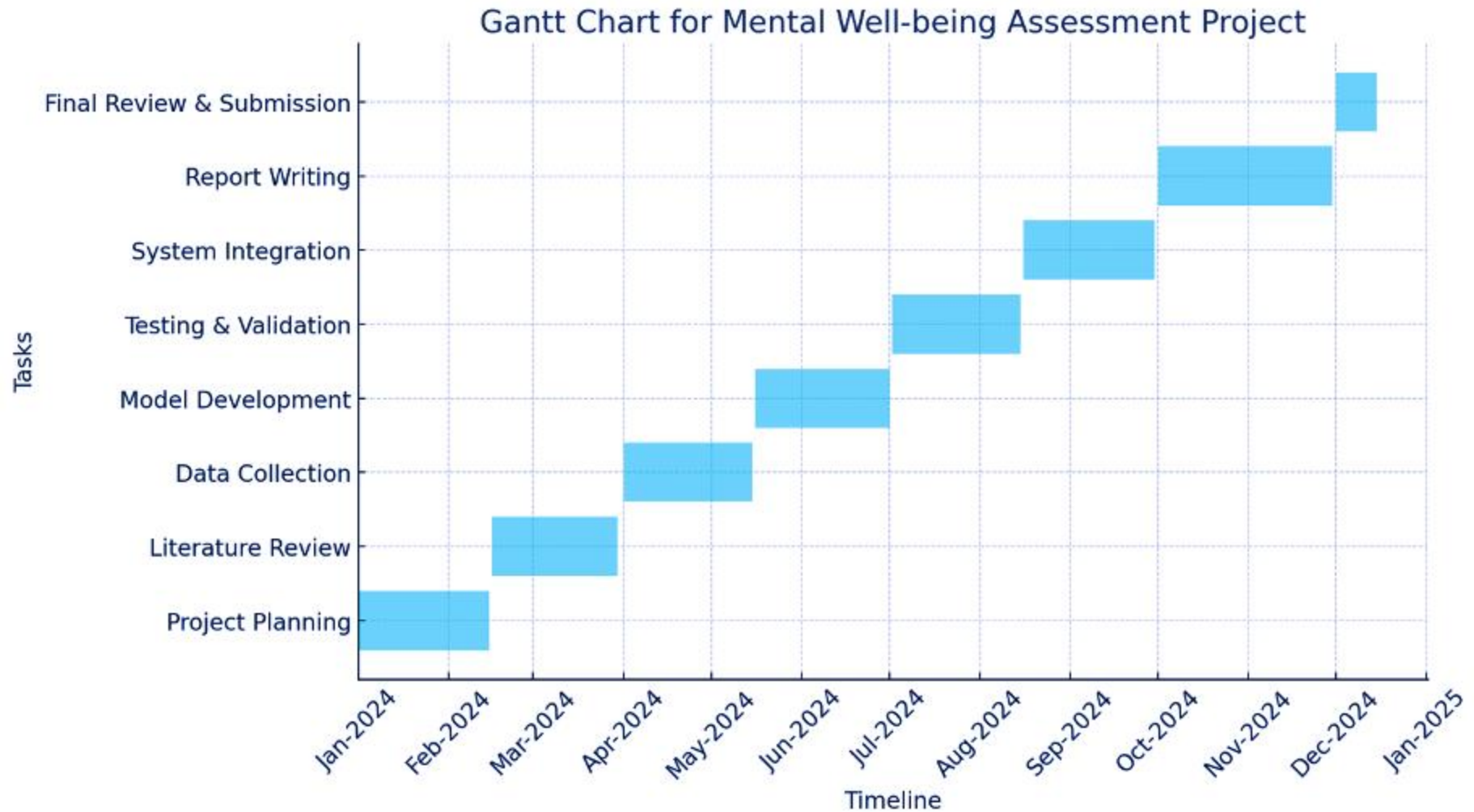
Partnerships:

- Collaborate with Ayurvedic practitioners and wellness centers.
- Partner with wearable tech companies for seamless integration.

System Diagram



Gantt Chart



Work Breakdown Chart

Project Planning & Requirements Gathering

- Define project scope, objectives, and team roles
- Develop project timeline and milestones

Data Collection & Preprocessing

- Identify data sources
- Collect facial expression dataset
- Preprocess and clean dataset

AI Model Development

- Select suitable model architecture
- Train emotion detection model
- Evaluate model performance

System Development

- Develop web/mobile application
- Integrate AI model with system
- Ensure real-time processing

Testing & Validation

- Test system with sample users
- Gather and analyze feedback
- Optimize system performance

Deployment & Maintenance

- Deploy final system
- Develop user manuals and tutorials
- Monitor system performance

REFERENCES

- Lad, Vasant. Textbook of Ayurveda: Fundamental Principles of Ayurveda. Ayurvedic Press, 2002.
- Yang, Guang-Zhong, et al. "The future of wearable technologies for personalized healthcare." Nature Electronics 1.6 (2018): 302-312.
- McCall, Timothy. Yoga as Medicine: The Yogic Prescription for Health and Healing. Bantam, 2007.
- Bonato, Paolo. "Wearable sensors and systems." IEEE Engineering in Medicine and Biology Magazine 29.3 (2010): 25-36.

CONCLUSION

- This system combines AI-driven emotion detection with Ayurvedic well-being to provide real-time stress monitoring and personalized self-care.
- ensures accuracy, privacy, and accessibility, empowering users to manage their mental health effectively.



IT21190520 | KAUSHALYA H.G.B



AI-BASED AYURVEDIC MEDICINAL PLANT IDENTIFICATION USING LEAF IMAGE ANALYSIS.

Information Technology (IT)

Research Progress Overview

Complete Chapters:

- Chapter 1 – Introduction
- Chapter 2 – Literature Review
- Chapter 3 – Technology
- Chapter 4 – Research Design

Next Phase:

- Chapter 5 – Analysis
- Chapter 6 – Result & Discussion
- Chapter 7 – Conclusion & Recommendations



Figure 1: Research Progress Roadmap

BACKGROUND & MOTIVATION

- **Background** = Similar to how the ITSM slide explained the shift from ITIL → AI-driven ITSM, here you explain the shift from **traditional/manual plant identification** → **AI-powered real-time plant diagnosis with Ayurvedic treatments**.
- **Motivation** = The reason *why* this matters (ex: difficulty in manual identification, risk of misdiagnosis, lack of digital Ayurvedic tools, farmers needing quick solutions, etc.).

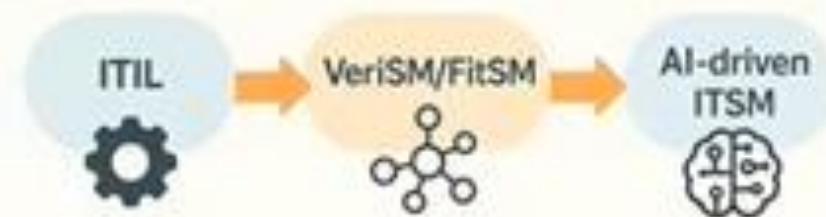
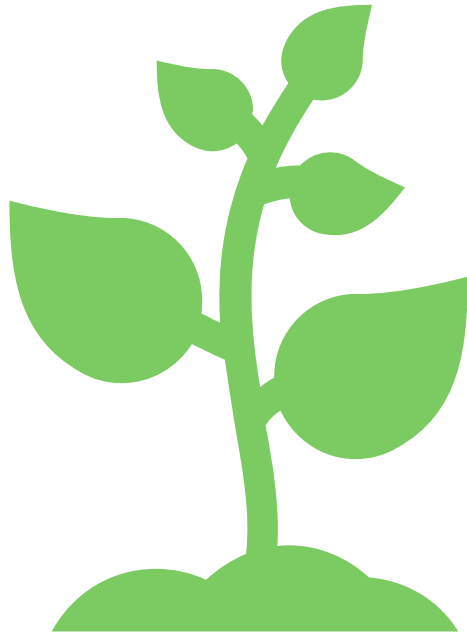


Figure 2: Evolution from ITIL
→ VeriSM/FitSM → AI-driven ITSM



Aim & Objectives

Aim:

Develop an AI-powered plant identification system that accurately recognizes medicinal plant leaves and provides **personalized Ayurvedic treatment guidance** for selected diseases.

Objectives:

- Develop and train an accurate AI model for plant identification
- Build a comprehensive medicinal plant database with Ayurvedic treatments
- Enable offline accessibility for rural users
- Integrate personalized Ayurvedic guidance into the mobile app
- Design a user-friendly mobile interface for easy plant recognition
- Improve model accuracy using diverse leaf images



Problem & Proposed Solution

Problem

- Difficult to identify medicinal plants accurately
- Morphological similarities & seasonal variations cause confusion
- Limited availability of Ayurvedic experts in rural areas
- Existing plant ID apps lack Ayurvedic medicinal insights

Proposed Solution

- AI-powered mobile app **Ayushasthra**
- Uses **leaf image analysis (CNN models)** for plant identification
- Provides Ayurvedic medicinal knowledge (uses, dosage, contraindications)
- Offline functionality for rural accessibility
- Improves accuracy, usability, and healthcare empowerment



Key Framework Solutions

- ITIL → Reliable but rigid, not flexible for plant ID
- VeriSM → Flexible & adaptive, fits Ayurveda + AI
- FitSM → Lightweight & quick, but limited scalability
- AI Tools → Automated, accurate, but with ethical challenges

Research Gap Analysis

- Few AI-based plant identification systems focus on **Ayurvedic medicinal plants**
- Existing apps → classify plants but **lack medicinal knowledge integration**
- **Dataset limitations** → small, not diverse, limited expert validation
- No strong **offline solutions** for rural communities
- Limited studies combining **AI + Ayurveda** for real-world healthcare impact

Focus of Study	Research Approach/Method	Gap Identified	Example
AI in plant identification	Image-based analysis using CNNs	Limited to taxonomy; lacks Ayurvedic medicinal integration	Prior plant ID apps (PlantNet, LeafSnap)
Dataset creation for medicinal plants	Small, public datasets (PlantCLEF, Flavia)	Low diversity, no expert validation	Open datasets only cover general plants
Ayurveda + AI integration	Conceptual discussions only	No empirical systems linking plant ID with Ayurvedic knowledge	Few academic papers, no apps
Accessibility in rural areas	Mobile/online plant ID apps	Require internet; no offline solutions	Current AI apps need connectivity
Validation & usability	Limited user studies	Few evaluations with Ayurveda doctors/botanists	Research lacks expert-backed validation

Table: Research Gap Analysis

Technology be used

- Python, TensorFlow, Keras
- OpenCV (image processing)
- MySQL (database)
- Flutter (mobile app)
- Android Studio (deployment)



Modified Function: Plant Identification

- Uses **leaf image analysis** (CNN models)
- Identifies selected Ayurvedic plants (Pawatta, Nika, Beli)
- Provides **Ayurvedic medicinal insights** (uses, dosage, contraindications)
- Works in **offline mode** for rural accessibility
- **Validated** by Ayurvedic doctors & botanists



Research Design

- **Dataset Creation** → 1,500 leaf images (healthy & diseased)
- **Preprocessing** → resizing, normalization, background removal, contrast enhancement
- **AI Model Training** → CNN (ResNet, VGG, MobileNet), transfer learning, augmentation
- **Integration** → Ayurvedic medicinal knowledge (uses, dosha, contraindications)
- **Mobile Deployment** → Flutter , TensorFlow Lite, offline support
- **Validation** → Expert reviews (Ayurveda doctors, botanists), user testing



Conceptual Framework



INDEPENDENT: AI MODELS (CNNs), IMAGE PREPROCESSING, DATASET CREATION



DEPENDENT: ACCURACY OF PLANT IDENTIFICATION, AYURVEDIC KNOWLEDGE DELIVERY, USER SATISFACTION



MODERATORS: EXPERT VALIDATION, OFFLINE ACCESSIBILITY, USABILITY TESTING

Research Gap

Table 1-4. Research Gap Analysis

Features	Study 1: Leaf Image Classificati on [9]	Study 2: Transfer Learning Approach es [10]	Study 3: Mobile Plant Identificati on Apps [11]	Study 4: AI + Ayurveda Integratio n [12]	Study 5: Preprocessi ng & Dataset Studies [13]	Propose d Approac h
User-Friendly System	X	X	✓ (Medium usability)	X	X	✓ (High usability)
Mobile Solution	X	X	✓	X	X	✓
Integration with Ayurvedic Knowledge	X	X	X	✓	X	✓
Diverse Dataset for Training	X	✓	X	X	✓	✓
Robust Image Preprocessing	X	X	X	X	✓	✓
Lightweight, Offline-Compatible Model	X	X	X	X	X	✓
Expert Validation (Ayurveda + Botany)	X	X	X	✓ (Partial)	X	✓

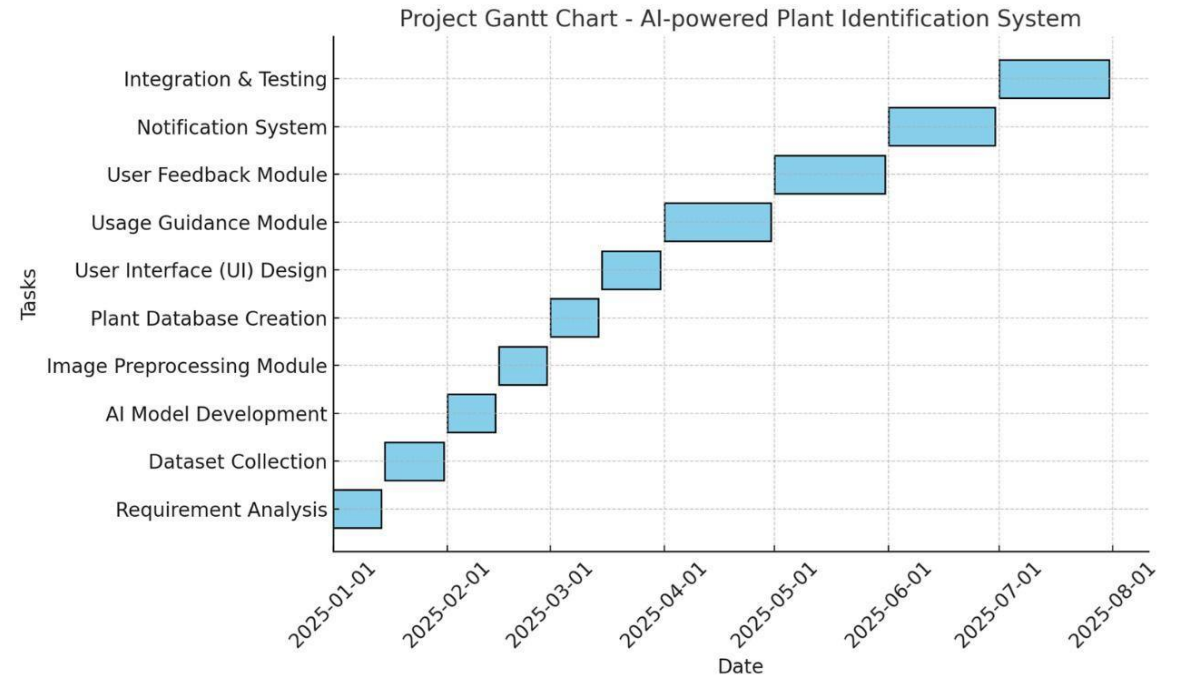
Model: "EffNetB0_MT"

Layer (type)	Output Shape	Param #	Connected to
image_0to1 (InputLayer)	(None, 300, 300, 3)	0	-
effnet_preprocess (Lambda)	(None, 300, 300, 3)	0	image_0to1[0][0]
rescaling_6 (Rescaling)	(None, 300, 300, 3)	0	effnet_preproces...
normalization_3 (Normalization)	(None, 300, 300, 3)	7	rescaling_6[0][0]
rescaling_7 (Rescaling)	(None, 300, 300, 3)	0	normalization_3[...
stem_conv_pad (ZeroPadding2D)	(None, 301, 301, 3)	0	rescaling_7[0][0]
stem_conv (Conv2D)	(None, 150, 150, 32)	864	stem_conv_pad[0]...
stem_bn (BatchNormalizatio...	(None, 150, 150, 32)	128	stem_conv[0][0]
stem_activation (Activation)	(None, 150, 150, 32)	0	stem_bn[0][0]
block1a_dwconv (DepthwiseConv2D)	(None, 150, 150, 32)	288	stem_activation[...
block1a_bn (BatchNormalizatio...	(None, 150, 150, 32)	128	block1a_dwconv[0...
block1a_activation (Activation)	(None, 150, 150, 32)	0	block1a_bn[0][0]
block1a_se_squeeze (GlobalAveragePooli...	(None, 32)	0	block1a_activati...

Layers



Timeline – Gantt chart



Achievement

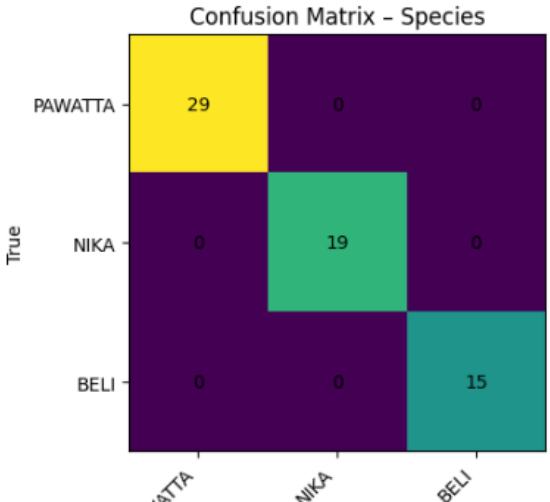
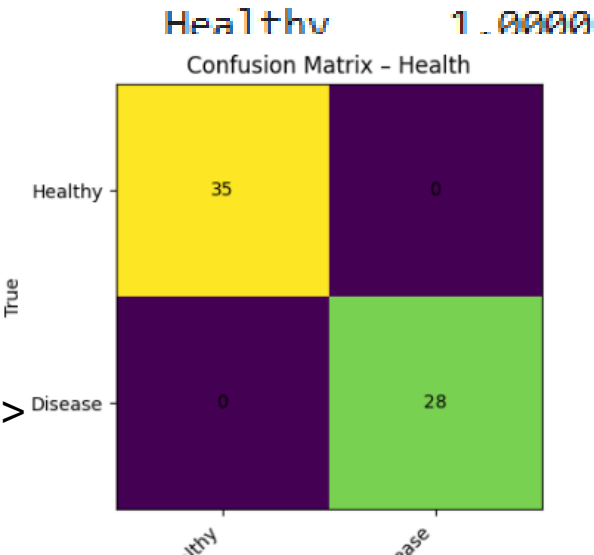


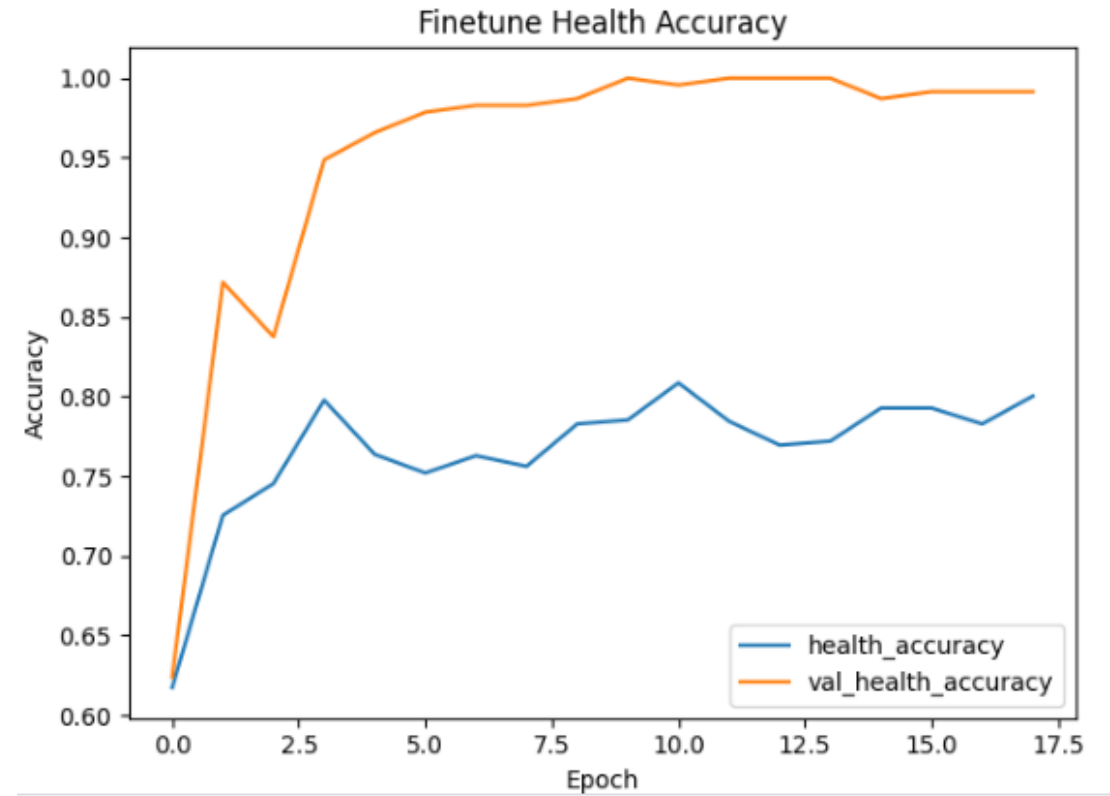
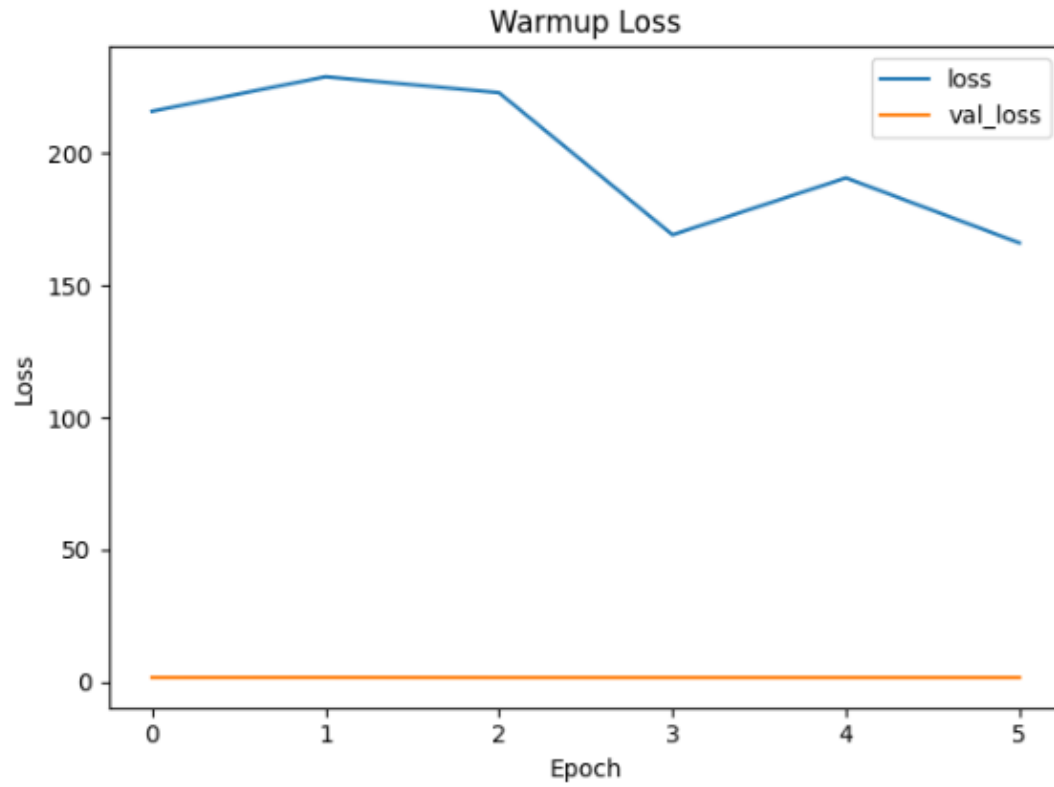
- Completed **Chapters 1–4** (Introduction, Literature Review, Methodology, Design)
- Created **dataset** of selected medicinal plants (Pawatta, Nika, Beli)
- Applied **preprocessing techniques** (resizing, background removal, enhancement)
- Built **prototype CNN model** for plant identification
- Designed initial **Ayushasthra mobile app prototype**

Confusion Matrix

Species report:				
	precision	recall	f1-score	support
PAWATTA	1.0000	1.0000	1.0000	29
NIKA	1.0000	1.0000	1.0000	19
BELI	1.0000	1.0000	1.0000	15
accuracy			1.0000	63
macro avg	1.0000	1.0000	1.0000	63
weighted avg	1.0000	1.0000	1.0000	63

Health report:				
	precision	recall	f1-score	support
Healthy	1.0000	1.0000	1.0000	35





Testing Validation Accuracy & Lost

```

Epoch 1/6
51/51 ————— 59s 659ms/step - health_accuracy: 0.4399 - health_loss: 98.8439 - loss: 235.0756 - species_accuracy: 0.3755 - species_loss: 137.2213 - val_health_accuracy: 0.6325 - val_health_loss: 0.6330 - val_loss: 1.6862 - val_species_accuracy: 0.4744 - val_species_loss: 1.0656 - learning_rate: 0.0010
Epoch 2/6
51/51 ————— 16s 319ms/step - health_accuracy: 0.4950 - health_loss: 92.8730 - loss: 252.2424 - species_accuracy: 0.3678 - species_loss: 159.2113 - val_health_accuracy: 0.6111 - val_health_loss: 0.6490 - val_loss: 1.7119 - val_species_accuracy: 0.4872 - val_species_loss: 1.0626 - learning_rate: 0.0010
Epoch 3/6
51/51 ————— 16s 304ms/step - health_accuracy: 0.4731 - health_loss: 92.6364 - loss: 233.8956 - species_accuracy: 0.3989 - species_loss: 141.4363 - val_health_accuracy: 0.6410 - val_health_loss: 0.6231 - val_loss: 1.6749 - val_species_accuracy: 0.4786 - val_species_loss: 1.0604 - learning_rate: 0.0010
Epoch 4/6
51/51 ————— 16s 304ms/step - health_accuracy: 0.4916 - health_loss: 56.6672 - loss: 167.1257 - species_accuracy: 0.3902 - species_loss: 110.5412 - val_health_accuracy: 0.6282 - val_health_loss: 0.6264 - val_loss: 1.6710 - val_species_accuracy: 0.4829 - val_species_loss: 1.0524 - learning_rate: 0.0010
Epoch 5/6
51/51 ————— 16s 305ms/step - health_accuracy: 0.5378 - health_loss: 54.1771 - loss: 161.1778 - species_accuracy: 0.4060 - species_loss: 106.8695 - val_health_accuracy: 0.6880 - val_health_loss: 0.6000 - val_loss: 1.6729 - val_species_accuracy: 0.4359 - val_species_loss: 1.0753 - learning_rate: 0.0010
Epoch 6/6
50/51 ————— 0s 298ms/step - health_accuracy: 0.4578 - health_loss: 69.6958 - loss: 162.0012 - species_accuracy: 0.4134 - species_loss: 92.3045
Epoch 6: ReduceLROnPlateau reducing learning rate to 0.0005000000237487257.
51/51 ————— 16s 306ms/step - health_accuracy: 0.4587 - health_loss: 69.6250 - loss: 162.1575 - species_accuracy: 0.4132 - species_loss: 92.4181 - val_health_accuracy: 0.6709 - val_health_loss: 0.6084 - val_loss: 1.6773 - val_species_accuracy: 0.3974 - val_species_loss: 1.0834 - learning_rate: 0.0010
Epoch 1/25
51/51 ————— 141s 1s/step - health_accuracy: 0.5630 - health_loss: 0.6954 - loss: 1.7495 - species_accuracy: 0.4535 - species_loss: 1.0530 - val_health_accuracy: 0.6239 - val_health_loss: 0.6308 - val_loss: 1.5042 - val_species_accuracy: 0.6880 - val_species_loss: 0.8928 - learning_rate: 1.0000e-04
Epoch 2/25
51/51 ————— 21s 413ms/step - health_accuracy: 0.7233 - health_loss: 0.5538 - loss: 1.4047 - species_accuracy: 0.6278 - species_loss: 0.8499 - val_health_accuracy: 0.8718 - val_health_loss: 0.4213 - val_loss: 0.9692 - val_species_accuracy: 0.8333 - val_species_loss: 0.5604 - learning_rate: 1.0000e-04

```

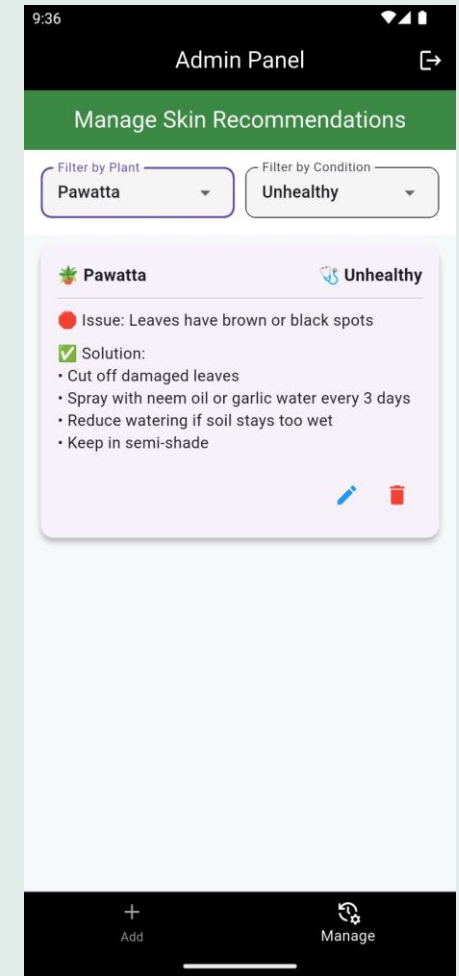
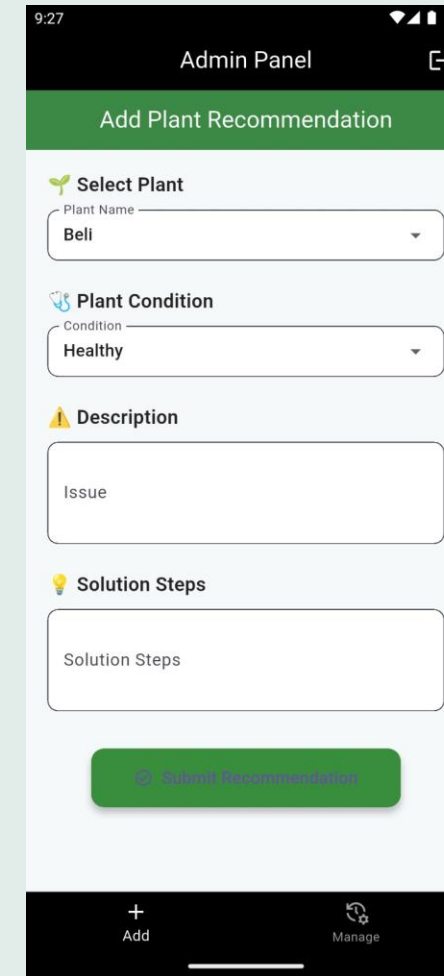
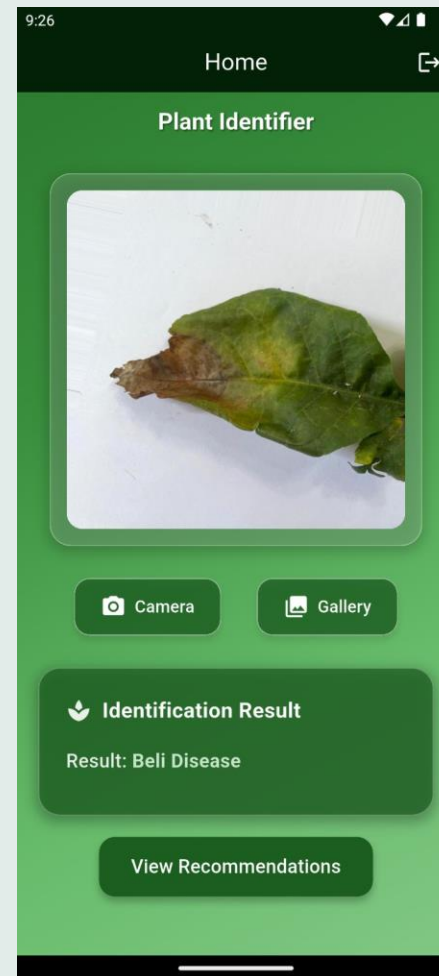
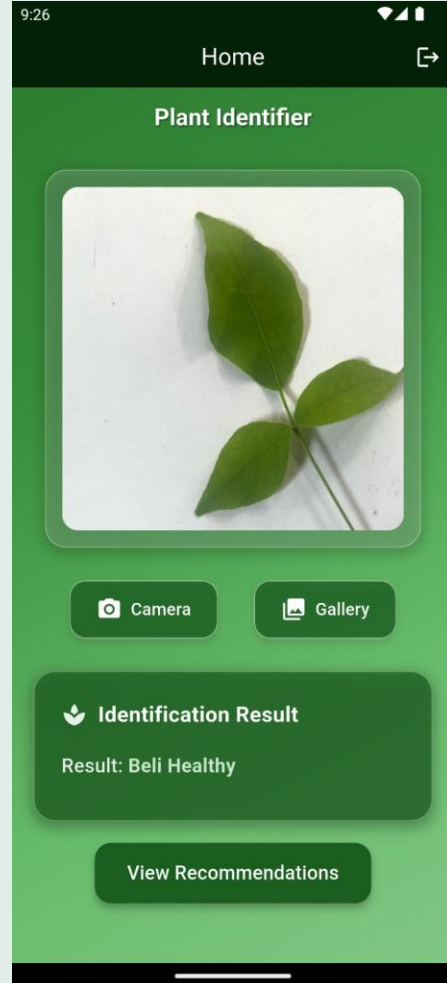
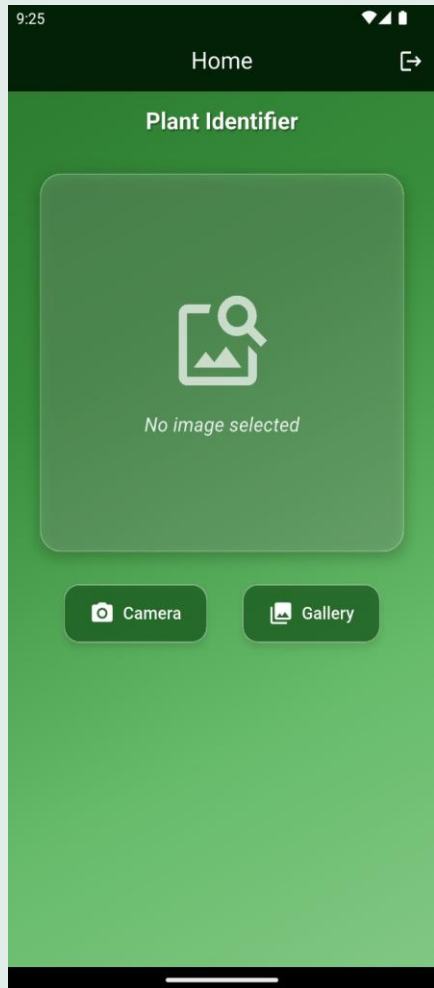
```

Epoch 13/25
51/51 ————— 20s 391ms/step - health_accuracy: 0.7957 - health_loss: 0.4300 - loss: 1.1081 - species_accuracy: 0.6950 - species_loss: 0.6772 - val_health_accuracy: 1.0000 - val_health_loss: 0.1340 - val_loss: 0.3203 - val_species_accuracy: 1.0000 - val_species_loss: 0.1856 - learning_rate: 1.0000e-04
Epoch 14/25
51/51 ————— 20s 392ms/step - health_accuracy: 0.7875 - health_loss: 0.4311 - loss: 1.0927 - species_accuracy: 0.7138 - species_loss: 0.6613 - val_health_accuracy: 1.0000 - val_health_loss: 0.1380 - val_loss: 0.3328 - val_species_accuracy: 1.0000 - val_species_loss: 0.1942 - learning_rate: 1.0000e-04
Epoch 15/25
51/51 ————— 20s 392ms/step - health_accuracy: 0.7871 - health_loss: 0.4279 - loss: 1.0748 - species_accuracy: 0.7248 - species_loss: 0.6458 - val_health_accuracy: 0.9872 - val_health_loss: 0.1484 - val_loss: 0.3333 - val_species_accuracy: 1.0000 - val_species_loss: 0.1855 - learning_rate: 1.0000e-04
Epoch 16/25
50/51 ————— 0s 387ms/step - health_accuracy: 0.7996 - health_loss: 0.3938 - loss: 1.0136 - species_accuracy: 0.7385 - species_loss: 0.6189
Epoch 16: ReduceLROnPlateau reducing learning rate to 4.999999873689376e-05.
51/51 ————— 20s 394ms/step - health_accuracy: 0.7994 - health_loss: 0.3942 - loss: 1.0147 - species_accuracy: 0.7379 - species_loss: 0.6197 - val_health_accuracy: 0.9915 - val_health_loss: 0.1424 - val_loss: 0.3240 - val_species_accuracy: 1.0000 - val_species_loss: 0.1825 - learning_rate: 1.0000e-04
Epoch 17/25
51/51 ————— 20s 392ms/step - health_accuracy: 0.7865 - health_loss: 0.4233 - loss: 1.0640 - species_accuracy: 0.7394 - species_loss: 0.6395 - val_health_accuracy: 0.9915 - val_health_loss: 0.1426 - val_loss: 0.3260 - val_species_accuracy: 1.0000 - val_species_loss: 0.1839 - learning_rate: 5.0000e-05
Epoch 18/25
51/51 ————— 20s 392ms/step - health_accuracy: 0.8001 - health_loss: 0.4101 - loss: 1.0498 - species_accuracy: 0.7275 - species_loss: 0.6388 - val_health_accuracy: 0.9915 - val_health_loss: 0.1414 - val_loss: 0.3239 - val_species_accuracy: 1.0000 - val_species_loss: 0.1828 - learning_rate: 5.0000e-05

```

Accuracy – 100%

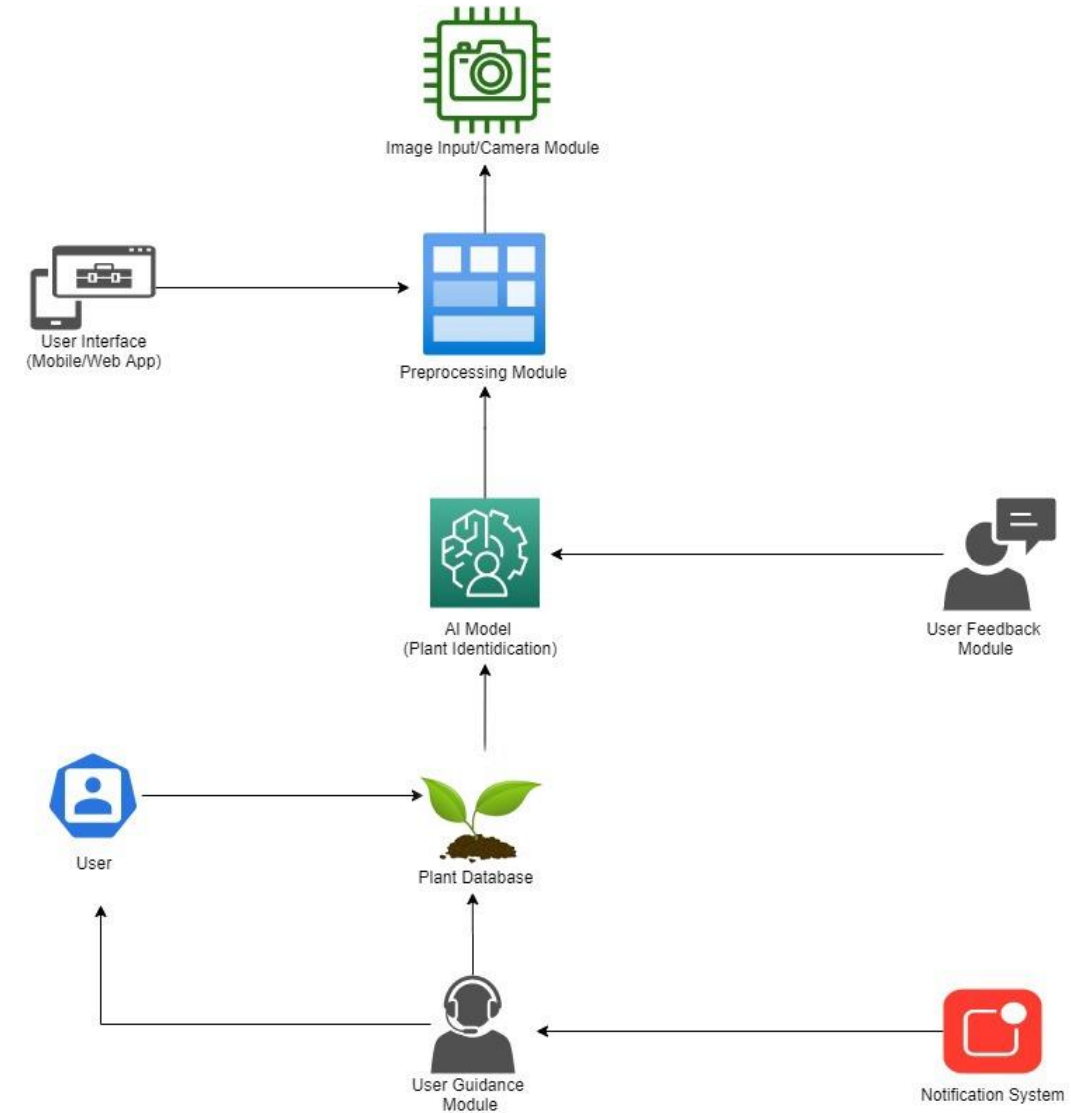
Ui design prototype



METHODOLOGY

System Diagram

- **1. Leaf Image Capture:**
The user uploads a clear image of the medicinal plant leaf via the mobile app.
- **2. Image Preprocessing:**
The system enhances the image quality and segments the leaf to focus on critical features.
- **3. AI-Based Plant Identification:**
The refined image is analyzed by the AI model, which accurately determines the plant species.
- **4. Result Display:**
The app presents the plant name, detailed medicinal properties, usage guidance, and—in cases where the leaf appears diseased—tailored treatment recommendations and preventive measures.



WBS - AI-Powered Plant Identification and Usage Guidance System

Data Collection

Collect plant images in real-time

Annotate images with plant names and medicinal properties

Ensure data quality and consistency

Store data securely

AI Model Development

Design and build an AI model for plant identification

Train and test the model with the dataset

Optimize for speed and accuracy

Plant Database Creation

Develop a comprehensive database of Ayurvedic plants

Link each plant to its medicinal benefits and applications

Integrate real-time plant identification with the database

User Interface Design

Design a user-friendly interface for plant image uploads

Display real-time identification results

Provide easy-to-understand usage guidance based on the plant

Usage Guidance Module

Provide users with Ayurvedic treatments based on plant identification

Suggest applications and usage based on medical conditions

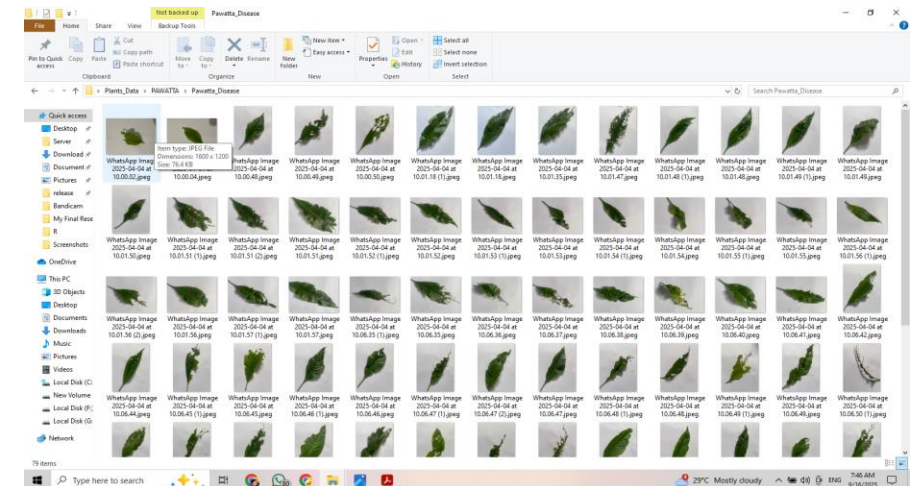
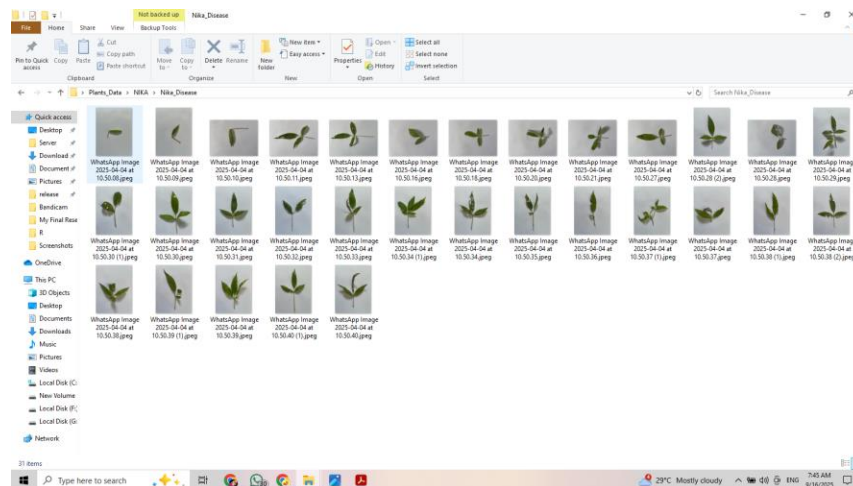
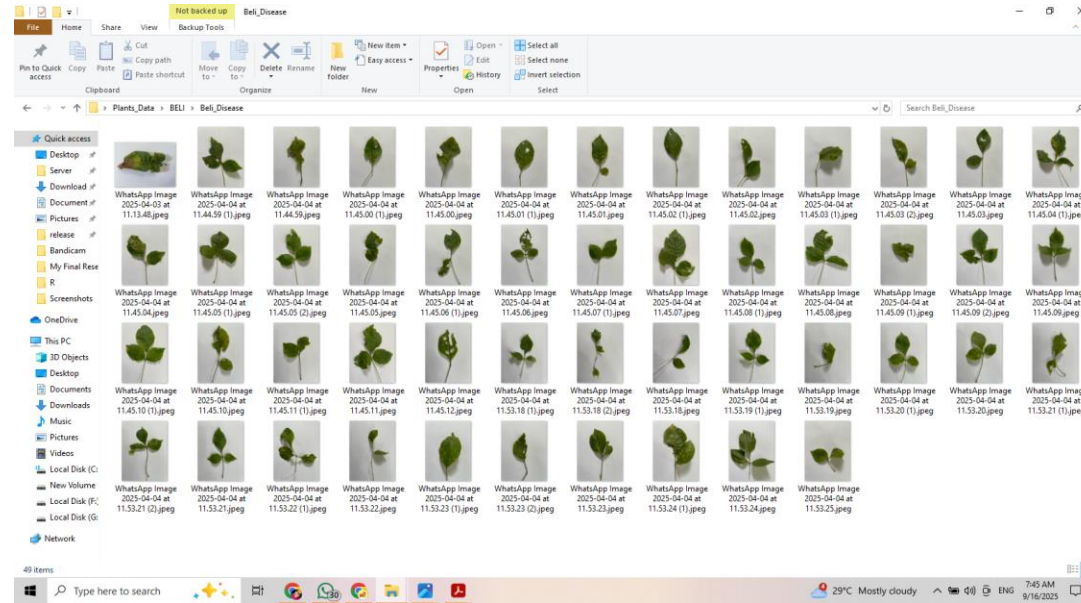
Testing and Validation

Test for accuracy with real plant images

Validate guidance with Ayurvedic experts

Collect feedback and improve the system

Successful Data Collection for Model Training



Current Progress & Next Steps

-  **Completed: Chapters 1–4**
-  **Next:**
 - Frontend-developed
 - Integrate Ayurvedic knowledge base
 - Mobile app deployment
 - Write Chapters 5–7



Figure 9: Research Progress Tracker

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